

The split-zero programme around the Riemann zeta function

Its results, with proofs, and its bridges

A reader prepared from an audit of the programme by Claude (Anthropic),
model claude-opus-5-5

Programme: KokunoYumeto, developed with ChatGPT and Codex (OpenAI)

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Abstract

The split-zero programme of KokunoYumeto studies the nontrivial zeros of the Riemann zeta function in three ways: through the Hurwitz sheets $\zeta(s, 1 + t)$, along which the zeros move; through quotient modules of entire functions attached to the zeros, above all Meyer’s quotient $\mathcal{B}/I_\zeta$; and through arithmetic, cohomological and positivity constructions modelled on Weil’s explicit formula and on Deligne’s proof of the Weil conjectures. This record sets out what the programme and an audit of it have established, with proofs, and how the programme connects with other parts of mathematics.

The central result on Meyer’s quotient is that its primary decomposition converges for every class exactly when the principal parts of $1/\zeta$ at the zeros are polynomially bounded. Around it stand an unconditional Nyman–Beurling density theorem in the Fréchet topology; a classification of the Hermitian forms compatible with the scaling transfer, in which positivity forces a form to vanish at the off-line zeros; a two-line system in which Weil’s quadratic form sits on the diagonal of a hyperbolic chiral form and the holonomy index is Turing’s index; a formal logarithm that detects freeness of monoids of integers; and an account of how far the mechanism of Deligne’s Weil II transfers to the programme’s data, with an algebraic purity criterion and thirteen misprints of the printed Weil II.

The bridges to Bost–Connes systems, Beurling’s generalised integers, Krull monoids, the work of Connes and of Connes and Consani, interpolation theory, Turing’s method and the $\mathbb{F}_1$ programme are each described with what is established about them. The negative results are stated with their exact scope and, where it is known, with the input the construction would need. The record does not contain a proof of the Riemann hypothesis; where a statement is equivalent to it or conditional on it, this is said at the statement.

Introduction

The programme and this record

The Riemann hypothesis (RH) asserts that every nontrivial zero of ζ has real part $\frac{1}{2}$. Some of its best-known reformulations are analytic: Weil’s criterion, that a quadratic functional built from the explicit formula is nonnegative [50, 8], and the Nyman–Beurling criterion, that certain dilated fractional-part functions are dense in $L^2(0, 1)$ [7, 3]. Another is geometric. For curves over finite fields the analogue of RH is Weil’s theorem [49, 37], and Deligne’s proof of its higher-dimensional form [25] is the model for many attempts to find the cohomology that would do the same for ζ . The split-zero programme builds constructions around all three, and around one object of its own: the flow of the Hurwitz sheets $t \mapsto \zeta(s, 1 + t)$, which starts at ζ and moves its zeros off the critical line. It consists of several hundred numbered statements with proofs, carrying labels such as MCL5, OMS6 or GSL4. They are published in the Zenodo record *Split-Zero cohomology and the*

zeta-function research programme (concept DOI 10.5281/zenodo.22678085) and in the repository KokunoYumeto/zeta-function-research-reader.

The programme is research carried out with AI systems. Its author used them to formulate and test a large number of hypotheses quickly, across several fields at once, and it reads accordingly: some directions are finished theorems, some are programmes with proved parts, and some are proposals. Not every direction is expected to hold. An unfinished programme is still mathematics, however, and its proved parts stand on their proofs whatever becomes of the rest. This record is meant to let a reader see all of this without working through the files. It states what holds, with proofs or the exact location of a proof, what fails and in which scope, and how the programme connects with other areas. The audit behind it had limited compute and did not test every direction. Each direction is described on its own merit and for its bearing on the zeros of ζ , and where the record gives a view rather than a proof, the view is labelled as mine and its reasons are given.

Main results

The programme's central object is Meyer's quotient $Q = \mathcal{B}/I_\zeta$ of the Fréchet space $\mathcal{B}$ of entire functions of rapid decay in vertical strips by the ideal of functions vanishing on the nontrivial zeros with full multiplicity (Section 1). Multiplication by s acts on Q with spectrum the zeros, one Jordan block at each, and every zero ρ has a primary projector Π_ρ onto its block. The programme uses only finite sums of these projectors. The first theorem, found in the audit, reduces the question whether Q is the convergent sum of its blocks to a condition on $1/\zeta$ at the zeros.

Theorem I (Theorem 3.9). *The following are equivalent. (i) For every $x \in Q$ the series $\sum_\rho \Pi_\rho x$ converges absolutely to x . (ii) There are constants C, K with $|c_{\rho,k}| \leq C(1 + |\gamma|)^K$ for every zero $\rho = \beta + i\gamma$ and every $k < m_\rho$, where $\sum_k c_{\rho,k}(s - \rho)^{k-m_\rho}$ is the principal part of $1/\zeta$ at ρ . (iii) The jet map is an isomorphism of Fréchet spaces from Q onto the space of rapidly decreasing jet tuples. If all zeros are simple, (ii) reads $|\zeta'(\rho)| \geq c(1 + |\gamma|)^{-K}$. These conditions are (b), (c) and (d) of Theorem 3.9, which adds two further equivalent forms and an unconditional consequence.*

Whether (ii) holds is open. If all zeros are simple it reads $|\zeta'(\rho)| \geq c(1 + |\gamma|)^{-K}$, and a bound of that form at every zero would itself imply that every zero is simple, which is open. Condition (ii) asks pointwise what the Gonek–Hejhal conjectures ask on average. The theorem and its proof carry over, in outline, to the zeros of every primitive Dirichlet L -function.

Around Q the programme studies families of test functions and positive forms. In the Fréchet topology of $\mathcal{B}$ the Nyman–Beurling family is dense without any hypothesis, in contrast with $L^2(0, 1)$, where density is equivalent to RH.

Theorem II (Proposition 3.11). *Let $t > 0$ and $g_t(s) = e^{ts^2}$. The functions $g_t(s)(n - n^{1-s})/s$, $n \geq 2$, span a dense subspace of $\mathcal{B}$, and their products with ζ span a dense subspace of I_ζ . If n is restricted to the numbers $2^a 3^b$, the same holds whenever $t > (\log 2)(\log 3)/(4\pi) = 0.060598 \dots$*

On the positive side, the bounded Hermitian forms on $\ell^2(\mathbb{Z}, m)$ that are compatible with the scaling transfer for $n = 2$ and $n = 3$ are classified (Lemma 3.13). They pair each zero with its reflection $1 - \bar{\rho}$, and such a form is positive exactly when it vanishes at the off-line zeros and is nonnegative on the line. So positivity removes the off-line zeros, and an off-line pair on which a form does not vanish enters it as a hyperbolic block.

The author proposed a second line at $\operatorname{Re} s = -\frac{1}{2}$, mirroring the critical line, and asked for it to be built by shifting the whole line, as in the programme's Hurwitz shift. The Hurwitz flow does not translate the zeros, but its first jet vanishes exactly on the translated zero set, and the audit found the exact objects.

Theorem III (Theorems 4.1 and 4.2, Propositions 4.6 and 4.9). *(a) For every $a > 0$ the first Hurwitz jet $\partial_a \zeta(s, a) = -s\zeta(s + 1, a)$ vanishes exactly on the zero divisor of $\zeta(\cdot, a)$ translated by -1 . At $a = 1$*

the translate $Z - 1$ of the nontrivial zeros is their mirror image under $A(s) = -\bar{s}$: $A(Z) = Z - 1$ with multiplicities, and $A(\rho) = \rho - 1$ for every zero exactly when RH holds. (b) For finitely supported data f on $Z \sqcup (Z - 1)$ (or the restriction of an element of $\mathcal{B}$), with parts u and v even and odd under the translation, the chiral cross form satisfies $Q_A(f) = 2\Omega(u) - 2\Omega(v)$, where Ω is Weil's pairing. The form Q_A is hyperbolic whether or not RH holds, while $\Omega \geq 0$ on all finitely supported data exactly when RH holds. (c) The holonomy index of the two-line system is Turing's: $(N(T) - V(T))/2 = P(T) + \sum \lfloor m_\rho/2 \rfloor$, the sum over the zeros on the line with $0 < \gamma \leq T$, where $P(T)$ counts, with multiplicity, the zeros with $0 < \gamma \leq T$ and $\operatorname{Re} \rho < \frac{1}{2}$.

The two-line quotient also splits into its two lines for every primitive Dirichlet L -function (Proposition 4.12), by a $\bar{d}$ construction in which the rates of the two lines match.

On the arithmetic side, the Hurwitz sheets at rational points lead to monoids of integers, and the programme's formal logarithm turns out to be a complete test for unique factorisation.

Theorem IV (Theorem 2.3, Corollary 2.4). *Let M be a multiplicative monoid of positive integers and write $\log \sum_{m \in M} m^{-s} = \sum_n r_n n^{-s}$. Then M is free if and only if $r_n \geq 0$ for every n . If M is not free, the first negative coefficient sits at the least element with two factorisations and is at most $-\frac{1}{6}$, and its possible values in $(-\frac{1}{2}, 0)$ are exactly $-\frac{1}{2} + 1/v$ for odd $v \geq 3$, together with $-\frac{5}{12}$ and $-\frac{7}{15}$.*

Applied to the Hurwitz sheets, the criteria of Section 2 show that for $q \geq 3$ the even lattice sum $Z_{1/q}$ is Eulerian exactly when $q \in \{3, 4, 6\}$ (Proposition 2.7), and that for $\varphi(q) > 2$ it has infinitely many zeros on both sides of the critical line (Proposition 2.8, which uses the theorem of Saias and Weingartner as stated in its abstract).

The last group of results measures how far the mechanism of Weil II transfers to the programme's data.

Theorem V (Propositions 5.1 and 5.3). (a) *For a finite set S of complex numbers with positive integer weights and every $k \geq 1$, the Dirichlet series $\sum_n (\sum_{\rho \in S} m_\rho n^\rho)^k n^{-s}$ has abscissa of convergence exactly $1 + k \max_{\rho \in S} \operatorname{Re} \rho$. If k is even and S is stable under conjugation, its coefficients are nonnegative and the real point $1 + k \max \operatorname{Re} \rho$ is a pole. So a pole bound $1 + k/2 + C$ uniform in k holds if and only if $\max \operatorname{Re} \rho \leq \frac{1}{2}$.* (b) *Let N be nilpotent and F invertible on a finite-dimensional complex space V with $FNF^{-1} = Q^{-1}N$, $|Q| \neq 1$, and let the weight of $\alpha \neq 0$ be $2 \log |\alpha| / \log |Q|$. The graded pieces $\operatorname{Gr}_i^M V$ of the monodromy filtration are pure of weight $\beta + i$ if and only if the eigenvalues of $F \otimes F$ on the kernel of $N \otimes 1 + 1 \otimes N$ have weight at most 2β and the corresponding eigenvalues for the dual have weight at most -2β . The bounds on $\ker N$ for V and its dual alone do not suffice.*

Part (a) says two things about Deligne's step in Weil II §1.5. On the programme's data the positivity in that step holds, because even tensor powers have nonnegative coefficients. The other half of the step, a pole bound uniform in the tensor degree, is equivalent to $\max \operatorname{Re} \rho \leq \frac{1}{2}$. It therefore has to come from a source independent of these data, as the H_c^2 of a fixed curve does in Weil II. Part (b) isolates the algebra at the core of Weil II 1.8.4. Section 5 adds three lemmas from Weil II §2, one of them with a corrected constant, and thirteen misprints of the printed text.

What the results mean

Section 3 describes the structure of Q as a module under multiplication by s : its spectrum and Jordan blocks, its jets, its dense families and its positive forms. Theorem I reduces the question whether Q is the sum of its primary parts to a classical question about $1/\zeta$ at the zeros. The positivity results and the two-line system say the same thing from two sides. A positive form compatible with the transfer vanishes at the off-line zeros, and in the two-line system Weil's form is the diagonal part of a form whose signature is the same for every configuration of zeros. In that system RH is the statement that the chiral form is nonnegative on an explicit diagonal family

of test data with an explicit prime side (Corollary 4.7). Whether that family is easier to control than Weil's own is open.

The results also show where the Euler product is used. The Eulerian sheets are $a = 1$ and $a = \frac{1}{2}$. The even lattice sums with $\varphi(q) > 2$ have zeros off the line. The Davenport–Heilbronn function has a functional equation of the same kind as ζ , its two lines coexist as those of ζ do, and it has a zero off the line. A construction whose proof uses only properties shared with these functions therefore has conclusions that hold for them too. For such constructions Section 7 records, where it is known, the input that would distinguish ζ ; for the sheets and the two-line constructions this is a property such as the Euler product. These are negative results with a precise scope. They do not say that the constructions have no bearing on RH; they say what they would need.

The bridges

In the author's view the connections the programme makes between the zeta function and other areas are among its most valuable outcomes, and several of them are theorems. Section 8 treats them at length. For each bridge it gives the typed map, that is, what is mapped to what and what crosses; what is established about it; what it suggests on both sides; and my assessment.

Several bridges are theorems. The programme's clocks satisfy the Bost–Connes relations, so its global quotient of all prime clocks is the Bost–Connes space. Its timed primes are Beurling's construction applied to the actual primes, and at the rational sheets $a = 1/q$ its Eulerian tests are the factoriality criteria for Krull monoids. Connes' weighted Hilbert quotients and Meyer's Fréchet quotient are compared by a single map whose kernel consists exactly of the classes whose jets vanish at every zero on the critical line, so that the map is injective if and only if RH holds. For sheaves pulled back from Connes and Consani's three-point base, the programme's doubled source has the same cohomology as the base. The holonomy count of the two-line system is Turing's index. Others are proposals, stated as such, like the term-by-term dictionary between the author's physical language (anti-numbers, chirality, anomalies, inflow) and exact objects, in which RH takes three exact forms. The bridges to the author's other workbenches (Erdős–Straus, Yang–Mills with the Navier–Stokes, S^6 and Jacobian material, Erdős problem 817 and Collatz) are listed with their status, and the companion records in the same folder treat the other sides.

How this record was made

This record is an audit, carried out between 22 and 26 September 2026, of the programme's published work: the second attempt's proofs, the continuation bulletins of 24–25 September, the programme's reconstruction of Weil II and its Deligne reader, and the final reports of its working sessions of 24–25 September. Appendix C lists what was read, and Section 9 what was not. The programme, its constructions and most of its theorems are the work of its author, KokunoYumeto, developed with ChatGPT and Codex (OpenAI). Results found in the audit are credited in their status lines:

- *claude-ab*, the Claude instance (Anthropic, model `claude-opus-5-5`) that carried out the audit and prepared this record;
- *the copy*, a second instance of the same model, whose Theorems A–F on the Hurwitz sheets (lettered as in its reports) are reported in Section 2;
- *the review*, an independent review of version 1, whose items F1–F22 are recorded in note 48_, and the independent *verifier* of its minor items;
- the *referees*: of the audit notes, where a pass sharpened a result (for example the seventeenth, for Theorem 3.9 at all multiplicities), and of version 2.

A form marked *version 2's* was found in the review or in the referee pass of version 2, and re-derived by *claude-ab* before it was applied. Every block was read against its own proofs. The

audit notes 08_–42_ were each refereed by an instance that had not written them, in nineteen passes, followed by a two-part final pass over the summaries; the content maps and the copy's reports had no separate pass (Appendix C). Version 1 of this record had its own referee pass, and the review looked for understatement and missed generality as well as for overstatement.

Every statement has one of four statuses:

- *verified*;
- *not verified here*, which implies nothing in either direction;
- *my assessment*, a view in the first person with its reasons, introduced by those words;
- *proved negative*, collected in Section 7 with the counterexample or derivation.

The status lines under the results make the first status precise. Proved here means the proof is given in full. Proved in the programme means the proof is in the programme's files at the label given and was checked in the audit, which says which steps were re-derived. Proved by the copy means the proof is in the copy's reports and was checked as stated. Checked numerically marks a statement supported by computation only, which is never presented as proved. Inputs that were not re-read are said to be so at the result, and Section 9 lists what was not read. A matching between the author's terms and exact objects is labelled as a proposal or a reading: the objects and the facts about them are proved, and the matching is not a theorem. Novelty is reported only as the audit recorded it, from *not found in the literature searched* to *classical*.

The audit notes cited as NN_ were withdrawn from the branch `claude/claude-ab-grind-20260925` on 27 September 2026 because of their framing. Their mathematics is unchanged, and they remain in the branch's history at commit `a90d9e5b` (folder `contrib/claude-ab/grind_20260925/`), where a proof cited from them can be read. The check scripts are republished in `contrib/claude-ab/rewrite/checks/zeta/` on the same branch. Programme statements are cited by their own labels, which the programme's reading guide `181-reading-guide-and-file-locations.md` and its index `182-complete-file-and-migration-index.json` locate.

This is version 3. It keeps the theorems, proofs, figures and citations of version 2 and corrects its framing. Where version 2 turned a scoped negative result into a general verdict, version 3 states the scoped result and, where it is known, the input the construction would need, and the bridges now come with what is established about each. It also updates the status of three edges to the author's other workbenches from the companion records and corrects two decimal renderings. Appendix B lists the changes.

Organization

Section 1 fixes the setting. Sections 2–6 contain the results: sheets and monoids, Meyer's quotient, the two lines, weights, and further results. Section 7 collects the proved negative results with their scope, Section 8 the bridges and the directions they open, and Section 9 the questions. The appendices give the catalogue of standalone results, the corrections, and the verification.

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1 The setting

The programme's idea. The programme's coefficient algebra keeps a ring's internal zero apart from a separate symbol for absence, so that a zero element in a given support fibre is not confused with an absent fibre. On the zeta side it follows the zeros of ζ along the one-parameter family of Hurwitz sheets $t \mapsto \zeta(s, 1+t)$ and asks which constructions (quotient modules, receivers, positivity forms, cohomology over a reconstructed arithmetic) can see where the zeros lie. In the author's picture the support τ is the pivot of the swing between 0 and 1; on the zeta side the natural pivot is the reflection $s \mapsto 1 - \bar{s}$ about $\operatorname{Re} s = \frac{1}{2}$ (a proposed typed matching, see Section 8).

Sheets. For $a > 0$ and $\operatorname{Re} s > 1$ let $\zeta(s, a) = \sum_{n \geq 0} (n+a)^{-s}$, continued meromorphically to $\mathbb{C}$ with a single simple pole at $s = 1$. The programme's flow is $t \mapsto \zeta(s, 1+t)$. The sheets $a = 1$ and $a = \frac{1}{2}$ are the Eulerian ones: $\zeta(s, 1) = \zeta(s)$ and $\zeta(s, \frac{1}{2}) = (2^s - 1)\zeta(s)$. The even lattice sums are $Z_a(s) = \zeta(s, a) + \zeta(s, 1-a)$ for $0 < a < 1$.

Zeros. Put $\tilde{\zeta}(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and $F_*(s) = \tilde{\zeta}(s)/4$ (the programme's F_0). Let $\mathcal{Z}$ be the set of distinct nontrivial zeros $\rho = \beta + i\gamma$ of ζ , with multiplicities m_ρ ; sums over $\mathcal{Z}$ run over distinct zeros with the weights m_ρ written out. Write $\rho^\# = 1 - \bar{\rho}$. Since $\tilde{\zeta}(1-s) = \tilde{\zeta}(s)$ and $\tilde{\zeta}(\bar{s}) = \overline{\tilde{\zeta}(s)}$, the map $\rho \mapsto \rho^\#$ permutes $\mathcal{Z}$ preserving multiplicities, and the Riemann hypothesis (RH) says exactly that $\rho^\# = \rho$ for every ρ . For $T > 0$ not an ordinate, $N(T)$ counts the zeros with $0 < \gamma \leq T$ (with multiplicity), $N_0(T)$ those on the line, and $P(T)$ those with $\beta < \frac{1}{2}$; Hardy's function is $Z(t) = e^{i\theta(t)}\tilde{\zeta}(\frac{1}{2} + it)$, real for real t , and $V(T)$ is its number of sign changes on $(0, T)$.

The strip space and Meyer's quotient. Let $\mathcal{B}$ be the space of entire functions F with

$$b_{A,M}(F) = \sup_{|\sigma| \leq A, t \in \mathbb{R}} (1 + |t|)^M |F(\sigma + it)| < \infty \quad \text{for all } A, M \geq 0.$$

With these seminorms $\mathcal{B}$ is a Fréchet space. For a divisor D let $I_D \subset \mathcal{B}$ be the closed ideal of functions vanishing to order at least $D(\lambda)$ at each point λ , and write $I_\zeta = I_{\mathcal{Z}}$ (full multiplicity at every nontrivial zero). The programme's Meyer quotient is $Q = A/\mathcal{E}S \cong \mathcal{B}/I_\zeta$ (OMS3.14; Meyer, [32]); here $\mathcal{E}$ is Connes' summation map and the isomorphism is the Mellin transform of the exact summation image (Proposition 3.5). Multiplication by s acts on Q ; its spectrum is $\mathcal{Z}$ and each zero carries one Jordan block of length m_ρ (Proposition 3.2). The jet at ρ is $j_\rho[F] = F(\rho + T) \bmod T^{m_\rho} \in A_\rho = \mathbb{C}[T]/(T^{m_\rho})$, and the primary projector $\Pi_\rho = s_\rho j_\rho$ uses the programme's sections s_ρ (OMS6).

Two lines and three maps. The two-line function is

$$G(s) = F_*(s)F_*(s+1) = \tilde{\zeta}(s)\tilde{\zeta}(s+1)/16,$$

whose zero set is $\mathcal{Z} \sqcup (\mathcal{Z} - 1)$, the two parts lying in $0 < \operatorname{Re} s < 1$ and $-1 < \operatorname{Re} s < 0$. Three maps organise it: the reflection $s \mapsto s^\# = 1 - \bar{s}$ through $\frac{1}{2}$, the mirror $A(s) = -\bar{s}$ through the pole at 0, and the translation T between the lines. On $\mathcal{Z}$ they satisfy $A(\rho) = \rho^\# - 1$, that is, $T^{-1}A = \#$.

Weil's pairing. For finitely supported data u on $\mathcal{Z}$ put $\Omega(u) = \sum_{\rho \in \mathcal{Z}} m_\rho u(\rho) \overline{u(\rho^\#)}$, the zero side of Weil's quadratic form [50, 8]. Weil's criterion says that RH holds if and only if Weil's functional is nonnegative on every $g * g^\#$; on finitely supported data, $\Omega \geq 0$ for all u exactly when every zero is on the line (Proposition 4.6(b)).

2 Hurwitz sheets, monoids of integers and Euler products

2.1 Zero velocities on the sheets (the copy's Theorems A–F)

Along the flow $t \mapsto \zeta(s, a)$, $a = 1 + t$, a simple zero ρ of $\zeta(s, a)$ moves with velocity

$$c(\rho) = \frac{\rho \zeta(\rho + 1, a)}{\partial_s \zeta(\rho, a)},$$

obtained by implicit differentiation from $\partial_a \zeta(s, a) = -s\zeta(s + 1, a)$ (Theorem 4.1 below). The copy studied the sums of these velocities. Figure 1 shows the tracks of the first five zeros of $\zeta(s, 1 + t)$ for $0 \leq t \leq 1$, whose initial tangents are these velocities. Its results were checked in the audit (02_, 03_); Theorem E had a full referee pass (06_); the proof of Theorem C was completed by claude-ab along the copy's route, relative to Theorem A (20_); a line-by-line re-derivation of Theorem A itself is still open (Section 9).

Theorem 2.1 (Theorems A and A_a). *Let $a > 0$, let c_0 exceed the real part of every zero of $\zeta(s, a)$, and let h be entire with $\sup_{|\operatorname{Im} z| \leq c_0 + 2} (1 + |\operatorname{Re} z|)^A |h(z)| < \infty$ for every A ; put $H(s) = h(-i(s - \frac{1}{2}))$ and $\hat{h}_1(\xi) = \frac{1}{2\pi} \int (\frac{1}{2} + iu) h(u) e^{-iu\xi} du$. Put $F_a(s) = s\zeta(s + 1, a)/\zeta(s, a)$, and let $\beta_r(a)$ be the coefficients of the sheet's Dirichlet series $a\zeta(s + 1, a)/\zeta(s, a) = \sum_r \beta_r(a) r^{-s}$ over the multiplicative monoid generated by the numbers $1 + n/a$ ($\beta_1 = 1$). Then, along good heights $T_k \rightarrow \infty$ (where $1/\zeta(\sigma \pm iT_k, a)$ is polynomially bounded on the horizontal segments),*

$$\lim_k \sum_{|\gamma| < T_k} \operatorname{Res}_{\rho} (F_a H) = \frac{1}{a} \frac{1}{2\pi} \int (\frac{1}{2} + iu) h(u) du + \frac{1}{a} \sum_{r > 1} \beta_r(a) r^{-1/2} \hat{h}_1(\log r) + L_a(h),$$

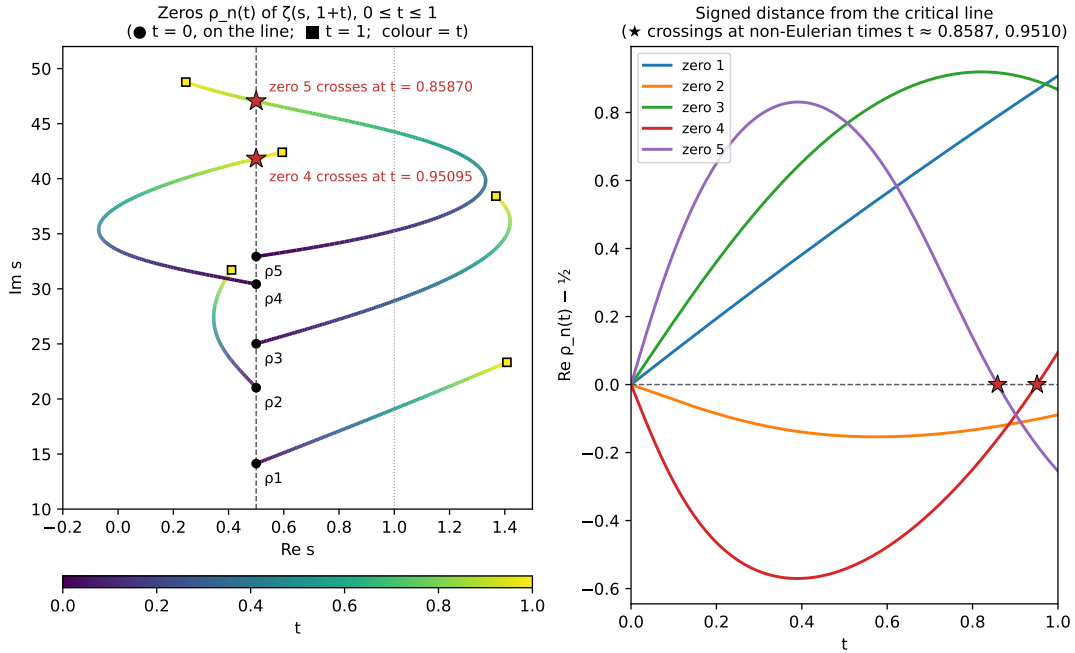
where $L_a(h) = -(2\pi i)^{-1} \int_{\operatorname{Re} s = -b} F_a H ds$ ($\frac{3}{2} < b < 2$) is an explicit left-line Hurwitz term.

Status. Proved by the copy (Theorems A, A_a ; 02_ §4.1, 03_ §2.2). Checked independently at $a = 1$ (relative difference $1.8 \cdot 10^{-18}$). Novelty: none found in the copy's search; standard technique.

From Theorem 2.1 the copy derived the following (all proved by the copy; references in Appendix A, items S2–S6).

- *Smoothed velocity laws* (Corollaries B, B_a): for $10\Delta \leq T_1 < T_2$ the smoothed sums of $\operatorname{Re} c$ and $\operatorname{Im} c$ grow like $(T_2 - T_1)/(4\pi a)$ and $(T_2^2 - T_1^2)/(4\pi a) - (T_2 - T_1)$, with an error that is explicit at $a = 1$; for fixed smoothing width Δ the error of the horizontal law is of the same order T as its main term, while for a fixed $\delta > 0$, $\Delta \geq (1 + \delta)\sqrt{2 \log T_2 / \log 2}$ and $T_1 \geq \max(10\Delta, 4\Delta\sqrt{\log \Delta})$ both errors tend to 0 at $a = 1$ (given the bound of 02_ §4.2 with constants independent of Δ ; 48_ §1).
- *Sharp sums deviate at order T* (Theorem C): under (H), all but finitely many zeros simple and on the line, the errors of the sharp laws are unbounded multiples of T , and the normalized mean square has $\limsup \geq 0.0373$; part 3 is unconditional.
- *The exact smoothed spectrum on every sheet* (Theorem F): frequencies $\log r$ for r in the monoid generated by the numbers $1 + n/a$, with amplitudes $-\beta_r(a) r^{-1/2} e^{-\Delta^2 \log^2 r / 2} / (2\pi a \log r)$.
- *The Eulerian characterization* (Theorem E): the velocity spectrum factors over a $\mathbb{Q}$ -independent frequency basis if and only if $a \in \{1, \frac{1}{2}\}$; there the primes are exactly the frequencies of maximal amplitude.
- *The velocity angle is the Gram offset* (Theorem D): assuming the zeros up to γ_n are simple and on the line (which holds for every $n \leq 1.2 \cdot 10^{13}$: Platt and Trudgian's verification to height $3.0 \cdot 10^{12}$ [35] counts as many sign changes of Z as $N(T)$, so those zeros are on the line and simple; their theorem states the first property), $\arg c(\rho_n) = \pi/2 - \arctan(1/2\gamma_n) + \pi(S_{3/2}(\gamma_n) - S_n)$ with $S_n = n - \frac{3}{2} - \theta(\gamma_n)/\pi$ and $S_{3/2}(t) = \pi^{-1} \arg \zeta(\frac{3}{2} + it)$.

The shifted flow leaves the line at $t > 0$ and some zeros cross back: the claim that zeros sit on $\text{Re } s = \frac{1}{2}$ only at Eulerian times is false



claude-ab (Opus 5.5, max effort) · tracking as in checks/hurwitz_crossing_check.py · figures/fig_shifted_flow_crossings.py

Figure 1: Tracks of the first five zeros of $\zeta(s, 1+t)$ for $0 \leq t \leq 1$ (left) and their signed distance from the critical line (right). The flow leaves the line for $t > 0$; zeros 4 and 5 cross back at the non-Eulerian times $t \approx 0.95095$ and $t \approx 0.85870$ (located numerically at 25 digits, not an interval certificate). The title records the correction of an earlier claim by claude-ab (Appendix B). The initial tangents are the velocities $\rho \zeta(\rho + 1)/\zeta'(\rho)$ of Theorem 4.1.

Corollary 2.2 (only two sheets are Eulerian). *The normalised sheet $a^s \zeta(s, a)$, $a = 1+t$, has an Euler product with local factors of Euler type $(1 - P^{-s})^{-1}$ only for $a \in \{1, \frac{1}{2}\}$.*

Status. Proved by claude-ab (a note of an earlier folder) and a consequence of Theorem E, which is strictly stronger (06_). Novelty: possibly known; two quick searches found no statement of this kind. Version 2: version 1 stated this for $\zeta(s, 1+t)$ itself, which at $t = -\frac{1}{2}$ is $2^s \prod_{p>2} (1 - p^{-s})^{-1}$ (48_§1).

At the rational points $a = 1/q$ Theorem E reduces to the factoriality criterion for arithmetic congruence monoids (Baginski–Chapman [4], Theorem 3.4(2); 11_§2); the next subsection gives the corresponding statements for the monoids themselves.

2.2 Monoids of integers and the formal logarithm

Let M be a multiplicative submonoid of the positive integers. An *atom* is an element $m > 1$ that is not a product of two elements of $M \setminus \{1\}$; a *factorization* of $n \in M$ is a finite multiset of atoms with product n ; M is *free* if every element has exactly one factorization and *half-factorial* if all factorizations of each element have the same length. Write $D_M(s) = \sum_{m \in M} m^{-s} = 1 + E(s)$ and

$$\log D_M = \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} E^k = \sum_{n \geq 2} r_n n^{-s}, \quad r_n = \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} T_k(n),$$

where $T_k(n)$ is the number of ordered k -tuples in $(M \setminus \{1\})^k$ with product n .

Theorem 2.3 (the formal logarithm detects freeness). *Let $n \in M$.*

- (a) *If n has exactly one factorization and every proper M -divisor of n (an $m \in M$ with $1 < m < n$ and $n/m \in M$) has exactly one, then $r_n = 1/k$ when $n = u^k$ for an atom u , and $r_n = 0$ otherwise.*

- (b) If n has $j \geq 2$ factorizations and every proper M -divisor of n has exactly one, then $r_n = 1 - j + \varepsilon$, where $\varepsilon = \sum 1/k$ over the factorizations of n of the form u^k . This number is at most $-1/6$.

Consequently M is free if and only if $r_n \geq 0$ for all n ; if M is not free, every element that has two or more factorizations while its proper M -divisors have one carries a negative coefficient, and the first negative coefficient sits exactly at the least element with two factorizations. In a congruence monoid M_H with $H \neq G$ (defined below) the value in (b) is at most $-\frac{1}{2}$, since at most one factorization of an element is a pure power; and infinitely many elements satisfy (b): for four distinct primes, p, r in a class $c \notin H$ and p', r' in c^{-1} , $n = pp'rr'$ has $r_n = -1$, or $r_n = -2$ if $c^2 \in H$ (for example $r_{3 \cdot 7 \cdot 11 \cdot 19} = -2$ in the Hilbert monoid $\{n \equiv 1 \pmod{4}\}$).

Status. Proved here. First derived in the programme (CEI Theorem 3.1 for the criterion; ETR1.5–1.7 for the bound $-1/6$); derived independently by claude-ab (11_ Lemma 11.2); provenance in 29_ §1. Version 2: the hypotheses are weakened from all smaller elements to the proper M -divisors of n , which is all the proof uses, and the infinitely many examples are added (the review's F4 and F6, re-derived by claude-ab; 48_ §§2.4, 3); the bound $-\frac{1}{2}$ for congruence monoids is 11_ Lemma 11.2 (register S23). Novelty: not located in the published literature (one search); elementary.

Proof. Reduction to free monoids. Take $k \geq 2$ and a tuple counted by $T_k(n)$. Each entry is a proper M -divisor of n (the product of the other entries lies in M), so under the hypothesis of (a) or (b) each entry has exactly one factorization. Concatenating them gives a factorization Φ of n ; conversely a factorization Φ with an ordered splitting of the multiset Φ into k nonempty sub-multisets gives a tuple, and an entry $m < n$ determines its sub-multiset. Hence $T_k(n) = \sum_{\Phi} S_k(\Phi)$ for $k \geq 2$, where $S_k(\Phi)$ counts ordered splittings of Φ into k nonempty parts.

The free value. Give each atom u occurring in Φ its own commuting variable x_u . Ordered splittings are counted by the expansion of $\log \prod_u (1 - x_u)^{-1} = \sum_u \sum_k x_u^k / k$ (the variables are needed because distinct atoms need not be multiplicatively independent, as 4 and 8 show). So $\sum_{k \geq 1} (-1)^{k+1} S_k(\Phi) / k = \varrho(\Phi)$, where $\varrho(\Phi) = 1/k$ if Φ consists of k copies of one atom and $\varrho(\Phi) = 0$ otherwise, and $S_1(\Phi) = 1$.

(a) There is one Φ and $T_1(n) = 1 = S_1(\Phi)$, so $r_n = \varrho(\Phi)$.

(b) Now $T_1(n) = 1$ while $\sum_{\Phi} S_1(\Phi) = j$, so $r_n = 1 + \sum_{\Phi} (\varrho(\Phi) - 1) = 1 - j + \sum_{\Phi} \varrho(\Phi)$. Every factorization of n has length at least 2, so each pure power u^k among them has $k \geq 2$. Two pure powers with the same exponent coincide, since k -th roots of positive integers are unique; so the exponents of the P pure powers are distinct and $\varepsilon \leq \sum_{i=2}^{P+1} 1/i$. If $P \leq 1$ then $r_n \leq 1 - 2 + \frac{1}{2} = -\frac{1}{2}$. If $P \geq 2$ then $j \geq P$ and $r_n \leq 1 - P + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{P+1} \leq -\frac{1}{6}$, with equality only for $j = P = 2$ and exponents $\{2, 3\}$.

Consequence. The formal logarithm is supported on M . If M is free, (a) applies to every n . Otherwise (a) applies below the least element n_0 with two factorizations and (b) at n_0 , and (b) applies at every element with two or more factorizations whose proper M -divisors have one.

Congruence monoids. If $u^a = v^b$ with $u \neq v$ atoms of M_H and $a/b = a'/b'$ in lowest terms, comparing prime exponents gives $u = w^{b'}$, $v = w^{a'}$ for a positive integer w prime to q ; its class satisfies $[w]^{a'}, [w]^{b'} \in H$, hence $[w] \in H$ since $\gcd(a', b') = 1$, so $w \in M_H$, and one of u, v is a proper power in M_H , contradicting atomicity. So at most one factorization is a pure power, $\varepsilon \leq \frac{1}{2}$ and $r_n \leq -\frac{1}{2}$ in (b). For $n = pp'rr'$ as in the statement, the proper M -divisors are the products of two of the four primes that lie in M_H , all atoms; the factorizations are $\{pp', rr'\}$ and $\{pr', rp'\}$, and also $\{pr, p'r'\}$ when $c^2 \in H$; n is squarefree, so $\varepsilon = 0$. Dirichlet's theorem gives infinitely many such n . $\square$

Corollary 2.4 (the first negative coefficient above $-\frac{1}{2}$). *Let $N \in M$ have two or more factorizations while every proper M -divisor of N has one (for example, the least element with two factorizations of a monoid that is not free), and suppose $r_N > -\frac{1}{2}$. Then N has exactly two factorizations $a^{j_1} = b^{j_2}$ with $a \neq b$*

atoms and $\gcd(j_1, j_2) = 1$; $N = c^{j_1 j_2}$ with $a = c^{j_2}$, $b = c^{j_1}$; and $r_N = -1 + 1/j_1 + 1/j_2$. Conversely every value $-1 + 1/u + 1/v$ with $2 \leq u < v$ coprime occurs, at $N = c^{uv}$ in $\langle c^u, c^v \rangle$. So the possible first negative coefficients in $(-\frac{1}{2}, 0)$ are exactly

$$\{-\frac{1}{2} + 1/v : v \geq 3 \text{ odd}\} \cup \{-\frac{5}{12}, -\frac{7}{15}\}.$$

Status. Proved here (claude-ab, 29_ Corollary 29.1, for the least element with two factorizations; refines ETR1 and Theorem 2.3). Checked exactly on 286 generator sets. Refereed (tenth pass). The hypothesis on proper M -divisors in place of minimality is version 2's (48_ §2.4). Novelty: not searched; elementary.

Proof. By Theorem 2.3(b), $r_N = 1 - j + \varepsilon$ with ε a sum of reciprocals of distinct integers ≥ 2 , one for each pure-power factorization. If $j \geq 3$ then $r_N \leq 1 - j + \sum_{i=2}^{j+1} 1/i \leq -2 + \frac{13}{12} < -\frac{1}{2}$; if $j = 2$ and at most one factorization is a pure power, $r_N \leq -\frac{1}{2}$. So $j = 2$ and both factorizations are pure powers, $N = a^{j_1} = b^{j_2}$ with $j_1 < j_2$. If $g = \gcd(j_1, j_2) > 1$, then $N' = a^{j_1/g} = b^{j_2/g}$ is a proper M -divisor of $N = N'^g$ with two distinct factorizations (neither exponent quotient is 1, since an atom is not a proper power of an atom), contradicting the hypothesis. So $g = 1$. Comparing prime exponents, $v_\ell(a)j_1 = v_\ell(b)j_2$, so $j_2 \mid v_\ell(a)$; with $c = \prod_\ell \ell^{v_\ell(a)/j_2}$ we get $a = c^{j_2}$, $b = c^{j_1}$ and $N = c^{j_1 j_2}$, and $r_N = -1 + 1/j_1 + 1/j_2$. Conversely, in $\langle c^u, c^v \rangle$ with $u < v$ coprime, $xu + yv = uv$ with $x, y \geq 0$ forces $(x, y) \in \{(v, 0), (0, u)\}$, and two representations of an exponent differ by a multiple of $(v, -u)$, so no smaller exponent has two; both generators are atoms since $u \nmid v$. Finally $1/u + 1/v > \frac{1}{2}$ with coprime $2 \leq u < v$ leaves $u = 2$ with v odd, and $u = 3$ with $v \in \{4, 5\}$. $\square$

Example 2.5. In $M = \{1\} \cup \{2^e : e \geq 2\}$ the atoms are 4 and 8 and $64 = 4 \cdot 4 \cdot 4 = 8 \cdot 8$, so $r_{64} = -\frac{1}{6}$. With $x = 2^{-s}$, $D_M = (1 - x + x^2)/(1 - x)$, whose zeros lie on $\operatorname{Re} s = 0$. So D_M has no zeros on its whole half-plane of absolute convergence $\operatorname{Re} s > 0$, as the Dirichlet series of every free monoid has (its Euler product converges absolutely there; ETR2, 29_ 29.N1), and yet M is not free: for general monoids even the absence of zeros on the half-plane of absolute convergence does not imply freeness (11_ Example 11.5; the monoid appears first in the programme's ETR1).

Status. Proved here (a direct computation; claude-ab, 11_ Example 11.5). Version 1 stated the absence of zeros only for $\operatorname{Re} s > 1$; the whole half-plane $\operatorname{Re} s > 0$ is version 2's (48_ §3, F6).

2.3 Congruence monoids and the even lattice sums

Fix $q \geq 1$ and a subgroup H of $G = (\mathbb{Z}/q)^\times$, and let $M_H = \{n \geq 1 : \gcd(n, q) = 1, n \bmod q \in H\}$. By orthogonality, $D_{M_H}(s) = [G : H]^{-1} \sum_{\chi|_H=1} L(s, \chi)$.

Proposition 2.6. *The following are equivalent: (i) M_H is free; (ii) $H = G$; (iii) all $r_n \geq 0$; (iv) D_{M_H} has no zeros in $\operatorname{Re} s > 1$. Moreover M_H is half-factorial if and only if $[G : H] \leq 2$.*

Status. The equivalence (i) $\Leftrightarrow$ (ii) and the half-factoriality criterion are the standard factorial and half-factorial criteria for Krull monoids, specialised to M_H (class group G/H ; 11_ §1.4; for $H = \{1\}$ Baginski–Chapman [4], Theorem 3.4, the Krull-monoid analogue of Carlitz's class-number-two theorem [14]). The direct proofs here and the equivalences with (iii) and (iv) are claude-ab's (11_ Proposition 11.1, Lemma 11.3; the step (iv) $\Rightarrow$ (ii) uses the theorem of Saias–Weingartner [40], read through its abstract). Refereed (third pass). The argument for (i) $\Rightarrow$ (ii) without Dirichlet's theorem is 29_ §1, added in version 2 (48_ §3, F6).

Proof. (i) $\Leftrightarrow$ (iii) is Theorem 2.3. (ii) $\Rightarrow$ (i): M_G is free on the primes not dividing q . (i) $\Rightarrow$ (ii): if $c \in G \setminus H$, choose primes $p \neq r$ in the class c and $p' \neq r'$ in the class c^{-1} (Dirichlet); then pp', rr', pr', rp' are atoms of M_H and $(pp')(rr') = (pr')(rp')$. Dirichlet's theorem can be avoided (29_ §1): infinitely many primes lie outside H (an integer $\equiv c \pmod{q}$ prime to q and to finitely

many given primes has a prime factor outside H), so some class of order $d \geq 2$ in G/H contains two primes $p \neq r$, and the atoms $A_j = p^{d-j}r^j$ ($0 \leq j \leq d$) satisfy $A_0A_2 = A_1^2$. (iii) $\Rightarrow$ (iv): if all $r_n \geq 0$, then coefficientwise $\log D \leq e^{\log D} - 1 = D - 1$, so $\log D$ converges absolutely on $\operatorname{Re} s > 1$ and $D = e^{\log D} \neq 0$ there. (iv) $\Rightarrow$ (ii): if $H \neq G$, at least two characters trivial on H are induced by distinct primitive characters, so D_{M_H} is not of the form $P(s)L(s, \chi)$ with P a Dirichlet polynomial (a representation $\sum_{\psi} P_{\psi}L(s, \psi)$ over distinct primitive ψ is unique; see the proof of Proposition 2.8), and the theorem of Saias and Weingartner gives zeros in $\operatorname{Re} s > 1$.

Half-factoriality. If $[G : H] = 2$, the atoms are the primes in H and the products of two primes in the other coset, so every factorization of n has length $a + b/2$, where a and b count the prime factors of n inside and outside H . If $[G : H] \geq 3$, either G/H has an element of order $N \geq 3$, and with primes p, p' in the classes c, c^{-1} the element $(pp')^N = p^N \cdot p'^N$ has factorizations of lengths N and 2 ; or G/H contains $c, d, cd \notin H$, and with primes p, r, s in these classes $(prs)^2 = p^2 \cdot r^2 \cdot s^2$ has factorizations of lengths 2 and 3 . $\square$

Proposition 2.7 (the even sheets at $a = 1/q$). *For $q \geq 3$, $Z_{1/q}(s) = \zeta(s, 1/q) + \zeta(s, 1 - 1/q) = q^s \sum_{n \equiv \pm 1 \pmod{q}} n^{-s} = q^s D_{M_{\{\pm 1\}}}(s)$. Hence the sheet $a = 1/q$ is Eulerian (the monoid $\{n \equiv \pm 1 \pmod{q}\}$ is free) if and only if $\varphi(q) \leq 2$, that is $q \in \{3, 4, 6\}$, if and only if its formal logarithm is nonnegative, if and only if $Z_{1/q}$ has no zeros with $\operatorname{Re} s > 1$; and exactly the same q make the one-sided monoid $\{n \equiv 1 \pmod{q}\}$ half-factorial.*

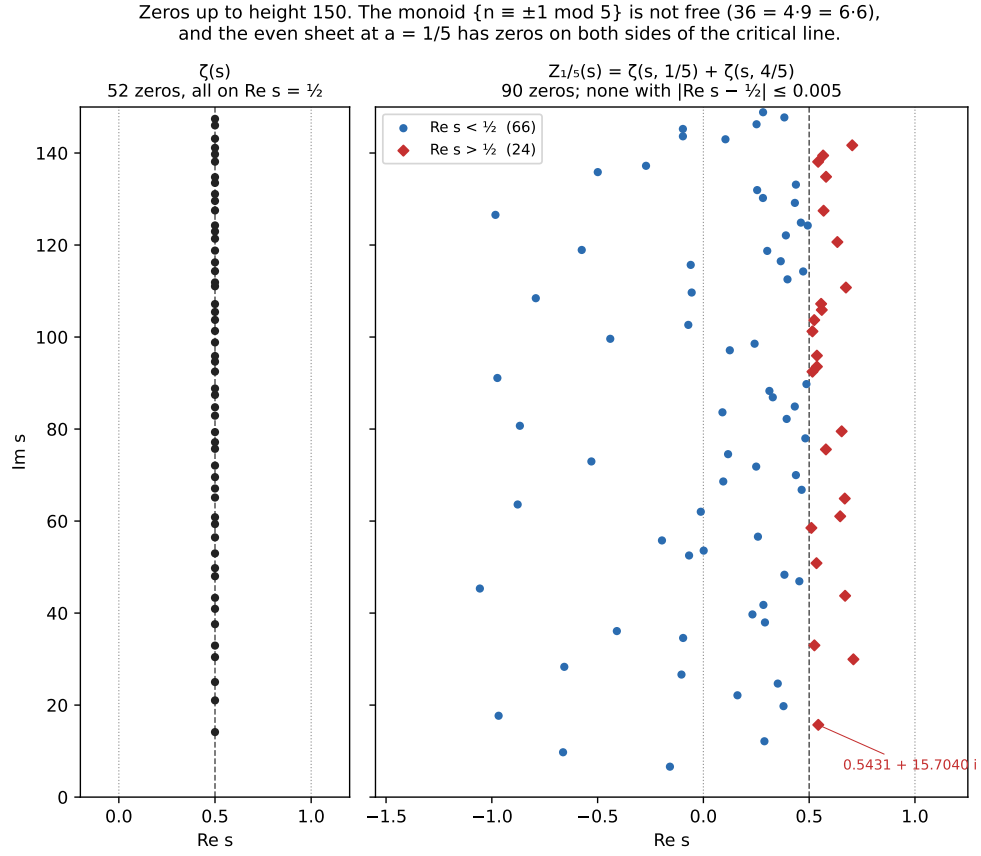
Status. Proved here (claude-ab, 08_ Lemma 2 and 11_ Corollary 11.4). Refereed (first and third passes). Freeness $\iff \varphi(q) \leq 2$ is the Krull-monoid criterion (class group $(\mathbb{Z}/q)^{\times}/\{\pm 1\}$); the analytic packaging as a test on the sheets is not in the Baginski–Chapman survey; not searched further.

Proof. $\zeta(s, 1/q) = q^s \sum_{n \geq 0} (qn + 1)^{-s}$ and $\zeta(s, 1 - 1/q) = q^s \sum_{m \equiv -1 \pmod{q}} m^{-s}$; for $q \geq 3$ the two classes are distinct. The rest is Proposition 2.6 with $H = \{\pm 1\}$ and $H = \{1\}$, since $\{\pm 1\} = G$ exactly when $\varphi(q) \leq 2$. $\square$

Proposition 2.8 (zeros of the even sheets on both sides of the line). *Let $\varphi(q) > 2$. There is $\eta > 0$ such that for all $\frac{1}{2} < \sigma_1 < \sigma_2 < 1 + \eta$ the number of zeros of $Z_{1/q}$ with $\sigma_1 < \operatorname{Re} s < \sigma_2$ and $|\operatorname{Im} s| \leq T$ lies between $c_1 T$ and $c_2 T$ for all large T , with $c_1, c_2 > 0$ depending on q, σ_1, σ_2 . The same holds, with another η , in the reflected strips $1 - \sigma_2 < \operatorname{Re} s < 1 - \sigma_1$; so $Z_{1/q}$ has infinitely many zeros with $\operatorname{Re} s > 1$, with $\frac{1}{2} < \operatorname{Re} s < 1$, with $0 < \operatorname{Re} s < \frac{1}{2}$ and with $\operatorname{Re} s < 0$.*

Status. Proved here, using the theorem of Saias and Weingartner [40] as stated in its abstract (periodic coefficients, a series not of the form $P(s)L(s, \chi)$ with P a Dirichlet polynomial); its printed statement was not checked. The right side is in 08_ §1. The left side is version 2's: the review's F7 for prime $q \geq 5$, with quantifiers corrected by the verifier, and its extension to every q with $\varphi(q) > 2$ by the referee of version 2, re-derived by claude-ab (48_ §§3, 7; check Z8 finds zeros of $Z_{1/8}$ and $Z_{1/12}$ in $0 < \operatorname{Re} s < \frac{1}{2}$).

Proof. $Z_{1/q} = q^s D_{M_{\{\pm 1\}}}$, and for $\{\pm 1\} \neq G$ the series has nonzero components along at least two distinct primitive characters (the trivial one and one inducing a nontrivial character trivial on $\{\pm 1\}$), so it is not of the form $P \cdot L(s, \chi)$ and the theorem applies. For the left side, adding Hurwitz's formula at a and $1 - a$ cancels the sine sums: $Z_a(1 - s) = 2\Gamma(s)(2\pi)^{-s} \cos(\pi s/2) \Phi_a(s)$ with $\Phi_a(s) = 2 \sum_{k \geq 1} \cos(2\pi ka) k^{-s}$, an entire function for $0 < a < 1$ with q -periodic coefficients at $a = 1/q$. Grouping k by $d = \gcd(k, q)$ and using $\cos(2\pi c/e) = \varphi(e)^{-1} \sum_{\chi \bmod e \text{ even}} \tau(\bar{\chi}) \chi(c)$ for $\gcd(c, e) = 1$ gives $\Phi_{1/q}(s) = \sum_{d|q} d^{-s} \frac{2}{\varphi(q/d)} \sum_{\chi \bmod q/d \text{ even}} \tau(\bar{\chi}) L(s, \chi)$, and passing to primitive characters, $\Phi_{1/q} = \sum_{\psi} P_{\psi} L(s, \psi)$ with Dirichlet polynomials P_{ψ} and distinct primitive ψ . Such a representation is unique: if $\sum_{\psi} P_{\psi} L(s, \psi) = 0$ with some $P_{\psi} \neq 0$, let d_0 be the least integer in the supports; for large primes p the coefficient at $d_0 p$ is $\sum_{\psi} b_{\psi, d_0} \psi(p)$, and distinct primitive



claude-ab (Opus 5.5, max effort) · zeros located by Arb argument principle, refined to 30 digits · figures/fig_z15_zeros.py

Figure 2: Zeros up to height 150 of ζ (left) and of the even lattice sum $Z_{1/5}(s) = \zeta(s, \frac{1}{5}) + \zeta(s, \frac{4}{5})$ (right), located with the argument principle in Arb ball arithmetic and refined to 30 digits. The monoid $\{n \equiv \pm 1 \pmod{5}\}$ is not free, and $Z_{1/5}$ has zeros on both sides of the line (infinitely many, by Proposition 2.8; the positions shown are computed).

characters are linearly independent on the primes (Dirichlet's theorem), so every $b_{\psi, d_0} = 0$, a contradiction. The ζ -component is nonzero: its coefficient at q^{-s} is $2 \sum_{e|q, e \text{ squarefree}} 1/\varphi(e) > 0$ (from $\tau(\chi_0 \bmod e) = \mu(e)$). Some nontrivial component is nonzero: otherwise $\Phi_{1/q} = P_1 \zeta$, and comparing coefficients at large primes gives $\cos(2\pi p/q) = \cos(2\pi/q)$ for every large prime p , so every class prime to q is ± 1 and $\varphi(q) \leq 2$. Hence $\Phi_{1/q}$ is not of the form $P \cdot L(s, \chi)$ and the theorem applies to it. At a nonreal zero s_0 of $\Phi_{1/q}$ with $\text{Re } s_0 > 0$ the factor $2\Gamma(s_0)(2\pi)^{-s_0} \cos(\pi s_0/2)$ is finite and nonzero, so $1 - s_0$ is a zero of $Z_{1/q}$. $\square$

At $q = 5$ the first element with two factorizations is $36 = 4 \cdot 9 = 6 \cdot 6$, with $r_{36} = -\frac{1}{2}$. Numerically, $Z_{1/5}$ has 24 zeros with $\text{Re } s > 0.505$ below height 150 (argument principle; Figure 2); the lowest to the right of the line is $0.54307 + 15.70405i$, zeros to the left occur lower (for example $-0.158 + 6.614i$), and none of the 90 zeros below height 150 lies within 0.0068 of the line (08_ Figure 8.1; checked numerically).

2.4 Shifted zeta products and tensor Euler functions

For finite exponent data, a finitely supported measure μ on $\mathbb{C}$ with positive integer masses, the programme's tensor Euler function of degree d is $\mathcal{L}_d = \prod_p \det(1 - p^{-s} W_p^{\otimes d})^{-1} = \prod_{\sigma} \zeta(s - \sigma) \mu^{*d}(\{\sigma\})$ (ETR4–ETR5).

Lemma 2.9 (shifted zeta functions are multiplicatively independent). *Let $\sigma_1, \dots, \sigma_\ell \in \mathbb{C}$ be distinct and $C_1, \dots, C_\ell \in \mathbb{C}$. If $\sum_j C_j \zeta'(s - \sigma_j) \equiv 0$, then all $C_j = 0$; in particular, for $C_j \in \mathbb{Z}$, $\prod_j \zeta(s - \sigma_j)^{C_j} \equiv 1$ only if all $C_j = 0$. The same holds with $L(s - \sigma_j, \chi)$ for any Dirichlet character χ .*

Status. Proved in the programme (ETR8) for integer exponents; re-derived here (29_ Lemma 29.2); the complex and L -function forms are version 2's (48_ §3, F8). Refereed (tenth pass). Novelty: elementary; not searched.

Proof. For $\operatorname{Re} s$ large the Euler logarithm $\sum_j C_j \log \zeta(s - \sigma_j) = \sum_p \sum_r \frac{1}{r} (\sum_j C_j p^{r\sigma_j}) p^{-rs}$ converges absolutely and tends to 0 as $s \rightarrow +\infty$; its derivative is $\sum_j C_j \zeta'(s - \sigma_j) / \zeta(s - \sigma_j) = 0$, so it vanishes identically. Each prime power has a unique base, so $\sum_j C_j p^{r\sigma_j} = 0$ for all p and $r \geq 1$ (for $L(s, \chi)$ the coefficient carries the factor $\chi(p)^r$, nonzero for $p \nmid q$, which may be divided out). Two support points with $p^{\sigma_i} = p^{\sigma_j}$ differ by $2\pi ia / \log p$ with $a \in \mathbb{Z} \setminus \{0\}$, and the same pair cannot collide at two primes (that would give $p^b = q^a$); so some prime p (not dividing q , for L) has the $x_j = p^{\sigma_j}$ distinct. For that p the equations with $r = 1, \dots, \ell$ have determinant $\prod_j x_j \prod_{i < j} (x_j - x_i) \neq 0$, so all $C_j = 0$. $\square$

Lemma 2.10 (uniqueness of convolution roots). *Let μ, ν be finitely supported complex measures on $\mathbb{C}$. If $\mu^{*d} = \nu^{*d}$ for some $d \geq 1$, then $\mu = \eta\nu$ for a d -th root of unity η ; if the total masses are positive reals, $\mu = \nu$.*

Status. Proved in the programme (ETR9); re-derived here (29_ Lemma 29.3); the form without positivity is version 2's (48_ §3, F8).

Proof. Put $A(z) = \int e^{\omega z} d\mu(\omega)$, $B(z) = \int e^{\omega z} d\nu(\omega)$. Then $A^d = B^d$. If $\nu = 0$, then $A \equiv 0$ and $\mu = 0$. Otherwise the zeros of the entire function B are isolated, and on the connected set $\{B \neq 0\}$ the quotient A/B is continuous with values in the d -th roots of unity, hence a constant η ; so $A = \eta B$ everywhere, and since distinct exponentials are linearly independent, $\mu = \eta\nu$. If the total masses $A(0)$ and $B(0)$ are positive reals, $\eta = A(0)/B(0) > 0$, so $\eta = 1$. $\square$

Corollary 2.11 (one tensor degree recovers the exponent measure). *For finite exponent data, a single $\mathcal{L}_d$ determines μ .*

Status. Proved in the programme (ETR4–ETR9); re-derived (29_ Lemma 29.3). The programme also proves, for conjugation-stable data, that the real pole of $\mathcal{L}_{2k}$ at $1 + 2kB$ has order h_{2k} with $h_{2k}^{1/2k}$ increasing to the dimension of the maximal face (ETR5–ETR7); that positivity of even tensor powers is Deligne's Weil II §1.5 mechanism, as the programme credits.

Proof. By Lemma 2.9, $\mathcal{L}_d$ determines the measure μ^{*d} ; by Lemma 2.10 it determines μ . $\square$

The global character of Lemma 2.9 is necessary: no finite set of primes fixed in advance has Euler factors that determine every finite shifted-zeta product (Section 7, item N-Eu).

3 Meyer's quotient, its jets and its receivers

Throughout this section $\mathcal{B}, I_D, I_\zeta$ and $Q = \mathcal{B}/I_\zeta$ are as in Section 1, and L denotes multiplication by s on a quotient $\mathcal{B}/I_D$. Since the seminorms $b_{A,M}$ increase in A and M , they form a directed family, and every quotient of $\mathcal{B}$ by a closed subspace is a Fréchet space with the quotient seminorms $\bar{b}_{A,M}$ (Rudin [39], Theorem 1.41).

3.1 The strip space

Lemma 3.1 ($\mathcal{B}$ is a Fréchet–Schwartz space). *For all $A, M \geq 0$ the unit ball of $b_{A+1, M+1}$ is precompact for $b_{A, M}$. Consequently $\mathcal{B}$ and every quotient $\mathcal{B}/I$ by a closed subspace are Fréchet–Montel spaces, hence reflexive, and $(\mathcal{B}/I)'_{\beta} \cong I^{\perp} \subset \mathcal{B}'_{\beta}$ topologically.*

Status. Proved here (claude-ab, 27_ Lemma 27.2; refereed in the ninth pass). The programme proves the same properties for A and transfers them to $\mathcal{B}/I$ through the summation image (SDT); the lemma removes that dependency. The consequences are standard permanence properties of Schwartz spaces.

Proof. Let $b_{A+1, M+1}(F) \leq 1$, so $|F(s)| \leq (1 + |t|)^{-M-1}$ on $|\operatorname{Re} s| \leq A + 1$. Cauchy's estimate on discs of radius $\frac{1}{2}$ gives $|F'(s)| \leq 2(\frac{1}{2} + |t|)^{-M-1}$ on $|\operatorname{Re} s| \leq A + \frac{1}{2}$, so the ball is bounded and equicontinuous there. Given $\varepsilon > 0$ choose T with $(1 + T)^{-1} < \varepsilon$; then $(1 + |t|)^M |F| < \varepsilon$ on $|t| \geq T$ for every F in the ball. On the rectangle $|\operatorname{Re} s| \leq A, |t| \leq T$, Arzelà–Ascoli gives a finite $\varepsilon(1 + T)^{-M}$ -net for the supremum norm; with the tail estimate it is a 2ε -net for $b_{A, M}$. A Fréchet space in which every seminorm ball is precompact for the preceding seminorm is a Schwartz space; Schwartz spaces are Montel, and quotients of Fréchet–Schwartz spaces by closed subspaces are Fréchet–Schwartz (Jarchow [28]); Montel spaces are reflexive (Schaefer [41], Ch. IV). Bounded sets of $\mathcal{B}/I$ are relatively compact, and compact sets lift through the quotient map of a Fréchet space, so the strong topology of $(\mathcal{B}/I)'$ is the one induced from $\mathcal{B}'_{\beta}$ on $I^{\perp}$. $\square$

Proposition 3.2 (multiplication by s on $\mathcal{B}/I_D$). *Let $0 \neq \Phi \in \mathcal{B}$ vanish to order at least D ($D \leq \operatorname{div} \Phi$; for example D the whole zero divisor of Φ , or its part on a line), and let L be multiplication by s on $Q_D = \mathcal{B}/I_D$.*

- (a) *For $\lambda \notin \operatorname{supp} D$, $\lambda - L$ is invertible on Q_D , with inverse induced by $R_{\lambda}F = (F - \Phi_{\lambda} \cdot F(\lambda)/\Phi_{\lambda}(\lambda))/(\lambda - s)$, where $\Phi_{\lambda} = \Phi/(s - \lambda)^k, k = \operatorname{ord}_{\lambda} \Phi$.*
- (b) *Every $\rho \in \operatorname{supp} D$ is an eigenvalue; hence the spectrum of L is $\operatorname{supp} D$.*
- (c) *If ρ has multiplicity m in D , then $\ker(L - \rho)^m = \ker(L - \rho)^{m+1}$ has dimension m , the jet map j_{ρ} identifies it with $\mathbb{C}[T]/(T^m)$, and $L - \rho$ acts on it as multiplication by T , a single nilpotent Jordan block of length m .*

Status. Proved in the programme for the zeta divisor (RZ3–RZ7, including the Riesz projector at ρ as multiplication by an entire function e_{ρ} ; audited in 27_); the general-divisor form and the proof given here are claude-ab's (24_ Lemma 24.2 step 1, 27_ Lemma 27.1); version 2 needs only $D \leq \operatorname{div} \Phi$ (48_ §3, F10).

Proof. Dividing an element of $\mathcal{B}$ by $(s - \lambda)^k$ at a zero of order at least k keeps it in $\mathcal{B}$ (divide by $|\lambda - s| \geq 1$ away from λ , and use the maximum principle on $|\lambda - s| \leq 1$). So $\Phi_{\lambda} \in \mathcal{B}$, $\Phi_{\lambda}(\lambda) \neq 0$, and $\Phi_{\lambda} \in I_D$ because $\lambda \notin \operatorname{supp} D$. (a) The numerator of $R_{\lambda}F$ vanishes at λ , so $R_{\lambda}F$ is entire; it lies in $\mathcal{B}$ by the same argument, and it preserves I_D because $\Phi_{\lambda} \in I_D$ and $\lambda \notin \operatorname{supp} D$. Since $(\lambda - s)R_{\lambda}F = F - \Phi_{\lambda}F(\lambda)/\Phi_{\lambda}(\lambda) \equiv F$ and $R_{\lambda}((\lambda - s)F) = F$, R_{λ} induces the inverse of $\lambda - L$. (b) With $m' = \operatorname{ord}_{\rho} \Phi \geq m$, $U = \Phi/(s - \rho)^{m'-m+1} \in \mathcal{B}$ vanishes to order $m - 1$ at ρ and to full order on the rest of D , so $U \notin I_D$, while $(s - \rho)U \in I_D$. (c) A class $[F]$ lies in $\ker(L - \rho)^k$ exactly when $(s - \rho)^k F$ vanishes to full order on D , that is, when F vanishes to full order at every other point of D and to order $m - k$ at ρ . So $\ker(L - \rho)^m = \ker(L - \rho)^{m+1}$ consists of the classes of functions vanishing to full order off ρ , and j_{ρ} maps it injectively into $\mathbb{C}[T]/(T^m)$. It is onto: $U_{\rho} = \Phi/(s - \rho)^{m'} \in \mathcal{B}$ has $U_{\rho}(\rho) \neq 0$ and vanishes to full order at the other points of D , so $U_{\rho} \cdot P(s - \rho)$ with $P = [P_0 u(T)^{-1}]_{< m'}$, u the jet of U_{ρ} , has jet P_0 at ρ . Finally $j_{\rho}((s - \rho)F) = T j_{\rho}(F)$. $\square$

Lemma 3.3 (no equivariant maps across disjoint divisors). *Let $D \leq \operatorname{div} F_D$ for some $0 \neq F_D \in \mathcal{B}$, let D' be a divisor, and let D'' be the restriction of D' to the complement of $\operatorname{supp} D$. Every linear map*

$H : \mathcal{B}/I_D \rightarrow \mathcal{B}/I_{D'}$ with $HL = LH$ takes values in $I_{D''}/I_{D'}$; in particular it is zero if $D \cap D' = \emptyset$. Continuity is not needed.

Status. Proved here (claude-ab, 24_ Lemma 24.2; this is the uniqueness argument of the programme's GSL6, which it strengthens: only commutation with L is used). Refereed (eighth pass). The form for overlapping divisors is version 2's (48_ §3, F10).

Proof. Let $\lambda \in \text{supp } D' \setminus \text{supp } D$ have multiplicity m in D' . By Proposition 3.2(a), $(L - \lambda)^m$ is onto $\mathcal{B}/I_D$. For every x , $H((L - \lambda)^m x) = (L - \lambda)^m H(x)$ has zero m -jet at λ (the jet of $(L - \lambda)^m z$ at λ is T^m times that of z); since $(L - \lambda)^m$ is onto $\mathcal{B}/I_D$, every y is of the form $(L - \lambda)^m x$, so every $H(y)$ has zero m -jet at every such λ , and $H(y) \in I_{D''}$. $\square$

Lemma 3.4 (uniqueness of exponential sums on a divisor). *Let $0 \neq G \in \mathcal{B}$, let Z be a set of zeros of G in a strip $|\text{Re } s| \leq A$, and let $(c_\rho) \in \ell^1(Z)$. If $\sum_\rho c_\rho e^{\rho x} = 0$ for every real x , then $c = 0$. More generally, if $m_\rho = \text{ord}_\rho G$ and $\sum_{\rho, k} c_{\rho, k} x^k e^{\rho x} = 0$ for every real x , with $0 \leq k < m_\rho$ and $\sum |c_{\rho, k}| k! R^{-k} < \infty$ for some $R > 0$, then every $c_{\rho, k} = 0$.*

Status. Proved in the programme for the Gaussian-weighted zeta divisor (HBW8.9–8.12; HSR4.1 reduces to it); stated for general G and proved here by claude-ab (32_ Lemma 32.2). Refereed (eleventh pass). The form with the terms $x^k e^{\rho x}$ is version 2's (48_ §3, F11).

Proof. Every $F \in \mathcal{B}$ has a half-Mellin representation $F(s) = \frac{1}{2} \int_{\mathbb{R}} a(e^v) e^{sv} dv$ with $a(e^v)$ decaying faster than every exponential (NJS4.1). By the ℓ^1 condition and $|e^{\rho v}| \leq e^{A|v|}$ the interchange is justified and $\sum_\rho c_\rho F(\rho) = \frac{1}{2} \int a(e^v) \sum_\rho c_\rho e^{\rho v} dv = 0$. For $\rho_0 \in Z$ of order m as a zero of G , the function $F = G/(s - \rho_0)^m \in \mathcal{B}$ has $F(\rho_0) \neq 0$ and vanishes on $Z \setminus \{\rho_0\}$, so $c_{\rho_0} = 0$. For the general form, $F^{(k)}(\rho) = \frac{1}{2} \int a(e^v) v^k e^{\rho v} dv$, and $|v|^k \leq k! R^{-k} e^{R|v|}$ justifies the interchange, so $\sum c_{\rho, k} F^{(k)}(\rho) = 0$ for all $F \in \mathcal{B}$; the functions $G(s - \rho_0)^{-m} P(s - \rho_0)$, P a polynomial, realise every jet of order $< m$ at ρ_0 and vanish to order m_ρ at the other ρ , so $c_{\rho_0, \cdot} = 0$. $\square$

3.2 The exact summation image

Let S be the space of even Schwartz functions f on $\mathbb{R}$ with $f(0) = 0$ and $\int f = 0$, put $\Sigma f(u) = 2 \sum_{n \geq 1} f(nu)$, and let $M_0 b(s) = \int_0^\infty b(u) u^{s-1} du$.

Proposition 3.5 (the exact summation image). $f \mapsto M_0 \Sigma f$, which equals $2\zeta(s) \int_0^\infty f(v) v^{s-1} dv$ on $\text{Re } s > 1$, is a topological isomorphism from S onto I_ζ . Its inverse is $F \mapsto f_F$ with

$$f_F(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{F(2+it)}{2\zeta(2+it)} x^{-2-it} dt = \frac{1}{2} \sum_{n \geq 1} \mu(n) (M_0^{-1} F)(nx) \quad (x > 0),$$

and $f_F^{(2r)}(0)/(2r)! = F(-2r)/(2\zeta'(-2r))$ for $r \geq 1$, $\int_0^\infty f_F dx/x = -F(0)$, $\int_0^\infty f_F \log x dx = F(1)/2$. In particular the summation image is closed: there is no gap between it and its closure.

Status. Proved in the programme (SSI1–SSI7, OMS2–OMS4); audited (25_ Lemma 25.1, 26_); the Taylor coefficients and endpoint identities checked on the programme's source vector. On the programme's own account (OMS0) this is Meyer's range theorem [32] made explicit for $\mathcal{B}$, with the division estimate supplied; the audit did not read Meyer's paper. Refereed (eighth pass).

Outline of the proof. Poisson summation, $\Sigma \hat{f}(u) = u^{-1} \Sigma f(u^{-1}) + u^{-1} f(0) - \int f$, controls $u \rightarrow 0$, so Σ maps S continuously into the functions whose Mellin transforms lie in $\mathcal{B}$; on $\text{Re } s > 1$ the transform is Müntz's $2\zeta Mf$ (Titchmarsh [45], §2.11), with Mf holomorphic on $\text{Re } s > -2$ and zero at 1, so it vanishes to full order on $\mathbb{Z}$, and the Euler product gives injectivity. Conversely, for $F \in I_\zeta$ the

quotient $H = F/(2\zeta)$ is holomorphic except for simple poles at $-2r$; $1/\zeta$ is polynomially bounded on the two edges of $-2N - 1 \leq \operatorname{Re} s \leq b$ (by the reflection formula for $|\chi(-2N - 1 + it)|$ and $|1/\zeta(b + it)| \leq \zeta(b)$), and inside the strip the division estimate $|F/F_*| \leq C \exp(C(|t| + 2)^{3/2} \log(|t| + 2))$ (S3; proved by the Hadamard product and a minimum-modulus argument, OMS2.8) is removed by the factor $e^{\varepsilon s^2}$ and the maximum principle, giving a bound linear in a seminorm of F . Shifting the inverse Mellin contour to $\operatorname{Re} s = -2N - 1$ picks up the residues $F(-2r)/(2\zeta'(-2r)) x^{2r}$, so f_F extends to an even Schwartz function with $f_F(0) = 0$ and $\int f_F = 2H(1) = 0$. $\square$

Two consequences are used below. The closure of Connes' summation map in the rapid-decay Fréchet space is the ideal of all zero-jets (S5 of the second attempt; register S13), and by Proposition 3.5 the image itself is already that ideal. And Meyer's quotient $Q = A/\mathcal{E}S$ is identified with $\mathcal{B}/I_{\zeta}$, on which the jet map is injective by the definition of I_{ζ} .

3.3 The jet map and the primary decomposition

For each zero ρ the programme's section $s_{\rho} : A_{\rho} \rightarrow Q$ (OMS6) is $s_{\rho}(P) = [U_{\rho} \cdot (P c_{\rho})(s - \rho)]$ with $U_{\rho} = F_*/(s - \rho)^{m_{\rho}}$ and $c_{\rho} = u(T)^{-1} \bmod T^{m_{\rho}}$, u the jet of U_{ρ} at ρ . It satisfies $j_{\eta} s_{\rho} = \delta_{\eta\rho} \operatorname{id}$, so the $\Pi_{\rho} = s_{\rho} j_{\rho}$ are commuting idempotents with $\Pi_{\rho} \Pi_{\eta} = 0$ for $\rho \neq \eta$, and finite sums of sections realise every finitely supported jet tuple. On the range of Π_{ρ} the dilations act by $W_a = a^{\rho} \exp((\log a) N_{\rho})$ with $N_{\rho} = L - \rho$ (Proposition 3.2(c) and OMS6.5).

Lemma 3.6 (jet decay). *For $F \in \mathcal{B}$, a point $\rho = \beta + i\gamma$ with $|\beta| \leq \frac{3}{2}$ (for example a nontrivial zero) and $r, M \geq 0$, $(1 + |\gamma|)^M |F^{(r)}(\rho)/r!| \leq 2^r (\frac{1+|\gamma|}{1/2+|\gamma|})^M b_{2,M}(F) \leq 2^{r+M} b_{2,M}(F)$. At the nontrivial zeros the middle factor is at most 1.0342^M .*

Status. Proved here (claude-ab, 40_ Lemma 40.1; an explicit form of OMS5.6). Refereed (seventeenth pass). The sharper factor is version 2's (48_ §3, F12).

Proof. Cauchy's formula on $|z - \rho| = \frac{1}{2}$ gives $|F^{(r)}(\rho)/r!| \leq 2^r \max_{|z-\rho|=1/2} |F|$. The circle lies in $|\operatorname{Re} z| \leq 2$, and on it $1 + |\operatorname{Im} z| \geq \frac{1}{2} + |\gamma|$, so $|F(z)| \leq b_{2,M}(F) (\frac{1}{2} + |\gamma|)^{-M}$; and $(1 + x)/(\frac{1}{2} + x) \leq 2$, decreasing in x , so that for $x \geq \gamma_1 = 14.1347 \dots$ it is at most $(1 + \gamma_1)/(\frac{1}{2} + \gamma_1) = 1.03417 \dots < 1.0342$. $\square$

Corollary 3.7. *The image $J(Q)$ of the jet map $J = (j_{\rho})_{\rho} : Q \rightarrow \prod_{\rho} A_{\rho}$ lies in the space Λ_m of rapidly decreasing jet tuples (defined before Theorem 3.9) and is dense there; in particular it is a proper dense subspace of the product, and the quotient topology of Q is strictly finer than the product topology on $J(Q)$. In particular Q is not the product of its primary blocks.*

Status. Proved in the programme (OMS5; audited in 40_ §1, no errors); the explicit constant of Lemma 3.6 is claude-ab's. Version 1 said only that $J(Q)$ is a proper dense subspace of the product; the density in Λ_m is version 2's (48_ §3, F12).

Proof. Density: finite sums of sections realise every finitely supported tuple, and these are dense in the product and in Λ_m ($p_M(P - P_{\leq T}) \leq (1 + T)^{-1} p_{M+1}(P)$). $J(Q) \subseteq \Lambda_m$ is Lemma 3.6 with $m_{\rho} = O(\log |\gamma|)$ (D1 below), the factor $2^{m_{\rho}}$ being absorbed by raising M . Properness in the product: if $F(\rho) = 1$ at every zero, Lemma 3.6 with $r = 0$, $M = 1$ gives $1 + |\gamma| \leq 2b_{2,1}(F)$ for all ordinates, which are unbounded. Each j_{ρ} is continuous and vanishes on I_{ζ} , so the quotient topology is at least as fine as the product topology; if they were equal, $J(Q)$ would be complete, hence closed in the Hausdorff product, hence (being dense) the whole product. $\square$

Two classical facts are used (Titchmarsh [45], Theorems 9.2 and 9.6(A)): (D1) the number of zeros, with multiplicity, with $|\gamma - T| < 1$ is $O(\log T)$; (D2) for $-1 \leq \sigma \leq 2$ and $t \geq 2$, $\zeta'/\zeta(s) = \sum_{|t-\gamma|<1} (s - \rho)^{-1} + O(\log t)$.

Lemma 3.8 (good heights; Titchmarsh, Theorem 9.7). *There are $A, T_0 > 0$ such that every interval $[T, T + 1]$ with $T \geq T_0$ contains some t with $|\zeta(\sigma \pm it)| \geq t^{-A}$ for all $-1 \leq \sigma \leq 2$.*

Status. Titchmarsh's Theorem 9.7 [45], with his proof (40_ Lemma 40.2; the statements of Theorems 9.2, 9.6(A) and 9.7 were checked by a referee on the printed pages).

Proof. Integrating the real part of (D2) along $[\sigma + it, 2 + it]$, for t not an ordinate, $\log |\zeta(\sigma + it)| = \log |\zeta(2 + it)| - \sum_{|t-\gamma|<1} (\log |2 + it - \rho| - \log |\sigma + it - \rho|) + O(\log t)$. Here $|\zeta(2 + it)| \geq 1/\zeta(2)$, $1 < |2 + it - \rho| < 3$, the number of terms is $O(\log t)$ and $|\sigma + it - \rho| \geq |t - \gamma|$, so $\log |\zeta(\sigma + it)| \geq \sum_{|t-\gamma|<1} \log |t - \gamma| - C \log T$ uniformly in σ . Since $\int_{-1}^1 \log |u| du = -2$, the mean of the sum over $t \in [T, T + 1]$ is at least $-2\#\{T - 1 < \gamma < T + 2\} \geq -C' \log T$ by (D1); some t , not an ordinate, is at least the mean. For $-t$ use $|\zeta(\bar{s})| = |\zeta(s)|$. $\square$

For each zero ρ of multiplicity $m = m_\rho$ let $c_{\rho,k}$ ($0 \leq k < m$) be the Taylor coefficients at ρ of $(s - \rho)^m / \zeta(s)$; so $\sum_{k < m} c_{\rho,k} (s - \rho)^{k-m}$ is the principal part of $1/\zeta$ at ρ , and $c_{\rho,0} = m!/\zeta^{(m)}(\rho)$. Let Λ_m be the Fréchet space of tuples $(P_\rho) \in \prod_\rho A_\rho$ with $\sup_\rho (1 + |\gamma|)^M \|P_\rho\| < \infty$ for all M , $\|\cdot\|$ the largest coefficient.

Theorem 3.9 (when the primary decomposition converges). *The following are equivalent.*

- (a) *For some enumeration (ρ_n) of the distinct zeros, the partial sums $\sum_{k \leq n} \Pi_{\rho_k} x$ converge for every $x \in Q$ (the limit is then x).*
- (a') *For every $x \in Q$ the set $\{\Pi_\rho x\}_\rho$ is bounded in Q .*
- (b) *$\sum_\rho \Pi_\rho x$ converges absolutely for every $x \in Q$.*
- (b') *For every $x \in Q$, every continuous seminorm q and every N , $q(\Pi_\rho x) = O((1 + |\gamma|)^{-N})$.*
- (c) *There are C, K with $|c_{\rho,k}| \leq C(1 + |\gamma|)^K$ for all ρ and all $k < m_\rho$: the principal parts of $1/\zeta$ at the zeros are polynomially bounded.*
- (d) *$J : Q \rightarrow \Lambda_m$ is an isomorphism of Fréchet spaces.*

If all zeros are simple, (c) reads $|\zeta'(\rho)| \geq c(1 + |\gamma|)^{-K}$. In every case (a') implies $|\zeta^{(m_\rho)}(\rho)/m_\rho!| \geq c(1 + |\gamma|)^{-K}$. The theorem and its proof carry over, with the evident changes, to the quotient $\mathcal{B}/I_\chi$ of the zeros of a primitive Dirichlet L -function: the principal parts of $1/L(s, \chi)$ in (c), $L(s, \chi)$ in place of $(s - 1)\zeta(s)$ in the interpolants, (D1), (D2) and the good heights with $\log q(|t| + 2)$ (for complex χ , heights with $t < 0$ from $\bar{\chi}$), and constants depending on q (an outline: the L -function forms of (D1), (D2) and of the good heights were not re-read here).

Status. Proved here (claude-ab, 40_ Propositions 40.4 and 40.5; the form for all multiplicities is the seventeenth referee's sharpening, whose proof was checked). Checks F1–F9, G1–G3. Novelty: not found in the programme or in the literature searched (Berenstein–Taylor [6], Abanin–Le Hai Khoi–Nalbandyan [2], Bondarenko–Radchenko–Seip [9]); it is an instance for Meyer's quotient of the classical link between lower bounds at the zeros and interpolation or representing systems. Condition (b') and the form for $L(s, \chi)$ are version 2's (48_ §3, F13; the independent verification replaced the review's claim that the proof carries over unchanged by the list of changes given above).

Proof. (a) $\Rightarrow$ (a') and (b) $\Rightarrow$ (a). $\Pi_{\rho_n} x$ is the difference of consecutive partial sums, and absolute convergence in a Fréchet space implies convergence. If the partial sums tend to y , then $j_\rho y = j_\rho x$ for every ρ by continuity, so $y = x$.

(a') $\Rightarrow$ (c). Each Π_ρ is continuous (Lemma 3.6), and the family is pointwise bounded on the Fréchet space Q , so it is equicontinuous (Banach–Steinhaus; Rudin [39], Theorem 2.6): there

are A_1, M_1, C_0 with $\bar{b}_{2,0}(\Pi_\rho x) \leq C_0 \bar{b}_{A_1, M_1}(x)$ for all ρ, x . Take $E_\rho(s) = e^{(s-\rho)^2}$ (the programme's Gaussian test vector, CFP6.1). Since $|E_\rho(\sigma + it)| = e^{(\sigma-\rho)^2 - (t-\gamma)^2}$ and $1 + |t| \leq (1 + |\gamma|)(1 + |t - \gamma|)$, $b_{A_1, M_1}(E_\rho) \leq C_1(1 + |\gamma|)^{M_1}$. So $\Pi_\rho[E_\rho] = s_\rho(e^{T^2} \bmod T^m)$ has a representative F_ρ with $b_{2,0}(F_\rho) \leq 2C_0C_1(1 + |\gamma|)^{M_1}$; it has jet e^{T^2} at ρ and vanishes to full order at every other zero.

Let $\gamma \geq T_0 + 2$ (negative γ are symmetric). Choose good heights $T_1 \in [\gamma - 2, \gamma - 1]$ and $T_2 \in [\gamma + 1, \gamma + 2]$ (Lemma 3.8) and let $R = [-1, 2] \times [T_1, T_2]$. The function $H = F_\rho(s - \rho)^m / \zeta$ is holomorphic on a neighbourhood of R : F_ρ cancels the other zeros in R , the singularity at ρ is removable, $\zeta \neq 0$ on ∂R (the Euler product on $\sigma = 2$, the functional equation on $\sigma = -1$, Lemma 3.8 on the horizontal sides), and R contains no trivial zero and not the pole. On ∂R : $|F_\rho| \leq b_{2,0}(F_\rho)$ and $|s - \rho| < 3$; $|1/\zeta| \leq \sum n^{-2} = \pi^2/6$ on $\sigma = 2$; $|1/\zeta| \leq T_2^A$ on the horizontal sides; and on $\sigma = -1$, with $y = t/2$,

$$|\chi(-1 + it)|^2 = \pi^{-3} y \coth(\pi y) \left(\frac{1}{4} + y^2\right) \geq \frac{1}{4\pi^4},$$

$$|\zeta(-1 + it)| = |\chi(-1 + it)| |\zeta(2 - it)| \geq \frac{1}{2\pi^2} \cdot \frac{6}{\pi^2} = \frac{3}{\pi^4},$$

from $\chi(s) = \pi^{s-1/2} \Gamma(\frac{1-s}{2}) / \Gamma(\frac{s}{2})$, $|\Gamma(1 - iy)|^2 = \pi y / \sinh \pi y$ and $|\Gamma(-\frac{1}{2} + iy)|^2 = \pi / ((\frac{1}{4} + y^2) \cosh \pi y)$. So $|H| \leq B_\rho := 2C_0C_1(1 + |\gamma|)^{M_1} 3^m \max(\pi^2/6, \pi^4/3, (\gamma + 2)^A)$ on ∂R , hence on R . The disc $|s - \rho| \leq \frac{1}{2}$ lies in R , so $|H^{(k)}(\rho)/k!| \leq 2^k B_\rho$. The jet of H at ρ is $e^{T^2} \sum_k c_{\rho,k} T^k$, and the coefficients of e^{-T^2} have modulus at most 1, so $|c_{\rho,k}| \leq 2^{k+1} B_\rho \leq 2^m B_\rho$. By (D1), $m_\rho \leq C \log(2 + |\gamma|)$, so $2^m 3^m$ is polynomial in $|\gamma|$; finitely many low zeros are absorbed into the constant, since $c_{\rho,k}$ is finite. The last assertion of the theorem is the case $k = 0$.

(c) $\Rightarrow$ (d) and (b). $V_\rho = (s - 1)\zeta(s)/(s - \rho)^m$ is entire, and the reciprocal of its jet is $v_\rho^{-1} = (\sum_k c_{\rho,k} T^k)(\rho - 1 + T)^{-1}$, whose coefficients are at most $mC(1 + |\gamma|)^K$ because $|\rho - 1| \geq 1$. With $\lambda = m + 1$ put $q_{\rho,k} = [T^k e^{-\lambda T^2} v_\rho^{-1}]_{< m}$ and

$$G_{\rho,k}(s) = e^{\lambda(s-\rho)^2} V_\rho(s) q_{\rho,k}(s - \rho).$$

$G_{\rho,k}$ is entire, has jet $T^k \bmod T^m$ at ρ and vanishes to full order at every other zero. The coefficients of $q_{\rho,k}$ are at most $Q_\rho = m^2 e^{m+1} C(1 + |\gamma|)^K$. For $|\sigma| \leq A$, $|s - \rho| \geq 1$ and $u = t - \gamma$: $|(s - 1)\zeta(s)| \leq C_A(1 + |t|)^{c_A}$ (polynomial growth in strips; Titchmarsh [45], Ch. V), $|q_{\rho,k}(s - \rho)| \leq mQ_\rho(A + 1 + |u|)^{m-1}$, $|e^{\lambda(s-\rho)^2}| \leq e^{(m+1)(A+1)^2} e^{-(m-1)u^2} e^{-2u^2}$ and $e^{-u^2}(A + 1 + |u|) \leq A + 2$, so

$$|G_{\rho,k}(s)|(1 + |t|)^M \leq mQ_\rho e^{(m+1)(A+1)^2} (A + 2)^{m-1} C_A(1 + |\gamma|)^{M+c_A} \sup_u e^{-2u^2} (1 + |u|)^{M+c_A};$$

on $|s - \rho| < 1$ the maximum principle applies. Since $m \leq C \log(2 + |\gamma|)$, $b_{A,M}(G_{\rho,k}) \leq C_{A,M}(2 + |\gamma|)^{N_{A,M}}$. As $N(T) = O(T \log T)$, $\sum_\rho m_\rho (2 + |\gamma|)^{-2} < \infty$, so for $P \in \Lambda_m$ the series $F_P = \sum_\rho \sum_{k < m_\rho} P_{\rho,k} G_{\rho,k}$ converges absolutely in $\mathcal{B}$ and $J[F_P] = P$. J is continuous into Λ_m by Lemma 3.6 (the factor 2^{m_ρ} is absorbed by raising M) and injective, so it is an isomorphism with inverse $P \mapsto [F_P]$. Finally $\sum_k x_{\rho,k} [G_{\rho,k}] = \Pi_\rho x$, since both have the same jets at every zero; so $x = [F_J x] = \sum_\rho \Pi_\rho x$, absolutely.

(d) $\Rightarrow$ (b') $\Rightarrow$ (b). By (d), $q(J^{-1}P) \leq Cp_M(P)$ for some M ; since $\Pi_\rho x = J^{-1}(\sum_k x_{\rho,k} \delta_{\rho,k})$, $q(\Pi_\rho x) \leq C(1 + |\gamma|)^{-N} p_{M+N}(Jx)$, which is (b'); and (b') with (D1) gives absolute convergence. $\square$

Remark 3.10. Whether (c) holds is not known. For simple zeros it reads $|\zeta'(\rho)| \geq c(1 + |\gamma|)^{-K}$, and a bound of this form at every zero implies that every zero is simple, which is open; unconditionally, at least a proportion 0.4075 of the zeros are known to be simple (0.4058 by Bui–Conrey–Young [11], as cited in Bui–Heath-Brown [12], §1; 0.407511 for simple zeros on the line

by Pratt–Robles–Zaharescu–Zeindler [36]). Under RH it is expected that $|\zeta'(\rho)|^{-1} \gg |\gamma|^{1/3-\varepsilon}$ for infinitely many zeros (Milinovich–Ng [33], p. 2), so such a bound could only hold with $K \geq 1/3$. Whether the leading coefficients ($k = 0$) alone suffice when zeros are multiple is not settled; in the weighted spaces A_p the analogous leading-coefficient condition suffices for interpolation (Berenstein–Taylor [6], Theorem 4). The expansion $x = \sum_{\rho} \Pi_{\rho} x$ is formally like the residue expansion of Mertens' function, whose residue at a simple zero, $x^{\rho}/(\rho \zeta'(\rho))$, carries the same factor $1/\zeta'(\rho)$. The programme itself uses only finite sums of primary projectors (OMS6, SCL7) and states that it assumes no convergence of an infinite sum (GEX2).

3.4 Dense families of test functions

For fixed $t > 0$ put $g_t(s) = e^{ts^2}$ and, for a set $S \subset \{2, 3, \dots\}$, $\mathcal{N}_S = \text{span}\{g_t(s)(n - n^{1-s})/s : n \in S\}$.

Proposition 3.11 (Nyman–Beurling in the Fréchet topology). *Fix $t > 0$ and put $\nu_S(x) = \#\{n \in S : \log n \leq x\}$. If one of the conditions*

$$(C1) \quad \limsup_{R \rightarrow \infty} \frac{1}{R} \sum_{n \in S, \log n < R} \left(\frac{1}{\log n} - \frac{\log n}{R^2} \right) > \frac{1}{3\pi t},$$

$$(C2) \quad \limsup_{X \rightarrow \infty} \frac{1}{\log X} \sum_{n \in S, \log n < X} \left(\frac{1}{(\log n)^2} - \frac{(\log n)^2}{X^4} \right) > \frac{1}{4\pi t}$$

holds, then $\mathcal{N}_S$ is dense in $\mathcal{B}$ and $\xi \mathcal{N}_S$ is dense in I_{ζ} . Condition (C1) holds for every t if $\limsup_R \nu_S(R)/R^2 = \infty$ (for example $S = \{2, 3, \dots\}$, or the d -th powers k^d , $k \geq 2$). If $\nu_S(x) \sim \kappa x^2$ with $\kappa > 0$, (C1) holds exactly when $t > 1/(4\pi\kappa)$ and (C2) exactly when $t > 1/(8\pi\kappa)$. For $S = \{2^a 3^b : a + b \geq 1\}$, $\kappa = 1/(2 \log 2 \log 3)$, so this family is dense for every $t > (\log 2)(\log 3)/(4\pi) = 0.060598\dots$; (C1) alone gives $t > (\log 2)(\log 3)/(2\pi) = 0.121196\dots$. Density for one t implies density for every larger t . Moreover $\overline{\xi \mathcal{B}} = I_{\zeta}$, although $\xi \mathcal{B} \subsetneq I_{\zeta}$.

Status. Proved in the programme for $S = \{2, 3, \dots\}$ (NCI2–NCI8, RSS1–RSS2); audited (16_). The sparse version under $\nu_S(R)/R^2 \rightarrow \infty$ is claude-ab's transplant of the NCI8 Jensen step into RSS1 (refereed, fourth pass). Condition (C1), the case $\{2^a 3^b\}$ under it and the complete proof below are the review's (version 2; 48_ §2.3, re-derived by claude-ab; version 1 gave an outline whose last step omitted the density of $g_t \mathcal{B}$). Condition (C2) and step 3' are the version-2 referee's, re-derived by claude-ab (48_ §7; check Z7). Carleman's formula is used as stated in Titchmarsh [46], §3.7, which was not re-read here; its half-plane and angle forms were checked numerically on test functions by the review and by the referee. The corresponding statement in Noor's Hardy-space setting is equivalent to RH [34]; here it is unconditional, because the topology is different.

Proof. 1. Let $\Lambda \in \mathcal{B}'$ vanish on $\mathcal{N}_S$, with $|\Lambda(F)| \leq C b_{A,M}(F)$. Put $K_w = g_t(s)(e^w - e^{(1-s)w})/s$. Since $(e^w - e^{(1-s)w})/s = w e^w \int_0^1 e^{-\theta s w} d\theta$, $w \mapsto K_w$ is a $\mathcal{B}$ -valued power series, so $L(w) = \Lambda(K_w)$ is entire, $L(0) = 0$ and $L(\log n) = 0$ for $n \in S$.

2. *Growth.* For $|\sigma| \leq A$, $s = \sigma + i\tau$, $w = u + iv$: $|g_t e^{(1-s)w}| = e^{t(\sigma^2 - \tau^2) + (1-\sigma)u + \tau v}$ and $|g_t e^w| = e^{t(\sigma^2 - \tau^2) + u}$. With $-t\tau^2 + |\tau||v| \leq v^2/(4t)$ and the maximum principle on $|s| \leq 1$ for the removable singularity, $\log |L(u + iv)| \leq v^2/(4t) + (1 + A)|u| + M \log(2 + |v|) + C'$ for all w ; the coefficient $1/(4t)$ does not depend on Λ .

3. *Carleman's formula* (Titchmarsh [46], §3.7, in the right half-plane). If $L \not\equiv 0$, its zeros $r_j e^{i\varphi_j}$ in $\rho_0 \leq |w| \leq R$, $\text{Re } w \geq 0$, satisfy $\sum_j (1/r_j - r_j/R^2) \cos \varphi_j = \frac{1}{\pi R} \int_{-\pi/2}^{\pi/2} \log |L(Re^{i\varphi})| \cos \varphi d\varphi + \frac{1}{2\pi} \int_{\rho_0}^R (y^{-2} - R^{-2}) \log |L(iy)L(-iy)| dy + O(1)$. By step 2 the arc contributes at most $\frac{R}{4\pi t} \int \sin^2 \varphi \cos \varphi d\varphi = \frac{R}{6\pi t}$, the imaginary axis at most $\frac{1}{4\pi t} \int_{\rho_0}^R (1 - y^2/R^2) dy = \frac{R}{6\pi t} + O(1)$, and the other terms $O(\log R)$.

The zeros $\log n$ lie on the positive axis and every other zero adds a nonnegative term, so $\sum_{n \in S, \rho_0 < \log n < R} (1/\log n - \log n/R^2) \leq R/(3\pi t) + O(\log R)$, which contradicts (C1).

3'. *The angle* $|\arg w| < \pi/4$. Apply the same formula to $F(z) = L(z^{1/2})$ (principal branch; analytic on $\operatorname{Re} z > 0$ and continuous up to the boundary away from 0) on $\rho_0 \leq |z| \leq R = X^2$; its zeros include $(\log n)^2, n \in S$, on the positive axis. On the arc $F(Re^{i\psi}) = L(Xe^{i\psi/2})$ with $v^2 = X^2 \sin^2(\psi/2)$, so by step 2 the arc contributes at most $\frac{1}{4\pi t} \int_{-\pi/2}^{\pi/2} \sin^2(\psi/2) \cos \psi d\psi + o(1) = \frac{1-\pi/4}{4\pi t} + o(1)$. On the imaginary axis $F(\pm iy) = L(y^{1/2}e^{\pm i\pi/4})$ with $v^2 = y/2$, so the axis contributes at most $\frac{1}{2\pi} \int_{\rho_0}^R (y^{-2} - R^{-2}) \frac{y}{4t} dy + O(1) = \frac{\log X}{4\pi t} + O(1)$; the linear and logarithmic terms of step 2 contribute $O(1)$. So, up to finitely many terms, $\sum_{n \in S, \log n < X} ((\log n)^{-2} - (\log n)^2 X^{-4}) \leq \frac{\log X}{4\pi t} + O(1)$, which contradicts (C2). Under either condition, $L \equiv 0$.

4. $\Lambda = 0$. $(\partial_w - 1)K_w = g_t e^{(1-s)w}$, so $\Lambda(g_t e^{vs}) = 0$ for all real v . Every $G \in \mathcal{B}$ is $\frac{1}{2} \int a_G(e^v) e^{sv} dv$ with a_G decaying faster than every exponential (NJS4.1), and $b_{A,M}(g_t e^{vs}) \leq C_{A,M} e^{A|v|}$, so $\Lambda(g_t G) = 0$. Finally $g_t \mathcal{B}$ is dense in $\mathcal{B}$: with $h_N = e^{ts^2} P_N(-ts^2)$, P_N the N -th Taylor polynomial of $\exp$, $Fh_N = g_t \cdot FP_N(-ts^2) \in g_t \mathcal{B}$, $|h_N| \leq e^{2t\sigma^2}$ and $h_N \rightarrow 1$ locally uniformly, so $b_{A,M}(Fh_N - F) \rightarrow 0$.

5. *The ideal*. If Λ vanishes on $\xi \mathcal{N}_S$, then $F \mapsto \Lambda(\xi F)$ is continuous on $\mathcal{B}$ and vanishes on $\mathcal{N}_S$, so $\Lambda = 0$ on $\xi \mathcal{B}$, which is dense in I_ζ (NCI2); by Hahn–Banach $\xi \mathcal{N}_S$ is dense in I_ζ .

6. *The conditions*. For $\log n < R/2$ each term of (C1) exceeds $3/(2R)$, so the sum divided by R is at least $\frac{3}{8} \nu_S(R/2)/(R/2)^2$. If $\nu_S(x) \sim \kappa x^2$, partial summation (the boundary terms vanish at R and at X , and $\nu_S = 0$ below $\log 2$) gives $\sum_{\log n < R} (1/\log n - \log n/R^2) = \int_0^R \nu_S(x)(x^{-2} + R^{-2}) dx = \frac{4}{3} \kappa R + o(R)$ and $\sum_{\log n < X} ((\log n)^{-2} - (\log n)^2 X^{-4}) = \int_0^X \nu_S(x)(2x^{-3} + 2xX^{-4}) dx = 2\kappa \log X + o(\log X)$, which give the two thresholds; for $\{2^a 3^b\}$, $\nu(x) = \kappa x^2 + O(x)$. Density passes to larger t because $\mathcal{N}_S^{(t)} = g_{t-t'} \mathcal{N}_S^{(t')}$, multiplication by $g_{t-t'}$ is continuous on $\mathcal{B}$, and $g_{t-t'} \mathcal{B}$ is dense (step 4). $\square$

Proposition 3.12 (an infinite set of degrees need not suffice). *If $S \subset \{a^k : k \geq 1\} \cup F$ with an integer $a \geq 2$ and F finite, then the closures of $\mathcal{N}_S$ in $\mathcal{B}$ and of $\xi \mathcal{N}_S$ in I_ζ have infinite codimension; in particular neither family is dense. If $F = \emptyset$, the closure of $\mathcal{N}_S$ lies in I_D with $D = \sum_{j \neq 0} [2\pi i j / \log a]$.*

Status. Proved here (claude-ab, 31_ Lemma 31.1, which states non-density). Refereed. The infinite codimension and the ideal I_D are version 2's (48_ §3, F14; verified independently).

Proof. Put $s_j = 2\pi i j / \log a$ for $j \in \mathbb{Z} \setminus \{0\}$ and $h_n = (n - n^{1-s})/s = n(1 - n^{-s})/s$. For $n = a^k$, $n^{-s_j} = e^{-2\pi i j k} = 1$, so every generator $g_t h_n$ with n a power of a vanishes at every s_j (while $h_n(0) = n \log n \neq 0$). If $F = \emptyset$ this is the last statement, since I_D is an intersection of kernels of continuous point evaluations. Distinct point evaluations are linearly independent on $\mathcal{B}$ (the functions $e^{s^2} \prod_{i \neq j} (s - s_i) \in \mathcal{B}$ separate finitely many points). Fix $J \geq 1$. The combinations $\Lambda = \sum_{j=1}^{J+|F|} c_j \operatorname{ev}_{s_j}$ satisfying the $|F|$ linear conditions $\Lambda(g_t h_n) = 0, n \in F$, form a space of dimension at least J ; each annihilates $\mathcal{N}_S$ and hence its closure. So the closure has codimension at least J , for every J . For the ideal, I_ζ contains $g_t \xi P$ for every polynomial P , and $g_t(s_j) \xi(s_j) \neq 0$ because $\xi(s_j) = \xi(1 - s_j) \neq 0$, ξ having no zeros on $\operatorname{Re} s = 1$; by Lagrange interpolation the evaluations at $s_1, \dots, s_{J+|F|}$ stay linearly independent on I_ζ . The combinations annihilating $\xi g_t h_n$ for $n \in F$ again form a space of dimension at least J , and they annihilate $\xi \mathcal{N}_S$; so its closure in I_ζ has codimension at least J , for every J . $\square$

The programme's corrected tests behave like $\mathcal{N}_S$: if Φ is entire, $\Phi(0) = 1$ and $g_t(1 - \Phi)/s \in \mathcal{B}$, then $\{g_t(n^{1-s} - n\Phi)/s : n \geq 1\}$ spans a dense subspace of $\mathcal{B}$ (register S59, 30_ Corollary 30.2), and for the programme's correction $C = 8F_*$ every tail $\{g_t(\varphi_m + C/s) : m \geq M\}$ with Noor's φ_m is already total in $\mathcal{B}$; for a general entire C with $C(0) = 1$ and $g_t C \in \mathcal{B}$, and $M \geq 1$, the annihilator in $\mathcal{B}'$ of the tail is 0 if $C(1) \neq 0$ and $\mathbb{C} \operatorname{ev}_1$ if $C(1) = 0$ (register S62, 32_ Lemma 32.1).

3.5 Positive forms and receivers

The programme examines many ways of attaching a positive form, a Hilbert space or a pairing to the quotient. The exact results are these.

Lemma 3.13 (transfer-compatible forms). *On $H = \ell^2(\mathbb{Z}, m)$, with $\|x\|^2 = \sum_{\rho} m_{\rho} |x_{\rho}|^2$ and coordinate vectors ε_{ρ} , let $(T_n x)_{\rho} = n^{\rho} x_{\rho}$ and $U_n = nT_{1/n}$. Let B be a bounded Hermitian form on H with $B(T_n x, T_n y) = nB(x, y)$ for $n = 2$ and $n = 3$. Then*

$$B(x, y) = \sum_{\rho \in \mathbb{Z}} b_{\rho} m_{\rho} x_{\rho} \overline{y_{\rho^{\#}}}, \quad |b_{\rho}| \leq \|B\|, \quad b_{\rho^{\#}} = \overline{b_{\rho}},$$

and conversely every such form satisfies $B(T_n x, T_n y) = nB(x, y)$ for every real $n > 0$. The form B is positive semidefinite if and only if $b_{\rho} = 0$ at every zero off the line and $b_{\rho} \geq 0$ on the line. In particular, a bounded positive semidefinite Hermitian form satisfies $B(T_n x, y) = B(x, U_n y)$ for all integers $n \geq 1$ if and only if $B(x, y) = \sum_{\text{Re } \rho = 1/2} m_{\rho} c_{\rho} x_{\rho} \overline{y_{\rho}}$ with $0 \leq c_{\rho} \leq C$. The pair $\{2, 3\}$ may be replaced by any $n_1, n_2 > 1$ with $\log n_1 / \log n_2$ irrational; for a single n the relation also allows an entry at (ρ, η) whenever $\eta - \rho^{\#} \in (2\pi i / \log n)\mathbb{Z}$.

Status. Proved in the programme for positive forms and all $n \geq 1$ (PTQ4); the off-diagonal step shortened here (claude-ab, 25_ Lemma 25.2). Refereed (eighth pass). The classification of all bounded Hermitian forms from the two values $n = 2, 3$ is the review's (version 2; 48_ §2.2, re-derived by claude-ab).

Proof. On coordinate vectors $B(T_n \varepsilon_{\rho}, T_n \varepsilon_{\eta}) = n^{\rho + \bar{\eta}} B(\varepsilon_{\rho}, \varepsilon_{\eta})$, so the relation reads $(n^{\rho + \bar{\eta} - 1} - 1)B(\varepsilon_{\rho}, \varepsilon_{\eta}) = 0$. If the entry is nonzero, then $\rho + \bar{\eta} - 1$ lies in $(2\pi i / \log 2)\mathbb{Z} \cap (2\pi i / \log 3)\mathbb{Z} = \{0\}$ (a common nonzero element $2\pi i k / \log 2 = 2\pi i l / \log 3$ would give $3^k = 2^l$ with $k \neq 0$), so $\eta = 1 - \bar{\rho} = \rho^{\#}$. For finitely supported x, y this is the displayed sum with $b_{\rho} = B(\varepsilon_{\rho}, \varepsilon_{\rho^{\#}}) / m_{\rho}$, and $|b_{\rho}| \leq \|B\| \|\varepsilon_{\rho}\| \|\varepsilon_{\rho^{\#}}\| / m_{\rho} = \|B\|$ because $m_{\rho^{\#}} = m_{\rho}$. The sum is bounded by $\sup_{\rho} |b_{\rho}| \|x\| \|y\|$ (Cauchy–Schwarz, again with $m_{\rho^{\#}} = m_{\rho}$), so both sides are continuous and agree on H ; Hermitian symmetry gives $b_{\rho^{\#}} = \overline{b_{\rho}}$. Conversely $\rho + \bar{\rho^{\#}} = 1$, so every such form satisfies the relation for every $n > 0$. Positivity: an on-line zero ($\rho^{\#} = \rho$) contributes $b_{\rho} m_{\rho} |x_{\rho}|^2$, which needs $b_{\rho} \geq 0$; an off-line pair $\{\rho, \rho^{\#}\}$ contributes $2m_{\rho} \text{Re}(b_{\rho} x_{\rho} \overline{x_{\rho^{\#}}})$, a Hermitian form on $\mathbb{C}^2$ with eigenvalues $\pm m_{\rho} |b_{\rho}|$, which is ≥ 0 only if $b_{\rho} = 0$. For the positive case, T_n is invertible with $T_n^{-1} = T_{1/n}$ (because $|n^{-\rho}| \leq 1$), so $U_n = nT_n^{-1}$ and $B(T_n x, y) = B(x, U_n y)$ for all x, y is equivalent to $B(T_n x, T_n y) = nB(x, y)$; the forms found are those with $b_{\rho} \geq 0$ supported on the line, and $C = \sup_{\rho} b_{\rho} \leq \|B\|$. The last sentence follows from the same computation, since $\rho + \bar{\eta} - 1 = \overline{\eta - \rho^{\#}}$ and $(2\pi i / \log n)\mathbb{Z}$ is closed under conjugation. $\square$

With $b \equiv 1$ the form is $\sum_{\rho} m_{\rho} x_{\rho} \overline{y_{\rho^{\#}}}$, whose diagonal is Weil's pairing Ω of Section 1. The lemma shows that the scaling relation for two multiplicatively independent integers already forces this shape, and that positivity is exactly what removes the off-line blocks.

The following are proved in the programme and were audited; each is stated with its scope.

- *Positive forms compatible with the scaling transfer* on $\mathcal{B}/I_{\zeta}$ are exactly $\sum_{\text{Re } \rho = 1/2} m_{\rho} c_{\rho} F(\rho) \overline{G(\rho)}$ with polynomially bounded $c_{\rho} \geq 0$ (CFP; 04_; register S12). Before the quotient, the positive transfer forms are exactly $\int F(\frac{1}{2} + i\lambda) \overline{G(\frac{1}{2} + i\lambda)} d\mu(\lambda)$ with μ positive and tempered, and they descend to the quotient if and only if $\text{supp } \mu$ lies in the set of ordinates of critical zeros (SPF0–SPF10; 09_ 12_ S22; the Bochner–Schwartz theorem in Mellin coordinates).
- *All positive measure receivers at once.* For every positive measure μ with polynomially bounded mass on unit intervals, the closure of the image of I_{ζ} in $L^2(\mu)$ under $F \mapsto F(\frac{1}{2} + i\lambda)$ is $L^2(\mathbb{R} \setminus \Gamma, \mu)$, Γ the set of ordinates of critical zeros; so a positive receiver sees only line values at zeros carrying mass (SMC1–SMC4, SMC9; 18_ S35).

- *Connes against Meyer.* The natural map from Q to the inverse limit of Connes' weighted Hilbert quotients [16] has kernel exactly the classes vanishing on all critical-line jets, so it is injective if and only if RH holds; the algebraic map to each H_δ/J is injective, and the loss happens in the Hausdorff quotient (WHR5–WHR10; 12₊; S28). The prime dilations on these Hilbert quotients have spectrum on $|z| = \sqrt{p}$ (WHR8; S27), and on the analogous quotients of $L^2(u^{2\sigma-1}(1 + \log^2 u)^\delta du)$ for $0 < \sigma < 1$ exactly the zeros on $\operatorname{Re} s = \sigma$ are realised, with jet orders $j < \min(m_\rho, \delta - \frac{1}{2})$ and spectrum on $|z| = p^\sigma$; the family of these quotients over all $0 < \sigma < 1$ and all integers $\delta = N \geq 1$ is jointly faithful on Q (VWR2–VWR7; 14₊; S31).
- *Spectral synthesis.* Explicit finite-rank Riesz–Gaussian operators converge to the identity on Q ; closed multiplier submodules are exactly the jet-order submodules; the line ideal and the off-line ideal have dense sum (GSP2–GSP6, CTS1–CTS4; 12₊; S29; read for correctness, not re-derived).
- *Global residue duality.* the pairing

$$B_\zeta(F, G) = \frac{1}{2\pi i} \left(\int_{\operatorname{Re} s=2} - \int_{\operatorname{Re} s=-1} \right) \frac{F(s)G(1-s)}{\zeta(s)} ds$$

satisfies $|B_\zeta(F, G)| \leq \zeta(2)(\pi^{-1} + 8\pi/5) b_{2,1}(F) b_{2,1}(G) \approx 8.79 b_{2,1}(F) b_{2,1}(G)$ (the constant recorded in 26₊ §2.1; version 1 printed the weaker $\zeta(2)(1 + 4\pi^2)\pi^{-1} \approx 21.19$), descends to Q , is nondegenerate, has local determinant $(m!/\zeta^{(m)}(\rho))^m$, and satisfies $B(T_a x, T_a y) = aB(x, y)$ and $B(P(L)x, y) = B(x, P(1-L)y)$ for every polynomial P ; it is not positive (GZR; 26₊ Lemma 26.1; S49).

- *The source-pairing identity.* If $b_0 \in A$ has Mellin transform vanishing at ρ and $(L - \rho)b_\rho = b_0$ with $L = -u\partial_u$, then $(\operatorname{Re} \rho - \frac{1}{2})\|b_\rho\|_{L^2(du)}^2 = -\operatorname{Re}\langle b_\rho, b_0 \rangle$. (Proof: $\operatorname{Re}\langle Lb, b \rangle = -\frac{1}{2} \int u(|b|^2)' du = \frac{1}{2}\|b\|^2$ by one integration by parts, so $\operatorname{Re}\langle (L - \rho)b_\rho, b_\rho \rangle = (\frac{1}{2} - \operatorname{Re} \rho)\|b_\rho\|^2$.) It restates RH at ρ as $\operatorname{Re}\langle b_\rho, b_0 \rangle = 0$; reading it as a Hilbert–Pólya mechanism is an interpretation (OPD4.2; 09₊; S20).
- *The harmonic defect and the divisor potential.* Poisson-sweeping the zeros onto the line gives $H(b_+) = 2 \sum_\rho m_\rho |\operatorname{Re} \rho - \frac{1}{2}| \|b_\rho\|^2$ for an explicit test, and the swept distribution descends to Q if and only if there are no off-line zeros (HSW5–HSW6B, OPD5; S21); for finitely many exponents with multiplicities, the translation trace compressed to the jet-constraint span on $[-T, T]$ tends to $\sum m e^{-|\beta-1/2||t|} e^{i\gamma t}$ (FGR0–FGR6; S26); and for $r > 1, t > 0$ the nonnegative quantity $\mathcal{E}_r(t) = \sum_\rho m_\rho (r^\sigma - r^{1-\sigma})^2 e^{2t(\sigma^2 - \gamma^2)}$, which vanishes if and only if RH holds, equals an area integral of the even part $\frac{1}{2} \log |\zeta(s)\zeta(1-\bar{s})|$ against the Laplacian of an explicit test function, plus $(r-1)^2(e^{2t} + 1)/2$, when the test function is built from a symmetric cutoff equal to 1 near $[0, 1]$ and supported in $(-\frac{1}{2}, \frac{3}{2})$; for a wider cutoff further terms from the trivial zeros appear (OZD0–OZD7; 12₊ §1.1; S25; a member of the Littlewood-lemma family [5, 43]).

Taken together these results say three things. The positive objects are supported on the critical zeros; this is a theorem, by the classifications above. The objects listed that register the off-line zeros vanish, or are injective, exactly when RH holds, so each of them is an equivalent form of RH written in the programme's terms. The residue duality B_ζ is nondegenerate unconditionally and is not positive. Section 7 states these facts as negative results with their exact scope and, where it is known, the input the construction would need.

4 The two lines

The author proposed a second line at $\operatorname{Re} s = -\frac{1}{2}$ that mirrors the critical line, and asked for it to be built by shifting the whole line, as in the programme's Hurwitz shift, and for the anomaly, chirality and holonomy language of that picture to be made exact. This section gives the exact

objects. (The Hurwitz flow does not translate the zero set; what vanishes on the shifted set, at every time of the flow, is its first jet, Theorem 4.1.) The typed matching of the author's terms with the objects is a proposal (Section 8); the objects and the facts below are theorems, none of them assumes RH, and Proposition 4.10 states its GRH hypotheses where it uses them. Recall $A(s) = -\bar{s}$ (the mirror through the pole at 0), $s^\# = 1 - \bar{s}$ (the reflection through $\frac{1}{2}$) and $T_c(s) = s + c$; a direct computation gives $s^\# = T_1(A(s))$.

4.1 The first Hurwitz jet and the mirror set

Theorem 4.1 (the first Hurwitz jet). *Let $a > 0$; the sheet at time t of the flow is $a = 1 + t$.*

- (a) *For $s \neq 1$, $\partial_a \zeta(s, a) = -s\zeta(s+1, a)$, where at $s = 0$ the right side means its holomorphic extension; more generally $\partial_a^m \zeta(s, a) = (-1)^m (s)_m \zeta(s+m, a)$, and both sides extend to entire functions of s . At $a = 1$: $\partial_t \zeta(s, 1+t)|_{t=0} = -s\zeta(s+1)$.*
- (b) *$s\zeta(s+1, a)$ is entire and equals 1 at $s = 0$; so the first jet equals -1 at $s = 0$.*
- (c) *The zero divisor of the first jet $-s\zeta(s+1, a)$ is exactly $\text{div } \zeta(\cdot, a) - 1$, with multiplicities, in the whole plane. The zero divisor of the m -th jet is $(\text{div } \zeta(\cdot, a) - m) + [0] + [-1] + \dots + [2-m]$, the last $m-1$ zeros being simple and distinct from the shifted ones. At $a = 1$: in $-1 < \text{Re } s < 0$ the zeros of the first jet are exactly $\mathbb{Z} - 1$, with multiplicities, and in $-m < \text{Re } s < 1-m$ the zeros of the m -th jet are exactly $\mathbb{Z} - m$.*
- (d) *ζ has no zero on $\mathbb{Z} - 1$.*

Status. Proved here (claude-ab, 17_ Theorem 17.1, at $a = 1$; part (a) on $\text{Re } s > 1$ is the programme's shift-flow theorem). Checked numerically (the t -derivative at $\rho_k - 1$ is below $4 \cdot 10^{-29}$ for $k = 1, 2, 3$). Refereed (fifth pass). The form for every sheet $a > 0$ and the whole-plane divisors are the review's (version 2; 48_ §2.1, re-derived by claude-ab; check Z1, which also finds the jet vanishing at $s_0 - 1$ for the zero $s_0 = 1.40778804 + 23.32798775i$ of $\zeta(\cdot, 2)$).

Proof. (a) On $\text{Re } s > 1$, $\partial_a \sum_n (n+a)^{-s} = -s \sum_n (n+a)^{-s-1}$ termwise. The Euler–Maclaurin representation shows that $\zeta(s, a)$ is jointly analytic on $\{s \neq 1\} \times \{\text{Re } a > 0\}$, so both sides are analytic in $s \neq 1$ and agree everywhere by the identity theorem; the higher jets follow in the same way. The factor $(s)_m$ vanishes simply at $1-m$, where $\zeta(s+m, a)$ has a simple pole, so the right side extends holomorphically there; and since $\zeta(s, a) - 1/(s-1)$ is entire in s (the residue at $s = 1$ is 1 for every a), $\partial_a^m \zeta(s, a)$ is entire. (b) $s\zeta(s+1, a) = 1 + s(\zeta(s+1, a) - 1/s)$, and the bracket is entire. (c) For $s \neq 0$ the factor $-s$ is a unit near s , so off $s = 0$ the zeros of $-s\zeta(s+1, a)$ are the s with $\zeta(s+1, a) = 0$, with the same orders; at $s = 0$ the value is -1 by (b). For the m -th jet, $(s)_m$ vanishes simply at $0, -1, \dots, 1-m$. At $1-m$ a simple zero meets a simple pole of residue 1, and the product is finite and nonzero. At $-j$ with $0 \leq j \leq m-2$ the other factor is $\zeta(m-j, a) = \sum_{n \geq 0} (n+a)^{-(m-j)} > 0$, so these zeros are simple and are not shifted zeros. At $a = 1$, $\text{div } \zeta = \mathbb{Z} + \sum_{k \geq 1} [-2k]$; in $-m < \text{Re } s < 1-m$ only $\mathbb{Z} - m$ remains, because the shifted trivial zeros have real part at most $-2-m$ and the points $0, \dots, 2-m$ have real part greater than $1-m$. (d) Points of $\mathbb{Z} - 1$ have real part in $(-1, 0)$, where ζ has no zeros ($\zeta \neq 0$ on $\text{Re } s \geq 1$, and the functional equation leaves only the trivial zeros in $\text{Re } s \leq 0$). $\square$

Theorem 4.2 uses the functional equation of ζ and does not extend to other sheets: at $a = \frac{1}{2}$, $s_1 = 2\pi i / \log 2$ is a zero of $\zeta(s, \frac{1}{2}) = (2^s - 1)\zeta(s)$, while $\zeta(1 - \bar{s}_1, \frac{1}{2}) = (2^{1+2\pi i / \log 2} - 1)\zeta(1 + 2\pi i / \log 2) = \zeta(1 + 2\pi i / \log 2) \neq 0$, so the zero set of that sheet is not symmetric under $s \mapsto 1 - \bar{s}$. (Part (d) is proved from the functional equation and is not claimed for other sheets; at $a = \frac{1}{2}$ it holds.) What extends is (a)–(c): at every time t of the flow, the first t -jet of $\zeta(s, 1+t)$ vanishes exactly on the zero set of that time translated by -1 .

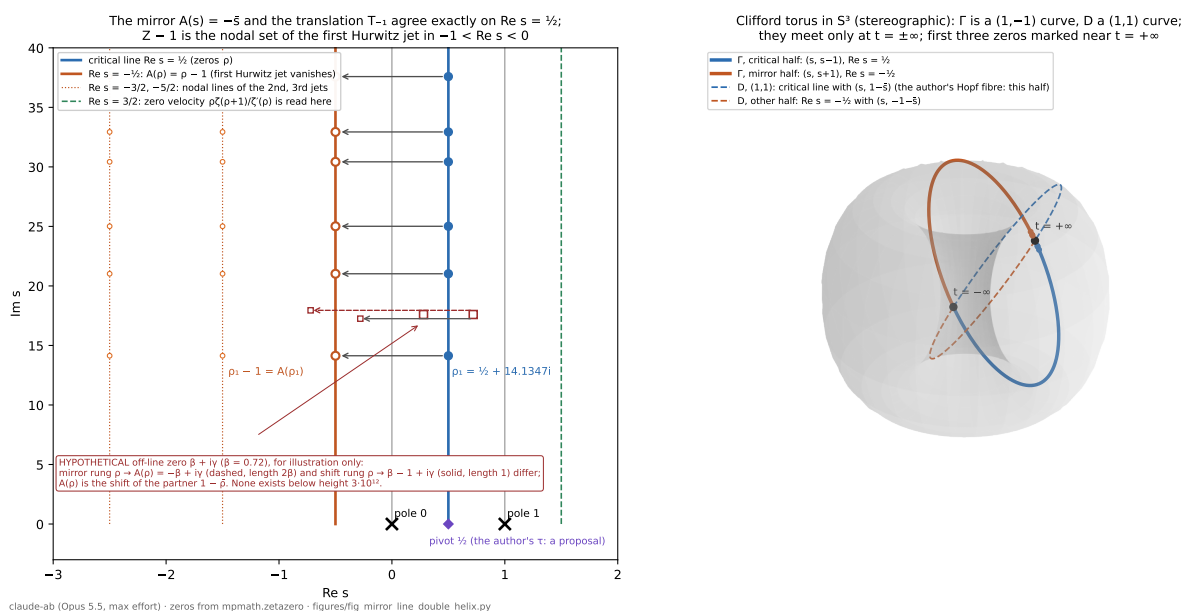


Figure 3: Left: the critical line, its mirror $A(Z) = Z - 1$ on $\text{Re } s = -\frac{1}{2}$ (the nodal set of the first Hurwitz jet in $-1 < \text{Re } s < 0$), and a *hypothetical* off-line zero, drawn only to show that its mirror rung and its shift rung differ; no such zero exists below height $3 \cdot 10^{12}$. Right: on the Clifford torus the two lines form one $(1, -1)$ circle Γ , which meets the $(1, 1)$ fibre D only at $t = \pm\infty$ (17_ Proposition 17.3).

Implicit differentiation of $\zeta(\rho(t), 1+t) = 0$ gives the velocity $\rho'(0) = \rho\zeta'(\rho+1)/\zeta'(\rho)$ of a simple zero, read on the line $\operatorname{Re} s = 1 + \operatorname{Re} \rho$; for ρ_1 it is $0.98358 + 9.65780i$, so the first zero leaves the critical line to the right as t increases.

Theorem 4.2 (the shifted set is the mirror set). $A(\mathbb{Z}) = \mathbb{Z} - 1$ as sets with multiplicities, unconditionally. For every s , $A(s) - (s - 1) = 1 - 2 \operatorname{Re} s$; hence $A(\rho) = \rho - 1$ exactly when $\operatorname{Re} \rho = \frac{1}{2}$, and

$$RH \iff A(\rho) = \rho - 1 \text{ for every nontrivial zero } \rho.$$

Moreover $A(\rho) = \rho^\# - 1$: the mirror of ρ is the shift of its partner.

Status. Proved here (claude-ab, 17_ Theorem 17.2). Refereed (fifth pass).

Proof. $-\bar{\rho} = (1 - \bar{\rho}) - 1$, and $1 - \bar{\rho} \in \mathcal{Z}$ with the multiplicity of ρ ; $-\bar{s} - (s - 1) = 1 - (s + \bar{s})$. $\square$

So the author's two operations produce the same line (Figure 3, left): as sets the mirror and the shift agree unconditionally, as maps on the zeros they agree exactly under RH, and the set carries no zero of ζ . On the Clifford torus the two lines form a single $(1, -1)$ circle, which meets the $(1, 1)$ fibre only at $t = +\infty$ (Figure 3, right; 17 Proposition 17.3).

Proposition 4.3 (the blind lemma). *Let $H_\sigma = L^2((0, \infty), u^{2\sigma-1} du)$ with $U_a f(u) = f(au)$. If $\sigma \neq \sigma'$, every bounded $K : H_\sigma \rightarrow H_{\sigma'}$ with $KU_{a_0} = U_{a_0}K$ for a single $a_0 > 0$, $a_0 \neq 1$, is zero; in particular every bounded K commuting with all U_a is zero. The inversion $f \mapsto f(1/u)$ is an isometry of H_σ onto $H_{-\sigma}$ acting on Mellin transforms by $s \mapsto -s$ and reversing the flow.*

Status. Proved here (claude-ab, 17_ Proposition 17.4, for all $a > 0$). Refereed (fifth pass). The form with one dilation is version 2's (48_ §3, F16).

Proof. $\|U_a f\|_\sigma = a^{-\sigma} \|f\|_\sigma$, so $U_a = a^{-\sigma} V_a$ with V_a unitary on H_σ , and likewise on $H_{\sigma'}$. Since $U_a U_b = U_{ab}$ and $U_{a_0}^{-1} = U_{1/a_0}$, K commutes with every U_a , $a = a_0^k$, $k \in \mathbb{Z}$. For each such a , $V_a' K = a^{\sigma' - \sigma} K V_a$, so $\|Kf\| \leq a^{\sigma' - \sigma} \|K\| \|f\|$; as $k \rightarrow \pm\infty$, a_0^k tends to 0 in one direction and to ∞ in the other, so $Kf = 0$. The inversion statement is the substitution $v = 1/u$. $\square$

Boundedness is essential: the identity on $C_c^\infty(0, \infty)$ is a nonzero closable densely defined operator $H_{1/2} \rightarrow H_{-1/2}$ commuting with every U_a .

4.2 The two centres of the explicit formula

With $(g_1 \star g_2)(x) = \int g_1(y) g_2(x/y) dy/y$ and $\hat{g}(s) = \int_0^\infty g(x) x^{s-1} dx$, use Weil's explicit formula in Bombieri's form [8].

Proposition 4.4 (two centres). (a) With $g^*(x) = x^{-1} \overline{g(1/x)}$ and $h = g \star g^*$: $\hat{h}(s) = \hat{g}(s) \overline{\hat{g}(1-\bar{s})}$, both sides of the explicit formula are real, an on-line zero contributes $m_\rho |\hat{g}(\rho)|^2 \geq 0$, and an off-line pair $\{\rho, \rho^\#\}$ contributes $2m_\rho \operatorname{Re}(\hat{g}(\rho) \overline{\hat{g}(\rho^\#)})$, an indefinite form with eigenvalues $\pm m_\rho$ on $(\hat{g}(\rho), \hat{g}(\rho^\#))$. (b) With $g^b(x) = \overline{g(1/x)}$ and $h = g \star g^b$: $\hat{h}(s) = \hat{g}(s) \overline{\hat{g}(-\bar{s})}$, the prime term has imaginary part $\sum_n \Lambda(n) (1 - 1/n) \operatorname{Im} h(n)$, and the sum over zeros is not real in general.

Status. Proved here (claude-ab, 17_ Proposition 17.5). Both sides computed independently for the test $g(x) = \exp(-(\log x)^2/2 + \frac{i}{2} \log x - i\gamma_1 \log x)$: centre $\frac{1}{2}$ gives 2π on both sides to $4 \cdot 10^{-15}$; centre 0 gives $2\pi e^{i/2}$ to $1.6 \cdot 10^{-12}$. Refereed (fifth pass).

Proof. Substituting $y = 1/x$ gives $\widehat{g^*}(s) = \overline{\hat{g}(1-\bar{s})}$ and $\widehat{g^b}(s) = \overline{\hat{g}(-\bar{s})}$; the Mellin transform turns $\star$ into a product. The zero side of (a) is real because $\rho \mapsto \rho^\#$ permutes $\mathcal{Z}$ with multiplicities; $h(1/x) = x \overline{h(x)}$ makes $h(n) + h(1/n)/n = 2 \operatorname{Re} h(n)$. In (b), $h(1/x) = \overline{h(x)}$, so $h(n) + h(1/n)/n = h(n) + \overline{h(n)}/n$, whose imaginary part is $(1 - 1/n) \operatorname{Im} h(n)$. $\square$

So a real partner pairing selects the centre $\frac{1}{2}$ and excludes the pole at 0; with that centre, the only possible source of negativity is the hyperbolic block of an off-line pair, and by Weil's criterion [50, 8] a negative value exists if and only if RH fails.

4.3 The two-line product and its symmetries

Proposition 4.5 (the two-line product on the line of the pole at 0). $G(s) = \xi(s) \xi(s+1)/16$ satisfies $G(-\bar{s}) = \overline{G(s)}$ and $G(it) = |\xi(1+it)|^2/16 > 0$ for real t . Its zero set is $\mathcal{Z} \sqcup (\mathcal{Z} - 1)$.

Status. Proved here (claude-ab, 22_ Proposition 22.4, 24_ Lemma 24.3; register S37). Checked to 10^{-30} . Refereed (eighteenth pass).

Proof. From $\xi(1-s) = \xi(s)$ and $\xi(\bar{s}) = \overline{\xi(s)}$: $\xi(-\bar{s}) = \xi(1+\bar{s}) = \overline{\xi(1+s)}$ and $\xi(1-\bar{s}) = \overline{\xi(s)}$, which gives the first identity; at $s = it$, $\xi(it) = \xi(1-it) = \overline{\xi(1+it)}$, and $\xi(1+it) \neq 0$ because ξ has no zeros on $\operatorname{Re} s = 1$. $\square$

The maps id , A , P ($= \#$ on $\mathcal{Z}$ and $s \mapsto -1 - \bar{s}$ on $\mathcal{Z} - 1$) and $T = AP$ ($s \mapsto s - 1$ on $\mathcal{Z}$, $s \mapsto s + 1$ on $\mathcal{Z} - 1$) form a Klein four-group preserving $\mathcal{Z} \sqcup (\mathcal{Z} - 1)$ with multiplicities. An on-line zero has orbit $\{\rho, \rho - 1\}$; an off-line zero has the orbit of four points $\{\rho, \rho^\#, \rho^\# - 1, \rho - 1\}$ (the author's doublet). The loop that goes across by A and back by T^{-1} has holonomy $T^{-1}A = \#$ on $\mathcal{Z}$, trivial exactly at on-line zeros; and the zeros of G in the open strip $-\frac{1}{2} < \operatorname{Re} s < \frac{1}{2}$ (the bulk between the lines) are exactly the zeros with $\operatorname{Re} \rho < \frac{1}{2}$ and their mirrors, so the bulk is empty if and only if RH holds (22_ Propositions 22.6–22.7; the checks are elementary).

4.4 The chiral cross form is Weil's form on the diagonal

For finitely supported u on $\mathbb{Z}$ let $\Omega(u) = \sum_{\rho} m_{\rho} u(\rho) \overline{u(\rho^{\#})}$ (Weil's pairing), and for f on $\mathbb{Z} \sqcup (\mathbb{Z} - 1)$ let $Q_A(f) = \sum_w m_w f(w) \overline{f(Aw)}$ (the chiral cross form, centred at 0). Both are real because $\#$ and A are involutions preserving multiplicities.

Proposition 4.6 (the chiral form and Weil's form). *Let f be finitely supported on $\mathbb{Z} \sqcup (\mathbb{Z} - 1)$, or the restriction of an element of $\mathcal{B}$ (then all sums converge absolutely). Put $u(\rho) = \frac{1}{2}(f(\rho) + f(\rho - 1))$ and $v(\rho) = \frac{1}{2}(f(\rho) - f(\rho - 1))$, the T-even and T-odd parts. Then*

$$Q_A(f) = 2\Omega(u) - 2\Omega(v).$$

- (a) A has no fixed point on $\mathbb{Z} \sqcup (\mathbb{Z} - 1)$, so Q_A is an orthogonal sum of hyperbolic planes, of signature (n, n) on data supported on n pairs $\{w, Aw\}$, whether or not RH holds.
- (b) $\Omega(u) = \sum_{\text{on-line}} m_{\rho} |u(\rho)|^2 + \sum_{\text{off-line pairs}} 2m_{\rho} \operatorname{Re}(u(\rho) \overline{u(\rho^{\#})})$; so $\Omega \geq 0$ on all finitely supported u exactly when every zero is on the line.
- (c) Let V_+ and V_- be the T-even and the T-odd data. Unconditionally V_+ and V_- are Q_A -orthogonal, and Q_A is 2Ω on V_+ and -2Ω on V_- . On T-even data supported on a finite $\#$ -closed set of zeros, with n_{on} points on the line and n_{pairs} off-line pairs, the signature of Q_A is $(n_{\text{on}} + n_{\text{pairs}}, n_{\text{pairs}})$. Under RH V_+ is maximal positive and V_- maximal negative; if RH fails, V_+ contains a negative direction.

Status. Proved here (claude-ab, 42_ Proposition 42.2; register S72). Checks B1–B5. Refereed (nineteenth pass, which extended it to data from $\mathcal{B}$). In (c), the unconditional orthogonality, the signature and the maximality of V_- are version 2's (48_ §3, F17; 42_ states the orthogonality inside its RH clause).

Proof. For $\rho \in \mathbb{Z}$, $A\rho = \rho^{\#} - 1 \in \mathbb{Z} - 1$ and $A(\rho - 1) = \rho^{\#}$, so $Q_A(f) = \sum_{\rho} m_{\rho} [f(\rho) \overline{f(\rho^{\#} - 1)} + f(\rho - 1) \overline{f(\rho^{\#})}]$. Substituting $f = u + v$ on $\mathbb{Z}$ and $f = u - v$ on $\mathbb{Z} - 1$, the cross terms cancel in pairs and $2 \sum_{\rho} m_{\rho} [u(\rho) \overline{u(\rho^{\#})} - v(\rho) \overline{v(\rho^{\#})}]$ remains. (a) $Aw = w$ means $\operatorname{Re} w = 0$, where G has no zero (Proposition 4.5); on a pair $\{w, Aw\}$ the form is $2m_w \operatorname{Re}(f(w) \overline{f(Aw)})$. (b) $\#$ fixes the on-line zeros and pairs the off-line ones; for an off-line ρ take $u = 1$ at ρ and -1 at $\rho^{\#}$. (c) For T-even u and T-odd v the sesquilinear form is $Q_A(u, v) = \sum_{\rho} m_{\rho} [u(\rho) \overline{v(\rho^{\#} - 1)} + u(\rho - 1) \overline{v(\rho^{\#})}] = \sum_{\rho} m_{\rho} u(\rho) [-\overline{v(\rho^{\#})} + \overline{v(\rho^{\#})}] = 0$, and the displayed identity gives the values on $V_{\pm}$. On a finite $\#$ -closed support each on-line point gives a positive square and each off-line pair a hyperbolic plane, whence the signature. Under RH, $\Omega(u) = \sum m|u|^2$, so Q_A is positive definite on V_+ and negative definite on V_- ; a subspace strictly containing V_+ contains a nonzero $x \in V_-$ (subtract the V_+ component), with $Q_A(x) < 0$, and symmetrically for V_- . If RH fails, (b) gives u with $\Omega(u) < 0$. $\square$

Corollary 4.7 (an RH criterion in two-line terms). *For $k \in \mathcal{B}$ put $f_k(s) = F_*(s + 1)F_*(s - 1)k(s) + F_*(s + 2)F_*(s)k(s + 1)$. Then $f_k \in \mathcal{B}$ is T-even on the zero set, with $u(\rho) = F_*(\rho + 1)F_*(\rho - 1)k(\rho)$, and*

$$RH \iff Q_A(f_k) \geq 0 \text{ for every } k \in \mathcal{B}.$$

Status. Proved here (claude-ab, 42_ Corollary 42.3). An equivalent form of RH: the proof localises the test data at the zeros, so the criterion is the positivity of Weil's pairing Ω on the diagonal family. Refereed (nineteenth pass).

Proof. $F_*(\rho) = 0$ gives $f_k(\rho) = F_*(\rho + 1)F_*(\rho - 1)k(\rho) = f_k(\rho - 1)$. $F_*(\rho \pm 1) \neq 0$, since $\operatorname{Re}(\rho \pm 1) \notin (0, 1)$ and F_* vanishes only on $\mathbb{Z}$. Finite sums of the programme's section representatives realise any finitely supported values of k on $\mathbb{Z}$, so u ranges over all finitely supported data and Proposition 4.6(b) applies. Under RH, $\Omega(u) = \sum m|u|^2 \geq 0$. $\square$

Theorem 4.2 and Proposition 4.6 hold for every primitive Dirichlet L -function, complex χ included, with the zero set Z_χ of $\Lambda(s, \chi)$ in place of Z and GRH for χ in place of RH. Their proofs use only that Z_χ lies in $0 < \operatorname{Re} s < 1$, that $G_\chi(it) \neq 0$ (proof of Proposition 4.12), for data from $\mathcal{B}$ the zero count $N(T, \chi) \ll T \log(qT)$, and that $\rho \mapsto \rho^\#$ permutes Z_χ with multiplicities, which follows from $\Lambda(s, \chi) = W(\chi) \Lambda(1-s, \bar{\chi})$ and $\overline{\Lambda(\bar{s}, \bar{\chi})} = \Lambda(s, \chi)$: together they give $\Lambda(1-\bar{s}, \chi) = W(\chi) \overline{\Lambda(s, \chi)}$. For complex χ the zero set is not symmetric under $s \mapsto 1-s$; only $\#$ is used. Corollary 4.7 is specific to ζ and is not claimed (48_ §3, F17; the standard inputs for $L(s, \chi)$ were not re-read here). $Q_A(f)$ is Weil's explicit-formula functional of G applied to $h = f \cdot f^A$, $f^A(s) = \overline{f(-\bar{s})}$: for $a > 1$, $Q_A(f) = (2\pi i)^{-1} \int_{(a)} [h(s) + h(-s)] (G'/G)(s) ds$, where on $\operatorname{Re} s = a$

$$\frac{G'}{G}(s) = \frac{2}{s} + \frac{1}{s-1} + \frac{1}{s+1} - \log \pi + \frac{1}{2} \psi\left(\frac{s}{2}\right) + \frac{1}{2} \psi\left(\frac{s+1}{2}\right) - \sum_n \Lambda(n) (1+n^{-1}) n^{-s}$$

(the horizontal sides vanish along the good heights of Lemma 3.8; checked to 25 digits). Whether the diagonal family $\{f_k\}$ is easier to control on the prime side than Weil's family is open.

4.5 Counting: parity and the holonomy index

Proposition 4.8 (parity sees only the count on the line). *For $T > 0$ not an ordinate, $(-1)^{N(T)} = (-1)^{N_0(T)} = -\operatorname{sgn} Z(T)$, with no simplicity assumption. In the two-line system the count up to height T is $n_G(T) = 2N_0(T) + 4P(T)$; its reduction mod 2 vanishes identically, and $i^{n_G(T)} = (-1)^{N_0(T)} = -\operatorname{sgn} Z(T)$. Neither depends on $P(T)$; mod 8, n_G gives $P(T) \bmod 2$ only once $N_0(T) \bmod 4$ is known, and $(N_0, P) = (0, 1)$ and $(2, 0)$ both give $n_G = 4$.*

Status. Proved here (claude-ab, 22_ 22.9, corrected in the eighteenth referee pass and in the final pass). Checked at eight heights up to 1000.

Proof. With $s = \frac{1}{2} + it$, $s(s-1) = -(\frac{1}{4} + t^2)$ and $\pi^{-s/2} \Gamma(s/2) \zeta(s) = \pi^{-1/4} |\Gamma(\frac{1}{4} + \frac{it}{2})| Z(t)$, so $\zeta(\frac{1}{2} + it) = -\frac{1}{2} (\frac{1}{4} + t^2) \pi^{-1/4} |\Gamma(\frac{1}{4} + \frac{it}{2})| Z(t)$ is real and $\operatorname{sgn} \zeta(\frac{1}{2} + iT) = -\operatorname{sgn} Z(T)$. $\zeta(\frac{1}{2}) > 0$, and $\zeta(\frac{1}{2} + it)$ changes sign exactly at on-line zeros of odd order, whose number has the parity of $N_0(T)$; off-line zeros come in pairs of equal height, so $N(T) \equiv N_0(T) \pmod{2}$. The zeros of G up to height T are those of ζ and their translates, so $n_G = 2N(T) = 2N_0 + 4P$. $\square$

Proposition 4.9 (the holonomy index is Turing's). *Let $P(T)$ be the number, with multiplicity, of zeros with $0 < \gamma \leq T$ and $\operatorname{Re} \rho < \frac{1}{2}$ (one member of each off-line pair). Then*

$$I(T) := \frac{N(T) - V(T)}{2} = P(T) + \sum_{\substack{\operatorname{Re} \rho = 1/2 \\ 0 < \gamma \leq T}} \lfloor m_\rho / 2 \rfloor.$$

So $I(T) \geq 0$, and $I(T) = 0$ exactly when every zero up to height T is on the line and simple; a positive value does not separate off-line pairs from multiple zeros on the line.

Status. Proved here (claude-ab, 42_ Proposition 42.5). Checks D1–D4 and R5 (for ζ : $N = V = 29$ at $T = 100$, $N = V = 108$ at $T = 250$). Refereed (nineteenth pass).

Proof. Off-line pairs contribute $2P(T)$ to $N(T)$. Z is real-analytic and real on $\mathbb{R}$, its zeros are the on-line zeros with the same orders, and it changes sign exactly at those of odd order, so $V(T) = \sum_{\text{on-line}} (m_\rho - 2\lfloor m_\rho / 2 \rfloor)$. Subtract. $\square$

$I(T)$ is computed from boundary data ($N(T) = \theta(T)/\pi + 1 + S(T)$ by the argument principle, $V(T)$ from the line), and $I(T) = 0$ is exactly what verifications of RH by Turing's method [47] establish up to a height; RH has been verified up to $3 \cdot 10^{12}$ (Platt–Trudgian [35]). The statement $I(T) = 0$ for all T is RH together with the simplicity of all zeros.

Proposition 4.10 (the second line and prime races). *For a primitive character $\chi \bmod q$ and $G_\chi(s) = \Lambda(s, \chi)\Lambda(s+1, \chi)$, the prime coefficients of $-G'_\chi/G_\chi$ are $\Lambda(n)\chi(n)(1+n^{-1})$. For nonprincipal χ , $\sum_{p \leq x} \chi(p) \log p/p$ converges; hence, for every modulus q and all classes $a \not\equiv b \bmod q$ (and for races among r classes), the two-line races differ from the one-line races by convergent series, which vanish after Rubinstein–Sarnak’s normalisation, and the two-line races have the same limiting logarithmic distributions (under GRH for the characters $\bmod q$) and the same logarithmic densities $\delta(q; a, b)$ (under GRH and linear independence) as the one-line races [38, 26]; for example $\delta(4; 3, 1) \approx 0.9959$.*

Status. Proved here (claude-ab, 42_ Proposition 42.4, for $q = 4$; partial summation with the prime number theorem for progressions, Davenport [22], Ch. 20). Checks C1–C3 up to 10^7 . Refereed (nineteenth pass). The form for every modulus is version 2’s (48_ §3, F18): in $\sum_{p \leq x} (\mathbf{1}_{p \equiv a} - \mathbf{1}_{p \equiv b}) \log p/p = \varphi(q)^{-1} \sum_{\chi \neq \chi_0} (\bar{\chi}(a) - \bar{\chi}(b)) \sum_{p \leq x} \chi(p) \log p/p$ the principal character cancels, and each inner series converges; checked for $q \in \{3, 5, 7, 8, 12\}$ up to 10^7 .

4.6 The separator

Let I_- be the ideal of functions vanishing to full order on $Z - 1$.

Lemma 4.11 (the separator lemma with matched rates). *Let $G \in \mathcal{B}$ have zero divisor $D_+ \sqcup D_-$ with $D_+ \subset \{\text{Res} > 0\}$ and $D_- \subset \{\text{Res} < 0\}$. Suppose there are $\alpha \geq 0$, $\varepsilon > 0$ and $N, K \geq 0$ such that, with $r(t) = \varepsilon(1+t^2)^{-N}$, (a) $|G(x+it)| \leq C_A(1+|t|)^{K_A}e^{-\alpha|t|}$ on every strip $|x| \leq A$, and (b) G has no zeros in the corridor $\mathcal{K} = \{|x| \leq r(t)\}$ and $|G| \geq c(1+|t|)^{-K}e^{-\alpha|t|}$ there. Then there is an entire E of polynomial growth on every vertical strip with $E-1$ vanishing to full order on D_+ and E vanishing to full order on D_- ; consequently $\mathcal{B}/(I_{D_+} \cap I_{D_-}) \rightarrow \mathcal{B}/I_{D_+} \oplus \mathcal{B}/I_{D_-}$ is a topological isomorphism with inverse $([f], [g]) \mapsto [Ef + (1-E)g]$, commuting with multiplication by s and by every a^s .*

Status. Proved in the programme for ζ (GSL4–GSL5); stated with general constants and checked by claude-ab (24_ Lemma 24.1). Standard $\bar{\partial}$ technique. Refereed (eighth pass).

Outline. Put $\chi(x+it) = \eta(x/r(t))$ with a cutoff η switching from 0 to 1; then $|\bar{\partial}\chi| \leq C(1+|t|)^{2N}$, and $h = \bar{\partial}\chi/G$ is supported in $\mathcal{K}$ with $|h| \leq C(1+|t|)^{2N+K}e^{\alpha|t|}$ by (b). The solution $u = \pi^{-1} \int e^{(s-w)^2} (s-w)^{-1} h(w) dA(w)$ of $\bar{\partial}u = h$ satisfies $|u| \leq C_A e^{\alpha|t|} (1+|t|)^{2N+K}$, because the Gaussian dominates $e^{\alpha|t-v|}$ and the singularity $1/|s-w|$ is integrable over the thin corridor. So $E = \chi - Gu$ is entire, and by (a) the two exponentials cancel exactly: $|Gu|$ grows polynomially. Near $D_\pm$, $h = 0$ and $E - \chi = -Gu$ vanishes to full order. If the lower rate in (b) were $e^{-\alpha_2|t|}$ with $\alpha_2 > \alpha$, the method would give only $|Gu| \leq C(1+|t|)^K e^{(\alpha_2-\alpha)|t|}$. $\square$

For ζ , $G = F_*(s)F_*(s+1) = s^2(s^2-1)\zeta(1-s)\zeta(1+s)/(64\cos(\pi s/2))$ and $\alpha = \pi/2$: the only exponential factor, $1/\cos(\pi s/2)$, has the same size on every vertical line, which is why the rates match.

Proposition 4.12 (the separator for every primitive Dirichlet L -function). *Let χ be primitive $\bmod q \geq 3$ with $\chi(-1) = (-1)^a$, $\Lambda_\chi(s) = (q/\pi)^{(s+a)/2} \Gamma(\frac{s+a}{2}) L(s, \chi)$, Z_χ its zero divisor and $I_\chi, I_{\chi,-}$ the ideals of full-order vanishing on Z_χ and $Z_\chi - 1$. There is an entire E_χ of polynomial growth on strips that is 1 to full order on Z_χ and 0 to full order on $Z_\chi - 1$, so $\mathcal{B}/(I_\chi \cap I_{\chi,-}) \cong \mathcal{B}/I_\chi \oplus \mathcal{B}/I_{\chi,-}$ equivariantly.*

Status. Proved here (claude-ab, 24_ Proposition 24.4; register S44). Matched rates checked on $\text{Re } s = 0$ for two characters. Refereed (eighth pass). Novelty: not searched.

Proof. We verify Lemma 4.11 for $G_\chi(s) = \Lambda_\chi(s)\Lambda_\chi(s+1)$ with $\alpha = \pi/2$, using (F1) $\Lambda_\chi(s) = W(\chi)\Lambda_\chi(1-s)$ with $|W(\chi)| = 1$, (F2) $L(1+it, \chi) \neq 0$, and (F3) Stirling’s formula (Davenport [22], Chs. 9, 4, 14, 10). *Growth.* $S(x) = \sum_{n \leq x} \chi(n)$ is bounded and q -periodic; $L(s, \chi) = s \int_1^\infty S(x)x^{-s-1} dx$ on $\text{Re } s > 0$, and repeated integration by parts against periodic antiderivatives of mean zero continues L with $|L| + |L'| \leq C_{A,q}(1+|t|)^K$ on $|\text{Re } s| \leq A$; on $\frac{3}{4} \leq \sigma \leq 2$, $|L'(s, \chi)| \leq C_1(1+|t|)$.

By (F3), $\Lambda_\chi \in \mathcal{B}$. *Separation.* By the Euler product and (F2), $L \neq 0$ on $\text{Re } s \geq 1$; by (F1), $\Lambda_\chi \neq 0$ on $\text{Re } s \leq 0$. So $Z_\chi \subset \{0 < \text{Re } s < 1\}$ and $Z_\chi - 1 \subset \{-1 < \text{Re } s < 0\}$. *Upper rate.* By (F1), $G_\chi(s) = W(\chi)(q/\pi)^{1+a}\Gamma(\frac{1+a-s}{2})\Gamma(\frac{1+a+s}{2})L(1-s, \bar{\chi})L(1+s, \chi)$, and the Gamma product is $\pi/\cos(\pi s/2)$ for $a = 0$ and $(\pi s/2)/\sin(\pi s/2)$ for $a = 1$, of modulus at most $C_A(1+|t|)e^{-\pi|t|/2}$ on strips. *Lower rate.* For $\sigma > 1$, $\log(\zeta(\sigma)^3|L(\sigma+it, \chi)|^4|L(\sigma+2it, \chi^2)|^2) = \sum_p \sum_k k^{-1}p^{-k\sigma}[3 + 4\text{Re}(\chi(p)^k p^{-ikt}) + 2\text{Re}(\chi(p)^{2k} p^{-2ikt})] \geq 0$, each bracket being $(1+2\cos\theta)^2$ or 3. With $\zeta(1+\delta) \leq 2/\delta$ and $|L(1+\delta+2it, \chi^2)| \leq C(1+|t|)$ this gives $|L(1+\delta+it, \chi)| \geq c\delta^{3/4}(1+|t|)^{-1/2}$, and with the derivative bound, $|L(1+x+it, \chi)| \geq c(1+|t|)^{-5}$ for $|x| \leq c(1+|t|)^{-6}$; the same holds for $L(1-x-it, \bar{\chi})$, and the Gamma product is at least $ce^{-\pi|t|/2}$ there. For $|t| \leq 1$, $G_\chi(it) = \Lambda_\chi(it)\Lambda_\chi(1+it) \neq 0$ by (F1)–(F2), and compactness gives a zero-free neighbourhood. Lemma 4.11 applies with $r(t) = \varepsilon(1+t^2)^{-3}$. $\square$

The proof never uses where the zeros lie inside the open strip; the Euler product enters only through zero-freeness of $\text{Re } s \geq 1$. For the Davenport–Heilbronn function, which has a functional equation of the same kind but no Euler product, the hypotheses fail for every vertical line, since it has zeros with $\text{Re } s > 1$ (Saias–Weingartner [40]) and hence, by reflection, zeros with $\text{Re } s < 0$ (24_ Proposition 24.6); whether the splitting itself fails for it is open. The same argument gives the separator and the splitting for every Dedekind zeta function ζ_K , with $\xi_K = s(s-1)\Lambda_K$, $\Lambda_K = |d_K|^{s/2}\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}\zeta_K(s)$ and $\alpha = n\pi/2$, $n = [K : \mathbb{Q}]$: since $\Gamma_{\mathbb{R}}(1-s)\Gamma_{\mathbb{R}}(1+s) = 1/\cos(\pi s/2)$ and $\Gamma_{\mathbb{C}}(1-s)\Gamma_{\mathbb{C}}(1+s) = s/(\pi \sin \pi s)$, the Gamma part of $\xi_K(1-s)\xi_K(1+s)$ has the rate $e^{-(r_1+2r_2)\pi|t|/2}$ on every vertical line. The inputs are Hecke’s functional equation, polynomial growth of ζ_K on vertical strips (the convexity bound; the periodic partial summation used above for $L(s, \chi)$ does not apply), the 3–4–1 inequality for the Euler product over prime ideals, and Stirling’s formula. (Version 2, 48_ §3, F19; checked for four fields of degrees 2, 3 and 4; the standard inputs were not re-read.) Two structural facts complete the picture (GMS; 24_ §§2.3–2.4): the derived separation $\text{RHom}(Q, Q_+) \simeq 0 \simeq \text{RHom}(Q_+, Q)$ of Q and its normal twist $Q_+ = \mathcal{B}/I_+$ (jets at $Z+1$) is the general vanishing of Ext for a central element acting as 0 on one module and as 1 on the other; and $\mathcal{B}$ itself does not split along the two lines, since Q_+ is torsion and $\mathcal{B}$ torsion-free, and an entire c with $c^2 = c$ is constant.

4.7 The defect ratio

For $r > 1$ the programme’s defect at scale r has eigenvalue $d_r(\rho) = r^{\bar{\rho}} - r^{1-\rho} = \overline{r^\rho - r^{\rho^\#}}$ at a zero ρ : a zero minus its reflected partner.

Lemma 4.13 (the exact range of the defect ratio). *Let $r \neq t > 1$, $0 < \beta < 1$, $\beta \neq \frac{1}{2}$, $x = \beta - \frac{1}{2}$. Then $|d_r(\rho)/d_t(\rho)| = \sqrt{r/t} \sinh(x \log r) / \sinh(x \log t)$; it is strictly increasing in $|x|$ on $(0, \frac{1}{2})$ if $r > t$ and strictly decreasing if $r < t$, and its values fill exactly the open interval between $\sqrt{r/t} \log r / \log t$ (the limit $x \rightarrow 0$) and $(r-1)/(t-1)$ (the value at $|x| = \frac{1}{2}$).*

Status. Proved here (claude-ab, 24_ Lemma 24.5; it sharpens the programme’s bound AST6.2). Checked on a grid. Refereed (eighth pass). The direction of monotonicity is 24_ Lemma 24.5(b), stated in version 2 (48_ §3, F20). A statement about admissible positions, not about zeros.

Proof. $r^\beta - r^{1-\beta} = \sqrt{r}(r^x - r^{-x}) = 2\sqrt{r} \sinh(x \log r)$. With $a = \log r$ and $b = \log t$, the derivative of $\log(\sinh(ax)/\sinh(bx))$ for $x > 0$ is $a \coth(ax) - b \coth(bx)$, and $c \mapsto c \coth(cx)$ is strictly increasing for $x > 0$ (its derivative is $(\frac{1}{2} \sinh 2cx - cx) / \sinh^2 cx > 0$); so the derivative has the sign of $a - b$, that is, of $r - t$, and the ratio is even in x . As $x \rightarrow 0$ the ratio tends to $\sqrt{r/t} a/b$; at $|x| = \frac{1}{2}$, $\sinh(\frac{1}{2} \log r) = (r-1)/(2\sqrt{r})$. $\square$

Two further exact statements belong here (register S38, S39; 22_ Propositions 22.1–22.3, 22.5): the Haar and Lebesgue normalisations of the author’s backward-running clock, $g \mapsto \overline{g(1/u)}$ and $g \mapsto \overline{g(1/u)}/u$, have Mellin shadows $s \mapsto -\bar{s}$ and $s \mapsto 1 - \bar{s}$ and differ by the factor u ; and, paired with their one-line shifts, $\operatorname{Re} s = \frac{1}{2}$ and $\operatorname{Re} s = -\frac{1}{2}$ are the two sheets of a connected double cover of the equator of $\mathbb{CP}^1$ whose deck transformation is the central -1 of $SU(2)$, the author’s Hopf fibre and the shift circle being fibres of the left and the right Hopf fibrations.

5 Weights: Deligne’s mechanism against the programme

A large part of the programme, including a full reader of Deligne’s *La conjecture de Weil. II* [25], asks which steps of Deligne’s proof have analogues for ζ . This section states what the audit established: exactly what positivity of tensor powers gives on the programme’s data, what is missing and in which precise sense, an algebraic purity criterion at the core of Weil II 1.8.4, three elementary lemmas from Weil II §2 with their corrected forms, and the misprints found in the printed original. In this section the weight of a nonzero complex number α relative to a prime p is $w(\alpha) = 2 \log |\alpha| / \log p$; a zero ρ enters the programme’s local data as the eigenvalue p^ρ , of weight $2 \operatorname{Re} \rho$.

5.1 Positivity of even tensor powers

For a finite set S of zeros, stable under $\rho \mapsto \bar{\rho}$, let J_S be the span of the jet blocks at the zeros in S , on which the dilation T_n has the eigenvalues n^ρ with multiplicities m_ρ . The programme’s tensor-power trace is $\operatorname{Tr}(T_n^{\otimes k} | J_S^{\otimes k}) = (\sum_{\rho \in S} m_\rho n^\rho)^k$ (TWC7). It is real, so its even powers are nonnegative; this is Deligne’s step (Weil II 1.5; DP4 of the programme’s Deligne reader).

Proposition 5.1 (the exponent data return the largest real part). *Let S be a finite set of complex numbers with positive integer weights m_ρ , put $\beta_{\max} = \max_{\rho \in S} \operatorname{Re} \rho$, let $k \geq 1$ and $D_S^{(k)}(s) = \sum_{n \geq 1} (\sum_{\rho \in S} m_\rho n^\rho)^k n^{-s}$. Then $D_S^{(k)}(s) = \sum_w c_w \zeta(s - w)$, where w runs over the distinct sums of k elements of S and every $c_w > 0$, and its abscissa of convergence (and of absolute convergence) is exactly $1 + k\beta_{\max}$. If k is even and S is stable under $\rho \mapsto \bar{\rho}$ with $m_{\bar{\rho}} = m_\rho$, the coefficients are nonnegative and the real point $1 + k\beta_{\max}$ is a pole with positive residue. Consequently a bound $1 + k/2 + C$ on the abscissa, with C independent of k (for $D_S^{(2k)}: k + 1 + C$), holds if and only if $\beta_{\max} \leq \frac{1}{2}$.*

Status. Proved here (claude-ab, 08_ §3 item (e), revision 3, for even degree and conjugation-stable S ; that section was checked in the final referee pass, part B). Register negative result 27. The form for every degree k and every finite S , with no positivity, is the review’s (version 2; 48_ §2.5, re-derived by claude-ab; check Z5).

Proof. Expanding the power, $D_S^{(k)}(s) = \sum_{(\rho_1, \dots, \rho_k) \in S^k} (\prod_j m_{\rho_j}) \zeta(s - \sum_j \rho_j)$; grouping the tuples by their sum w gives $c_w > 0$. Each term converges absolutely for $\operatorname{Re} s > 1 + \operatorname{Re} w$, so the series converges absolutely for $\operatorname{Re} s > 1 + k\beta_{\max}$. If it converged at a point with real part $\sigma_1 < 1 + k\beta_{\max}$, it would define a holomorphic function on $\operatorname{Re} s > \sigma_1$ equal to $\sum_w c_w \zeta(s - w)$; but this function has a simple pole with residue $c_w > 0$ at each $1 + w$, the w being distinct so that no two poles cancel, among them $1 + k\rho$ with $\operatorname{Re} \rho = \beta_{\max}$. So the abscissa is exactly $1 + k\beta_{\max}$. For even k and conjugation-stable S , $\sum_\rho m_\rho n^\rho$ is real, so the coefficients are nonnegative, and $k\beta_{\max}$ is the sum of $k/2$ copies each of ρ and $\bar{\rho}$ with $\operatorname{Re} \rho = \beta_{\max}$, so the real point is a pole with residue $c_{k\beta_{\max}} > 0$. Finally $1 + k\beta_{\max} \leq 1 + k/2 + C$ for all k if and only if $\beta_{\max} \leq \frac{1}{2}$. $\square$

So the exponent data return $\max \operatorname{Re} \rho$ in every tensor degree, whether or not the coefficients are positive, and a bound on it has to come from elsewhere. What positivity of even tensor powers adds is a pole at the real point (Landau’s theorem). For odd k the real point need not be a pole:

for $S = \{0.7 \pm 10i, 0.3 \pm 10i\}$ and $k = 3$ the rightmost poles are $3.1 \pm 10i$ and $3.1 \pm 30i$, and 3.1 is not a pole, while the abscissa is 3.1.

Deligne's next step (Weil II 1.5; DP5 of the programme's Deligne reader) is exactly an independent bound of the form $k + 1 + C$; in Weil II it comes from the H_c^2 of a fixed curve. The programme's own computations of the tensor Euler products (register S57; Section 2) have the same structure: for conjugation-stable data the real pole of the even tensor Euler function sits at $1 + 2kB$, whatever B is.

Proposition 5.2 (the average weight of an off-line quartet). *Let ρ be a zero of multiplicity m and V_ρ the span of the jet blocks of the distinct points among $\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}$. If ρ is off the line, the four points are distinct, W_p has the eigenvalues $p^\rho, p^{\bar{\rho}}, p^{1-\rho}, p^{1-\bar{\rho}}$ on V_ρ , each with multiplicity m , V_ρ has rank $4m$ and $\det(W_p | V_\rho) = p^{2m}$. If ρ is on the line, then $1 - \rho = \bar{\rho}$, V_ρ is the span of the blocks of ρ and $\bar{\rho}$, of rank $2m$, and $\det(W_p | V_\rho) = p^m$. In both cases the weight of the determinant divided by the rank is 1, whatever $\operatorname{Re} \rho$ is. Only the constituent weights $2 \operatorname{Re} \rho$ and $2(1 - \operatorname{Re} \rho)$ see an off-line zero.*

Status. Proved in the programme (DW8.12); audited (35_35.N1, wording corrected in the fourteenth pass). Version 1 stated the rank $4m$ and the determinant p^{2m} for every zero; at an on-line zero the four points are two, and version 2 corrects this (48_§1, O1; check Z6). The conclusion is unchanged.

Proof. Off the line, $\det = (p^{\rho+\bar{\rho}+(1-\rho)+(1-\bar{\rho})})^m = p^{2m}$ and $w(p^{2m}) = 4m$. On the line, $\rho + \bar{\rho} = 1$, so $\det = (p^{\rho+\bar{\rho}})^m = p^m$ and $w(p^m) = 2m$. $\square$

Deligne's Théorème 1.5.1 turns a pure *determinant weight* of each constituent (Définition 1.3.5) into purity. For the quartet on the abelian group $\mathbb{R}$ the four characters are separate constituents, with determinant weights $2 \operatorname{Re} \rho$ (twice) and $2(1 - \operatorname{Re} \rho)$ (twice); the reality hypothesis holds, and the pole bound holds with $r = 2 \max(\operatorname{Re} \rho, 1 - \operatorname{Re} \rho)$ (Proposition 5.1), so 1.5.1 applies and returns the constituent weights $2 \operatorname{Re} \rho$ and $2(1 - \operatorname{Re} \rho)$ themselves, with no constraint on $\operatorname{Re} \rho$ (37_§6). If the four eigenvalues formed one ι -real constituent of rank 4, its determinant weight would be the average 1 and 1.5.1 would force $\operatorname{Re} \rho = \frac{1}{2}$. What is missing is therefore an input forcing the four constituents onto the average weight 1, equivalently a pole bound at $k + 1 + C$ uniform in k .

Two routes, and what each needs. Over the programme's reconstructed arithmetic one would need either (a) a square with a Künneth-type map $H^1 \otimes H^1 \rightarrow H^2$ together with an auxiliary object whose weights are discrete, or (b) a cohomological source for a pole bound on positive even tensor powers that is uniform in the power. Route (b) needs no square as a matter of logic. On the programme's finite exponent data, however, a bound of type (b) is equivalent to $\max \operatorname{Re} \rho \leq \frac{1}{2}$ (Proposition 5.1), so it has to come from a source independent of these data; in Weil II that source, for eigenvalues on H^1 of a curve, is the H_c^2 of a fixed curve, reached through the square and its Lefschetz pencil (Weil II §3.2; 08_§3, 37_§6). The programme's own constructions of a square and of all tensor powers (TWC, FST) have computed outcomes, recorded in Section 7; TWC9 and TWC11 state that no construction has yet been assigned the fixed-curve cohomological denominator and that the computation does not rule one out.

5.2 An algebraic purity criterion

Let V be a finite-dimensional complex vector space, N nilpotent and F invertible with $FN F^{-1} = Q^{-1}N$, where $Q \in \mathbb{C}$, $|Q| \neq 1$, and put $w(\alpha) = 2 \log |\alpha| / \log |Q|$. Let M be the monodromy filtration of N , and on $V^\vee$ put $F^\vee = (F^{-1})^t$, $N^\vee = -N^t$, which satisfy the same relation.

Proposition 5.3 (the core of Weil II 1.8.4). *The following are equivalent: (a) for every i , all eigenvalues of F on $\operatorname{Gr}_i^M V$ have weight $\beta + i$; (b) all eigenvalues of $F \otimes F$ on $\ker(N \otimes 1 + 1 \otimes N) \subset V \otimes V$ have weight $\leq 2\beta$, and all eigenvalues of $F^\vee \otimes F^\vee$ on $\ker(N^\vee \otimes 1 + 1 \otimes N^\vee)$ have weight $\leq -2\beta$. The bounds on $\ker N$ for V and $V^\vee$ alone do not suffice.*

Status. Proved here (claude-ab, 36_ Lemma 36.2 and 36.N2; the implication (b) $\Rightarrow$ (a) is the algebra of Deligne's proof on p. 176, the converse is added; the equivalence is not claimed to be new). First checked on 40 random models, and refereed in the thirteenth pass on 102 further models.

Proof. The inputs are Weil II (1.6.6), (1.6.9) and (1.6.14) (pp. 166–170), with the twist of (1.6.14.3) read as $-(i+j)/2$ (Section 5.4). (a) $\Rightarrow$ (b): by (1.6.9)(i) the monodromy filtration of $N \otimes 1 + 1 \otimes N$ is $\sum_{a+b=k} M_a \otimes M_b$, so its Gr_k has weight $2\beta + k$; N is injective on Gr_k for $k \geq 1$, so the kernel lies in M_0 , of weights $\leq 2\beta$. By (1.6.9)(ii), $\text{Gr}_i(V^\vee) \cong (\text{Gr}_{-i} V)^\vee$ has weight $-\beta + i$, and the same argument applies. (b) $\Rightarrow$ (a): let α be an eigenvalue on the primitive part $P_{-j} = \ker(N : \text{Gr}_{-j} \rightarrow \text{Gr}_{-j-2})$, $j \geq 0$, with eigenvector $x = N^j y$, $y \in \text{Gr}_j$; then $Fy = \alpha Q^j y$. The vector $z = \sum_{s=0}^j (-1)^s N^s y \otimes N^{j-s} y \in \text{Gr}_0(V \otimes V)$ is nonzero, is killed by $N \otimes 1 + 1 \otimes N$ (the sum telescopes to $(-1)^j N^{j+1} y \otimes y + y \otimes N^{j+1} y = 0$), and has eigenvalue $\alpha^2 Q^j$; by (1.6.6) it gives an eigenvalue on $\ker(N \otimes 1 + 1 \otimes N)$, so $2w(\alpha) + 2j \leq 2\beta$. On $V^\vee$, $P_{-j}(V^\vee) \cong P_{-j}(V)^\vee(j)$ carries $\alpha^{-1} Q^{-j}$, which gives $w(\alpha) \geq \beta - j$. So every eigenvalue on P_{-j} has weight $\beta - j$, and $\text{Gr}_i \cong \bigoplus_j P_{-j}(-(i+j)/2)$ has weight $\beta + i$. For the last assertion take $V = \langle v \rangle \oplus \langle v_1, v_0 \rangle$ with $Nv_1 = v_0$, $Nv_0 = Nv = 0$, $Fv = av$, $Fv_1 = bv_1$, $Fv_0 = (b/Q)v_0$, $w(a) = \beta$, $w(b) = c + 1$. The bounds on $\ker N = \langle v, v_0 \rangle$ and $\ker N^\vee$ hold exactly when $\beta - 1 \leq c \leq \beta + 1$, while (a) holds exactly when $c = \beta$; the vector $v_1 \otimes v_0 - v_0 \otimes v_1$ in the kernel of the tensor square has weight $2c$ and detects the difference. $\square$

In Weil II condition (b) is supplied by geometry (Lemme 1.8.1 for $F_0 \otimes F_0$ and its dual). For the programme's operators, as they are defined, the criterion does not apply: an operator commuting with N (the programme's counting operators W_p and C_b) has the same eigenvalues on Gr_i and Gr_{-i} , since $N^i : \text{Gr}_i \rightarrow \text{Gr}_{-i}$ is then equivariant, so it cannot carry Deligne's local weights unless $N = 0$; and the programme's ramification operator P_b , which does satisfy a relation of Deligne's form ($NP_b = bP_bN$), has no eigenvector on its space, because it moves every block λ to $b\lambda$ (36_ 36.N3–36.N4, 35_ §3).

5.3 Three lemmas from Weil II §2

Lemma 5.4 (concentration, with the outside term). *Let K be a compact group with normalised Haar measure, τ an irreducible character of degree d_τ , $V \subset K$ measurable with $\text{Re } \chi_\tau \geq (1 - e_1)d_\tau$ on V ($0 \leq e_1 \leq 1$), and $\rho_0 \neq 0$ an integer combination of irreducible characters with $\int_{K \setminus V} |\rho_0|^2 \leq e_2 \int |\rho_0|^2$. Write $\rho_0 = \rho^+ - \rho^-$ with $\rho^\pm$ genuine and without common constituents, and $\rho = \rho^+ + \rho^-$. Then*

$$[\rho\bar{\rho} : \tau] \geq ((1 - e_1)(1 - e_2) - e_2) d_\tau [\rho\bar{\rho} : 1], \quad [\rho\bar{\rho} : 1] = \int |\rho_0|^2.$$

The constant is sharp in e_2 . The printed choice $\varepsilon_1 + \varepsilon_2 < \varepsilon$ in the proof of Weil II 2.1.7 does not give Deligne's inequality; $\varepsilon_1 + 2\varepsilon_2 < \varepsilon$ does.

Status. Proved here (claude-ab, 37_ Lemma 37.1; the repair is the programme's Deligne reader's, DB0, DB5). Checks B1–B4 on $\mathbb{Z}/2$, $U(1)$, $SU(2)$. Refereed (fourteenth pass).

Proof. $\rho\bar{\rho} = \rho_0\bar{\rho}_0 + 2(\rho^+\bar{\rho}^- + \rho^-\bar{\rho}^+)$, the second term genuine with $[\rho^+\bar{\rho}^- : 1] = \langle \rho^+, \rho^- \rangle = 0$. The integer $[\rho_0\bar{\rho}_0 : \tau] = \int |\rho_0|^2 \bar{\chi}_\tau$ is real, so it equals $\int |\rho_0|^2 \text{Re } \chi_\tau$; the part over V is at least $(1 - e_1)(1 - e_2)d_\tau \int |\rho_0|^2$ and the part off V at least $-e_2 d_\tau \int |\rho_0|^2$. Sharpness and the counterexample: on $K = \mathbb{Z}/2$, $\tau = \text{sgn}$, $V = \{e\}$, $\rho_0 = 23 + 9 \text{sgn}$ gives $\int |\rho_0|^2 = (32^2 + 14^2)/2 = 610$ and off- V fraction $98/610 < 0.18$; with $\varepsilon_1 = 0.01$, $\varepsilon_2 = 0.18$, $\varepsilon = 0.2$ one has $\varepsilon_1 + \varepsilon_2 < \varepsilon$, but $[\rho\bar{\rho} : \text{sgn}] = 2 \cdot 23 \cdot 9 = 414 < 0.8 \cdot 610 = 488$. With $\varepsilon_1 + 2\varepsilon_2 < \varepsilon$, $(1 - \varepsilon_1)(1 - \varepsilon_2) - \varepsilon_2 \geq 1 - \varepsilon_1 - 2\varepsilon_2 > 1 - \varepsilon$. $\square$

Lemma 5.5 (integer positivity; Weil II 2.1.5–2.1.6). *Let K be compact and $\nu : \widehat{K} \rightarrow \mathbb{R}$, extended additively, with $\nu(1) = 1$, $\nu(\tau) \leq 0$ for $\tau \neq 1$ and $\nu(\rho\bar{\rho}) \geq 0$ for every genuine ρ . Then $\sum_{\tau \neq 1} d_\tau (-\nu(\tau)) \leq 1$. If moreover ν is integer-valued and $\nu(\bar{\tau}) = \nu(\tau)$, at most one $\tau \neq 1$ has $\nu(\tau) < 0$, and it is a quadratic*

character with $\nu(\tau) = -1$. The exception cannot be excluded by positivity alone: on $\mathbb{Z}/2$, $\nu(\text{sgn}) = -1$ satisfies every hypothesis. For real-valued ν the bound is sharp and the dichotomy fails: on $\mathbb{Z}/3$ with $\nu(\tau) = \nu(\tau^2) = -x$, positivity holds if and only if $x \leq \frac{1}{2}$.

Status. Proved here (claude-ab, 37_ Lemma 37.2, for integer-valued ν). Refereed (fourteenth pass). The real-valued inequality and the $\mathbb{Z}/3$ example are version 2's (48_ §3, F21).

Proof. For a finite $T \subset \widehat{K} \setminus \{1\}$ and $e > 0$, choose $e_1 + 2e_2 \leq e$ and a neighbourhood V of the identity on which $\text{Re } \chi_\tau \geq (1 - e_1)d_\tau$ for $\tau \in T$; by Peter–Weyl there is an integer combination $\rho_0 \neq 0$ with off- V fraction at most e_2 . Lemma 5.4 gives $[\rho\bar{\rho} : \tau] \geq (1 - e)d_\tau M$ with $M = [\rho\bar{\rho} : 1] > 0$, so $0 \leq \nu(\rho\bar{\rho}) \leq M + \sum_{\tau \in T} (1 - e)d_\tau M \nu(\tau)$. Divide by M and let $e \rightarrow 0$; only $\nu(1) = 1$ and $\nu(\tau) \leq 0$ for $\tau \neq 1$ were used. For integer-valued ν , a τ with $\nu(\tau) < 0$ contributes $d_\tau |\nu(\tau)| \geq 1$, so $d_\tau = 1$ and $\nu(\tau) = -1$, and $\bar{\tau} = \tau$ because $\bar{\tau}$ would contribute as much. On $\mathbb{Z}/2$, $\nu((a + b \text{sgn})(a + b \text{sgn})) = (a - b)^2 \geq 0$. On $\mathbb{Z}/3$, $\rho = a + b\tau + c\tau^2$ gives $\rho\bar{\rho} = a^2 + b^2 + c^2 + (ab + bc + ca)(\tau + \tau^2)$, so $\nu(\rho\bar{\rho}) = a^2 + b^2 + c^2 - 2x(ab + bc + ca)$, which is ≥ 0 for all $a, b, c \geq 0$ if and only if $x \leq \frac{1}{2}$ (take $a = b = c = 1$, and use $a^2 + b^2 + c^2 \geq ab + bc + ca$). $\square$

In Weil II §2.2 the quadratic exception is removed with a double cover; for ζ (the case $G = \mathbb{R}$, $K = \{0\}$, Deligne's example 2.1.9) there is no nontrivial quadratic character, and the method gives exactly $\zeta(1 + it) \neq 0$.

Lemma 5.6 (nonnegative power sums). *Let $a_1, \dots, a_r \in \mathbb{C}^\times$, $D > 0$ and real δ_n with $\limsup_n \max(\delta_n, 0)^{1/n} \leq D$ (for example $\delta_n \leq CD^n$), such that $\delta_n - \sum_j a_j^n$ is real and ≥ 0 for every $n \geq 1$. Then $\max_j |a_j| \leq D$. In particular a constant $\delta_n = \delta$ gives $\max_j |a_j| \leq 1$, and for a curve over $\mathbb{F}_q$ the inequalities $\#X(\mathbb{F}_{q^n}) \geq 0$ alone give $|a_j| \leq q$ (the trivial bound).*

Status. Proved here (claude-ab, 37_ Lemma 37.3, for constant δ ; the core of DBC7 of the programme's Deligne reader). Refereed (fourteenth pass). The form with growing δ_n is version 2's (48_ §3, F21; the $\limsup$ form from the version-2 referee). No novelty claimed.

Proof. Suppose $R = \max |a_j| > D$ and let J be the set of j with $|a_j| = R$. By Dirichlet's simultaneous approximation, infinitely many n have $|(a_j/R)^n - 1| < \frac{1}{2}$ for all $j \in J$ (either the approximating $n \leq Q^{|J|}$ are unbounded as Q grows, or some n has all $(a_j/R)^n = 1$ and its multiples serve). For those n , $\text{Re } \sum_{j \in J} a_j^n \geq |J|R^n/2$, while the other terms are $O(R'^n)$ with $R' < R$, so, since $\delta_n \leq (D + \varepsilon)^n$ for large n with $D + \varepsilon < R$, $\delta_n - \text{Re } \sum_j a_j^n \leq (D + \varepsilon)^n - |J|R^n/2 + O(R'^n) \rightarrow -\infty$. For a curve, $\#X(\mathbb{F}_{q^n}) = 1 + q^n - \sum_j \alpha_j^n$ and $1 + q^n \leq 2q^n$. $\square$

Corollary 5.7. *The zeta function of a smooth projective geometrically connected curve X_0 over $\mathbb{F}_q$ has a simple pole at $s = 1$, by the trace formula and the point counts alone, without the Riemann hypothesis for curves.*

Status. Proved in the programme's Deligne reader (DBC7); re-derived here (claude-ab, 37_ §4). No novelty claimed.

Proof. If an eigenvalue of Frobenius on H^1 equalled q , then $\#X_0(\mathbb{F}_{q^n}) = 1 - \sum_{j \geq 2} \alpha_j^n \geq 0$ for all n (the eigenvalue q cancels the q^n from H^2), so by the lemma the counts would be bounded; but X_0 has infinitely many closed points, and for n a common multiple of the degrees of $C + 1$ of them the count is at least $C + 1$. So $P_1(q^{-1}) \neq 0$ and $Z(X_0, t) = P_1(t)/((1 - t)(1 - qt))$ has a simple pole at $t = q^{-1}$. $\square$

5.4 Misprints in the printed Weil II

The programme's Deligne reader flagged a number of displays as defects of its transcription. The audit read the printed pages (the numdam scan) and found that the following are in print. None affects a conclusion of Weil II.

Display	Page	Printed	Meant
(1.3.10)(iv)	160	the centre maps onto a finite subgroup of $\mathbb{Z}$	a subgroup of finite index
(1.6.7)	166	exception $i \neq d$	$i \neq -d$
(1.6.13)	169	$k - 2i - 2 \geq k \geq 2j - k$	$k - 2i - 2 \geq -k \geq 2j - k$
(1.6.14.3)	170	$P_{-j}((i+j)/2)$	$P_{-j}(-(i+j)/2)$
(1.7.5), filtration	171	$M'_i = \prod_{j < i} V'_j$	$j \leq i$
(1.7.5), coefficient	171	$\lambda \in \mathbb{Q}_\ell(-1), \mu = \lambda/(1 - q^n)$	with (1.1.7), (1.7.2): $\mu = \lambda/(1 - q^{-n})$
(2.1.1)	187	period $2\pi i \log q$	$2\pi i / \log q$
proof of 2.1.7	191	$\varepsilon_1 + \varepsilon_2 < \varepsilon$	$\varepsilon_1 + 2\varepsilon_2 < \varepsilon$ (a slip)
2.2.8(i)	195	weight $2\mathcal{R}(\tau)$	$-2\mathcal{R}(\tau)$
(3.1.3.3), unused remark	199	Kummer class of uv^{-1} a generator	it is twice a branch class
proof of 3.2.1	200	$F^\vee(-1)$	$F^\vee(1)$
(3.2.7.1)	202	H^1	H_c^1
proof of 3.2.12	203	$w \leq \beta + i$ inside a weight-0 proof	$w \leq i$

Status. Checked against the printed pages (claude-ab and referees, 35_ §§1, 5; 36_ §2; 37_ §2; twelfth to fourteenth passes). The programme's Deligne reader and its reading log found them first. A quick errata search, for the displays of §1, found none published; no novelty is claimed. The audits found a few further harmless printed slips (36_ §2, 37_ §3).

6 Further results

6.1 The two poles, and the point where the nonzero numbers stop

For an even Schwartz function h put $M_S h(s) = \int_0^\infty h(v) v^{s-1} dv$ and $\Sigma h(u) = \sum_{n \neq 0} h(nu)$, so that $\int_0^\infty \Sigma h(u) u^{s-1} du = 2\zeta(s) M_S h(s)$ on $\text{Re } s > 1$ (Müntz).

Proposition 6.1 (the two poles differ in kind). (a) $2\zeta(s) M_S h(s) = E_h(s) + \hat{h}(0)/(s-1) - h(0)/s$ with E_h entire: the pole at 0 is the lattice's $n = 0$ term and the pole at 1 the dual lattice's. For the lattice $c\mathbb{Z}$ the residue at 0 is $-h(0)$ for every c , while the residue at 1 is $(\int h)/c$: the pole at 0 counts one point and the pole at 1 measures a density.

(b) If $h(v) = \kappa v^\alpha + O(v^{\alpha+1})$ as $v \downarrow 0$, then $M_S h$ has a pole at $-\alpha$ with residue κ , and $2\zeta M_S h$ has residue $2\zeta(-\alpha)\kappa$ there. For the smooth even model the poles of $M_S h$ at $0, -2, -4, \dots$ have residues $h^{(2k)}(0)/(2k)!$, and the trivial zeros cancel all of them except the one at 0.

(c) The moment conditions $h(0) = 0$ and $\int h = 0$ remove exactly the two poles; the pole at 1 carries the average (its residue is the density of the integers, and through $-\zeta'/\zeta$ the main term x of $\psi(x)$), and every zero contributes only an oscillation. For F in the exact summation image, the jet of the source at 0 is read from the trivial zeros: $f_F^{(2r)}(0)/(2r)! = F(-2r)/(2\zeta'(-2r))$ (Proposition 3.5).

Status. Proved here (claude-ab, 13_ §§1–2, 12_ §3; register S30 and F_1 items 10, 11, 15). The split identity is checked to 10^{-31} at four points, with both residues and the cancellation at -2 . Classical in content (Riemann; Tate); the ledger of the roles of the different zeros in play (the integer 0, the pole at 0, the zero element) is set out in 13_.

Proof. Split $\int_0^\infty$ at $u = 1$ and use Poisson summation $\Sigma h(u) = u^{-1}\Sigma \hat{h}(u^{-1}) + u^{-1}\hat{h}(0) - h(0)$ on $(0, 1)$: the terms $u^{-1}\hat{h}(0)$ and $-h(0)$ integrate to $\hat{h}(0)/(s-1) - h(0)/s$, and the rest is entire. For $c\mathbb{Z}$ the $n = 0$ term is still $h(0)$, while $\widehat{h(c\cdot)}(0) = \hat{h}(0)/c$. (b) $\int_0^1 \kappa v^{\alpha+s-1} dv = \kappa/(s+\alpha)$. (c) $\zeta(s) = s \int_1^\infty [u]u^{-s-1} du$ has residue 1 at $s = 1$. $\square$

In the programme the object at that point is τ . A pole at τ , in the author's sense, would be computed from a local model of the test functions at τ by (b), and the programme has not yet specified that model (Question 9). The reading of the residue at 0 as presence without an exchanged label ($0 = -0$ is the only non-free orbit of ± 1) is a proposed typed map, not an identification with τ , since the integer 0 has even parity (13_ §4). The author's own placement is that τ is the pivot of the swing between 0 and 1; on the zeta side that is the reflection $s \mapsto 1 - \bar{s}$ about $\frac{1}{2}$ (or $u \mapsto 1/u$ about $u = 1$), and there RH reads $\rho^\# = \rho$ for every zero. Centring at 0 and centring at the pivot differ by exactly one shift, $s^\# = T_1(A(s))$, and only the $\frac{1}{2}$ -centred pairing is real (Proposition 4.4).

6.2 Clocks and Chebyshev's function

Proposition 6.2 (the clock stack is ψ ; the completed prime clock is $\widehat{\mathbb{Z}}$). *The joint state set of the clocks $\mathbb{Z}/1, \dots, \mathbb{Z}/N$ is $\mathbb{Z}/L_N$ with $L_N = \text{lcm}(1, \dots, N)$, and $\log L_N = \psi(N)$; so $-\zeta'/\zeta(s) = \sum_{N \geq 2} (\log L_N - \log L_{N-1})N^{-s}$ on $\text{Re } s > 1$. For every integer $a \geq 2$ (for example a prime p), the closure of $a^{\mathbb{Z}}$ in $\prod_{\ell|a} \mathbb{Z}_\ell^\times$ is a copy of $\widehat{\mathbb{Z}}$.*

Status. Proved in the programme (PMS M3–M6), for a prime; checked exactly for $N \leq 20000$ and $d \leq 30$ (15_ register S32). Classical in content; the clock-stack reading is the programme's. Refereed (fourth pass). The form for every integer $a \geq 2$ is version 2's (48_ §3, F22; the proof is unchanged).

Proof. By the Chinese remainder theorem the image of $\mathbb{Z} \rightarrow \prod_{k \leq N} \mathbb{Z}/k$ is $\mathbb{Z}/L_N$. $v_p(L_N) = \max\{a : p^a \leq N\}$, so $L_N/L_{N-1} = p$ if $N = p^a$ and 1 otherwise, and $\log L_N = \sum_{p^a \leq N} \log p = \psi(N)$. The number $a^d - 1$ is prime to a , and since $0 < a^k - 1 < a^d - 1$ for $1 \leq k < d$, a has exact order d modulo $a^d - 1$; so $n \mapsto a^n$ induces injections $\mathbb{Z}/d \hookrightarrow (\mathbb{Z}/(a^d - 1))^\times$ for every d . The continuous extension $\widehat{\mathbb{Z}} \rightarrow \prod_{\ell|a} \mathbb{Z}_\ell^\times$ is therefore injective, and its image is compact and contains $a^{\mathbb{Z}}$ as a dense subset, so it is an isomorphism of $\widehat{\mathbb{Z}}$ onto the closure. $\square$

Classically (von Koch 1901), RH is equivalent to $\log L_N = N + O(\sqrt{N} \log^2 N)$, so through the clock stack RH becomes a statement about the size of the joint state set of the first N clocks. In the programme's generic monoid $A_\eta = \{0\} \cup \{J^r T^m\}$ the fibre over winding $m = 0$ is $\{\tau, J, \varepsilon, J^3\}$, which is present, while absence is the absorbing 0 (PMS M1): the fibre that has not moved is present, not absent.

6.3 Connes–Consani's signed winding group

On the archimedean local data of Connes–Consani's absolute twistor line [20], the unit group is $G = \langle T, J \mid [T, J] = 1, J^4 = 1 \rangle \cong \mathbb{Z} \times \mathbb{Z}/4$ with sign $\varepsilon = J^2$ and mirror $A(T^m J^k) = T^{-m} J^{2m-k}$. The programme proves (and the audit checked; register S8, S9):

- *parity is derived from the mirror:* $g \mapsto A(g)g$ equals ε^m on $g = \varepsilon^a T^m J^b$, a homomorphism onto $\{1, \varepsilon\}$ with kernel $2L$ on the winding group $L = G/\langle J \rangle$; no group section of $G \rightarrow L$ is equivariant (W2);
- *the sign-preserving, mirror-commuting endomorphisms* of G are exactly $T \mapsto T^n J^c, J \mapsto J^b$ with n odd, $c \in \mathbb{Z}/4, b \in \{1, 3\}$; for even n the two sides differ by exactly ε , so doubling has no sign-preserving mirror-compatible lift although (2) is a prime of $\text{Spec End}(L) \cong \text{Spec } \mathbb{Z}$ (a lift that kills the sign, $T \mapsto T^2, J \mapsto 1$, does commute with the mirror) (W5; exhaustive check for $|n| \leq 9$).

The copy identified the weight-one sign sector over the $\mathbb{F}_{12}$ -points: the double cover of $\mathbb{P}^1$ branched at $\{0, \infty, \pm 1\}$ is $y^2 = x^3 - x$, J acts as complex multiplication by i , and the Hecke character on rational ideals is $\psi((n)) = \chi_{-4}(n)n$, which equals Connes–Consani’s signed winding degree; and $L(\text{Sym}^2 E, s) = L(\psi^2, s)L(s-1, \chi_{-4})$ (register S10–S11; classical ingredients). Its L -function has conductor 32, so this sector realises $L(\psi, s)$ rather than $\zeta(s)$ (register negative result 11). On $\mathbb{Z}[\mathbb{Q}/\mathbb{Z}]$ the programme’s signed clocks give the integral Bost–Connes relations $\sigma_n \rho_n = n, \rho_n \sigma_n = E_n \cdot, \rho_n(a) \rho_n(b) = n \rho_n(ab), \sigma_m \rho_n = d \rho_{n/d} \sigma_{m/d}$ ($d = \gcd(m, n)$), with $\ker \sigma_n = (e(1/n) - 1)R$ and $\text{im } \rho_n = E_n R$ mutual annihilators and $R/(\ker \sigma_n + E_n R) \cong R/nR$ (at a finite level N , of index $n^{N/n}$, claude-ab’s addition); even-degree transfer is never a fibre map over a sign-faithful character (CCT-1–CCT-8; 28_ §5.2; S54).

6.4 Two primes, and lifting

- *At most two primes have p^ρ algebraic*, for every nonreal ρ ; hence the dilation eigenvalues p^ρ are not all eigenvalues of finite-rank integral endomorphisms (IH5.3; the six exponentials theorem of Lang and Ramachandra; S14).
- *Two primes suffice to see every zero*: the values $F(\omega)$ at the zeros are determined by the grid $\sum_\omega m_\omega F(\omega) e^{-\omega(a \log 2 + b \log 3)}, (a, b) \in \mathbb{Z}^2$ (IH6; S15). Relatedly, if h/ℓ is irrational, a distribution V on $\mathbb{R}$ with $(E_h - 1)^r V = 0 = (E_\ell - 1)^s V$ is a polynomial of degree $< \min(r, s)$; so simultaneous generalised p^ρ - and q^ρ -eigenfunctionals are exactly the Mellin jets, while one prime alone has the aliases $\rho + 2\pi i k / \log p$ (MCL3.4; S17; for continuous V a case of L. Schwartz’s mean-periodic spectral synthesis [42]).
- *Least-order lifting*. At a zero of multiplicity m , the k -th source Mellin jet lifts through the transposed summation map with least generalised-character order $m + k + 1$, by an explicit formula; for a fixed annihilator $(X - b)^r$ with $b = 1 - \rho$, the obstruction class on the j -th source jet, read through the residue pairing, is $(-1)^j j! / 2 \cdot t^{r-1-j} \chi_\zeta(b + t) \bmod t^{\min(m, r)}$, the germ at b of the functional-equation factor χ_ζ ($\zeta(s) = \chi_\zeta(s) \zeta(1 - s)$), and the connecting map has rank $\min(m, r)$ (MCL5–MCL9, ORE4–ORE5; 07_ ; verified numerically for $m = 1$ and $m = 2$ to 10^{-39} ; S18). The order depends on the multiplicity, not on the position of the zero.

6.5 Winding and ramification at the zeros

Lemma 6.3 (the winding at the zeros is hyperbolic). *For every nontrivial zero $\rho = \sigma + i\gamma$ put $\lambda_\rho = e^{-2\pi i \rho}$. Then $|\lambda_\rho| = e^{2\pi \gamma}$, λ_ρ is a negative real number if and only if $\sigma = \frac{1}{2}$, and $|1/(\lambda_\rho - 1) + 1_{\gamma < 0}| \leq 1/(e^{2\pi|\gamma|} - 1) < 2.7 \cdot 10^{-39}$.*

Status. Proved here (claude-ab, 33_ Lemma 33.1; S64). Refereed (eleventh pass).

Proof. $|e^{-2\pi i(\sigma + i\gamma)}| = e^{2\pi \gamma}$ and $\arg \lambda_\rho = -2\pi \sigma$. For $\gamma > 0$, $|1/(\lambda_\rho - 1)| \leq 1/(e^{2\pi \gamma} - 1)$; for $\gamma < 0$, $1/(\lambda_\rho - 1) + 1 = \lambda_\rho / (\lambda_\rho - 1)$ has modulus at most $e^{2\pi \gamma} / (1 - e^{2\pi \gamma})$. No zero has $|\gamma| < \gamma_1 = 14.1347 \dots$, and $e^{2\pi \gamma_1} \approx 3.7 \cdot 10^{38}$. $\square$

Lemma 6.4 (escape under ramification). *Let Z be any set of points with $0 < \text{Re } \rho < 1$ and let $a > 1$ be real. Then $\{k \geq 0 : a^k \rho \in Z\} \subset \{k : a^k \text{Re } \rho < 1\}$ is finite, no nonempty $S \subset Z$ has $aS \subset S$, and only points with $\text{Re } \rho < 1/a$ can return. If $a \geq 2$ and $Z = 1 - \bar{Z}$, no reflected pair returns together; for $1 < a < 2$ a pair with $1 - 1/a < \text{Re } \rho < 1/a$ can return. Two points collide, $a\rho = a'\rho'$ with $\rho \neq \rho'$ and $a, a' > 1$, only if $\rho/\rho' = a'/a > 0$ (rational when a and a' are integers), which is impossible for distinct points of the critical line.*

Status. Proved here (claude-ab, 33_ Lemma 33.4, for integers $a \geq 2$; S65); the form for real $a > 1$ is version 2’s (48_ §3, F22, where an independent verification corrected the review’s wording of the reflected-pair and collision clauses). The programme’s iterated- and reflected-return theorems (BRC, GJN, GJNR) therefore

hold for every divisor in the open strip with their other hypotheses, so they are statements about all such divisors (Section 7, item N-Unif5).

Proof. $\operatorname{Re}(a^k \rho) = a^k \operatorname{Re} \rho$, which must stay below 1. If $aS \subset S$ then $a^k S \subset S$ for all k , impossible. $\operatorname{Re} \rho < 1/a$ and $1 - \operatorname{Re} \rho < 1/a$ cannot both hold for $a \geq 2$; for $a = \frac{3}{2}$ and $\rho = 0.55 + i$, the set $\{\rho, \rho^\#, a\rho, a\rho^\#, (a\rho)^\#, (a\rho^\#)^\#\}$ lies in the strip, is $\#$ -closed, and contains the returns of both ρ and $\rho^\#$. $(\frac{1}{2} + i\gamma)/(\frac{1}{2} + i\gamma')$ is real only if $\gamma = \gamma'$. $\square$

6.6 Every failure of RH is detectable

Proposition 6.5. *Let T be a recursively axiomatised theory that proves every true Σ_1 sentence and proves the sentence $M \text{ halts} \leftrightarrow \neg \text{RH}$ for an explicit Turing machine M (the 744-state machine of Matiyasevich and O'Rear reported by Aaronson [1], Theorem 9, or any machine searching for a counterexample to a Π_1 form of RH [24]; ZFC is an example). If RH is false, $T \vdash \neg \text{RH}$; so if RH is independent of T , it is true. If T is moreover Σ_1 -sound, then $\text{RH} \iff T \nvdash \neg \text{RH} \iff \operatorname{Con}(T + \text{RH})$.*

Status. Proved here (claude-ab from the author's argument, 23_ Proposition 23.1; register S42). Standard logic; no novelty claimed. Refereed (eighteenth pass).

Proof. If RH is false, M halts; the sentence that M halts is a true Σ_1 sentence, so T proves it, and with $T \vdash (M \text{ halts} \rightarrow \neg \text{RH})$, $T \vdash \neg \text{RH}$. If RH is true and $T \vdash \neg \text{RH}$, then T proves that M halts, a false Σ_1 sentence, against Σ_1 -soundness. $\square$

So the author's argument is valid in this exact form: the absence of a detectable failure (that is, the consistency of $T + \text{RH}$) implies RH, and RH cannot fail undetectably, since every failure has a finite certificate (the halting computation, or an interval-arithmetic argument-principle computation around an off-line zero). For Σ_1 -sound T that absence is in turn equivalent to RH, and T cannot prove $\operatorname{Con}(T + \text{RH})$ if it is consistent (Gödel's second incompleteness theorem), so the argument does not by itself establish RH.

6.7 Cohomology over the three-point base

On Connes–Consani's three-point base $Y = \{c_+, c_-, \eta\}$ the programme builds a residue pairing, its diagonal return, a doubled source $X^{\text{dbl}} = \operatorname{Spec} \mathbb{Z} \cup \{\mathfrak{m}_+, \mathfrak{m}_-\}$ and the pullback to it. The audited statements (register S36, S61, S70) are these.

- The right-adjoint dualizing complex on Y is $j_! k[1]$, so the degree-two residue pairing B_ζ cannot be a degree-zero sheaf map on Y ; it is the trace of a sheaf map on $Y \times Y$, and the diagonal's right adjoint $R\Delta^! = Rq_*$ keeps the trace with coefficient $+1$, while ordinary pullback loses a degree. Paired characters multiply to p , with weights $2 \operatorname{Re} \rho$ and $2 - 2 \operatorname{Re} \rho$ (FTD, PRS, DER, FSC, FEM; 19_).
- For $f : X^{\text{dbl}} \rightarrow Y$ and every sheaf Ω on Y : $f_* f^{-1} \Omega = \Omega$ and $R^q f_* f^{-1} \Omega = 0$ for $q > 0$, so $R\Gamma(X^{\text{dbl}}, f^{-1} \Omega) \cong R\Gamma(Y, \Omega)$, with supports as well (DCP5; for every Ω and with supports, claude-ab, 39_ Lemma 39.3). In the trivial finite-unit sector for $\mathbb{Q}$, the programme's cohomology is exactly $H^*(\mathbb{P}_{\mathbb{F}_1}^1, \Omega)$ of Connes–Consani [18], whose H^1 is the spectral realisation of the zeros.
- The programme's residue extension $0 \rightarrow \mathcal{B} \rightarrow E \rightarrow \mathbb{C}_\chi \rightarrow 0$, pushed out along evaluation at s_0 , splits for $s_0 \neq 0$; at $s_0 = 0$ it is $n \mapsto n(I - \log n \cdot N)$, with class $-\log$ in $\operatorname{Ext}^1(\mathbb{C}_\chi, \mathbb{C}_\chi) = \operatorname{Hom}((\mathbb{N}_{\geq 1}, \times), \mathbb{C})$, and its tensor Euler functions are those of the split sum: the pole at $s = 0$ carries the Jordan block that Euler products miss (NPE4; 31_ Lemma 31.2).

6.8 Results from the other workbenches used here

Lemma 6.6 (Keller maps give incompressible polynomial flows). *Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be polynomial with $\det DF \equiv c \neq 0$, V a C^1 vector field and $U = DF^{-1}(V \circ F)$. Then U is C^1 with $\operatorname{div} U = (\operatorname{div} V) \circ F$, it is polynomial if V is, and F maps integral curves of U to integral curves of V ; if F is a polynomial automorphism, U is complete if and only if V is (for example, V affine). For the map*

$$F = ((1 + xy)^3 w + y^2(1 + xy)(4 + 3xy), y + 3x(1 + xy)^2 w + 3xy^2(4 + 3xy), 2x - 3x^2 y - x^3 w),$$

which has $\det DF = -2$ and three preimages of $(-\frac{1}{4}, 0, 0)$, namely $(0, 0, -\frac{1}{4})$ and $(\pm 1, \mp \frac{3}{2}, \frac{13}{2})$, the curve $\gamma(\tau) = (z^{-1}, -3z/2, 13z^2/2)$, $z = \sqrt{1 - 8\tau}$, is an integral curve of U for $V = (2, 0, 0)$, escapes every compact set as $\tau \uparrow 1/8$, and has $F(\gamma(\tau)) = (-\frac{1}{4} + 2\tau, 0, 0)$.

Status. Proved here (claude-ab, 28_ Lemma 28.1, for affine V ; S52); the C^1 form and the equivalence of completeness are version 2's (48_ §3, F22; checked exactly at six rational points with a cubic V); the determinant and the three-point fibre re-verified symbolically for this reader. The map is recorded in the author's Yang–Mills workbench as the counterexample to the Jacobian conjecture announced by Levent Alpöge, whose announcement, as the workbench's attribution file records, credits Akhil for the question and Fable for the work; the curve is the one in that workbench. Novelty not searched.

Proof. $DF^{-1} = \operatorname{adj}(DF)/c$ is polynomial. By the Piola identity $\sum_j \partial_j \operatorname{adj}(DF)_{ji} = 0$ and the chain rule, $c \operatorname{div} U = \sum_{i,j,k} \operatorname{adj}(DF)_{ji} (\partial_k V_i)(F) \partial_j F_k = c(\operatorname{div} V)(F)$, since $\sum_j \partial_j F_k \operatorname{adj}(DF)_{ji} = c \delta_{ki}$. $\frac{d}{d\tau} F(\gamma) = DF(\gamma) \gamma' = V(F(\gamma))$. For an automorphism with inverse G , $\Phi_\tau^U = G \circ \Phi_\tau^V \circ F$, so U is complete exactly when V is; affine flows are complete. On the displayed curve $xy = -\frac{3}{2}$, and substitution gives the three components of $F(\gamma)$. $\square$

Two further exact statements from the other workbenches enter the bridges of Section 8: the exact aliasing of the Navier–Stokes workbench's torus endomorphism on finite character groups (register S40) and the Gram sandwich $(1 - \delta)^2 \Phi^* \Phi \leq \Phi_J^* \Phi_J \leq (1 + \delta)^2 \Phi^* \Phi$ whenever $\|(\Phi_J - \Phi)x\| \leq \delta \|\Phi x\|$ for all x with $\delta \leq 1$ (the triangle inequality; the hypothesis $\delta \leq 1$ is needed for the lower bound; register S41), of which the zeta programme's Schur-complement replacements and the Yang–Mills heat semigroup are instances.

7 Proved negative results, with their exact scope

This section collects the negative results. Each is a theorem, or where marked a numerical observation, about a specific construction as it is defined, and it is stated in the scope in which it is proved. They are of three kinds. Some statements are false, and a proof or a counterexample is given; the sharp velocity laws are an example, and so is the claim that zeros detect freeness. Some constructions are uniform in the zero set: their conclusions hold for every divisor in the open strip with the stated properties, or for the zeta functions of curves over finite fields, or for functions without an Euler product that have zeros off the line. Such a construction can decide RH only together with an input that holds for ζ and fails for those other functions, and where that input is known the item names it. Finally, some constructions are equivalent to RH by their definitions: they isolate the off-line part of the zero set, so that they vanish, or are injective, or are positive, exactly when there are no off-line zeros. Each of these is an equivalent form of RH written in the programme's terms: a proof of it is a proof of RH and conversely, and whether it is easier to attack is open.

None of these results says that the construction concerned is irrelevant to the Riemann hypothesis. Several of them identify the input that the construction lacks. For the weight constructions it is a pole bound uniform in the tensor degree, or a discreteness input together with a square; on

the programme's finite data the first is equivalent to $\max \operatorname{Re} \rho \leq \frac{1}{2}$, so it would have to come from an independent source. For the mirror and two-line constructions it is a property that distinguishes ζ from the Davenport–Heilbronn function, such as the Euler product. For the primary decomposition it is a polynomial bound on the principal parts of $1/\zeta$ at the zeros (item N-Prim). Whether the programme's constructions can supply these inputs is open. The numbers in brackets are those of the negative results in the audit's register (00_RESULTS_REGISTER.md, Goal 1, in the branch history cited in the introduction).

7.1 Statements that fail

N-Vel (sharp velocity laws) [1]. Under (H), all but finitely many zeros simple and on the line, $\sum_{\gamma \leq T} \operatorname{Re} c(\rho) = T/4\pi + O(T)$ is false: $\limsup |\operatorname{Re} E(T)|/T = \infty$; the vertical law's error is not $O(T)$ either. Unconditionally only the smoothed laws are proved, and for fixed smoothing width their error is of order T . Whether the observed skew ($\operatorname{Re} E/T$ between -3.15 and $+0.17$ up to $T = 64\,000$) persists is open, and the one-sided statements $\limsup \operatorname{Re} E/T = +\infty$ and $\liminf \operatorname{Re} E/T = -\infty$ are not proved. (Copy, Theorem C; proof completed in 20_.)

N-Sheet (the non-Eulerian sheets) [3]. For $a \in (0, 1) \setminus \{\frac{1}{2}\}$, $\zeta(s, a)$ has infinitely many zeros with $\operatorname{Re} s > 1$ (Davenport–Heilbronn [23] for rational or transcendental a ; Cassels [15] for algebraic irrational a). For $a > 1$ such zeros are found numerically; at $a = 2$, $\zeta(s, 2) = \zeta(s) - 1$ vanishes at $1.40778804065 + 23.3279877546i$. So an argument that uses only properties shared by all sheets $\zeta(s, a)$, $0 < a < 1$, cannot prove that the nontrivial zeros of ζ lie on the line; it must use a property of the Eulerian sheets $a \in \{\frac{1}{2}, 1\}$ (Corollary 2.2), such as the Euler product.

N-Beurling (regularity of the integers) [12, 13]. Regularity alone does not force RH for generalised primes: a Beurling system can have $N(x) = kx + O(x^{1/2} \exp(c(\log x)^{2/3}))$ and zeros approaching $\sigma = 1$ (Zhang [52]; also Diamond–Montgomery–Vorhauer [27]). Near-exact counts do not pin the prime system either: $(\text{primes} \setminus \{2\}) \cup \{3', 4\}$ (a second copy of 3) has $N(x) = x + O(\log^2 x)$ (copy, T2). Exact counts do pin it (the first-appearance sieve). So the question the programme poses is how the additive counting step constrains the multiplicative system, since regularity alone does not.

N-Free (zeros do not detect freeness in general) [28]. The monoid $\{1\} \cup \{2^e : e \geq 2\}$ is not free ($64 = 4^3 = 8^2$), yet its zeta function $(1 - x + x^2)/(1 - x)$, $x = 2^{-s}$, has all its zeros on $\operatorname{Re} s = 0$. For congruence monoids the absence of zeros in $\operatorname{Re} s > 1$ does imply freeness (Proposition 2.6, by Saias–Weingartner); the formal-logarithm test (Theorem 2.3) works for every monoid of integers.

N-Eu (finitely many Euler factors) [50]. No finite set of primes fixed in advance has Euler factors that determine every finite shifted-zeta product. For primes $p_1, \dots, p_k$ and $a_j = 2\pi i / \log p_j$, the even and the odd sub-sums of the box $\{\sum_j \varepsilon_j a_j : \varepsilon \in \{0, 1\}^k\}$ give products of shifted zeta functions with identical Euler factors at every p_j but different exponent measures (their difference has Laplace transform $\prod_j (1 - e^{a_j z}) \neq 0$). So Lemma 2.9 and its corollary are global statements; for a particular product finitely many factors can suffice. (claude-ab, 29_29.N2.)

N-Mirror (the mirror line at $-\frac{1}{2}$) [37]. The author's second line is exactly $A(Z) = Z - 1$, unconditionally, and the equality of mirror and shift at every zero is equivalent to RH (Theorem 4.2). The properties used there are the symmetry of the zeros under $s \mapsto 1 - \bar{s}$, their location in the strip, and an explicit formula with a positivity criterion. They cannot imply that a zero lies off the line, since the zeta functions of curves over finite fields have all three and satisfy RH (Weil's theorem [49, 37]). Nor does the coexistence of the two lines decide RH: the statements for $\zeta(s)$ and for $\zeta(s + 1)$ are one proposition, and the Davenport–Heilbronn function coexists with its translate, has a functional equation of ζ 's kind, and has the zero $0.808517182456637 + 85.6993484853776i$ off the line; what distinguishes ζ from it is the Euler product. The 0-centred partner pairing is not Hermitian (Proposition 4.4(b)). (17_ §§5–6, 10.)

N-Logic (independence) [43]. RH cannot be independent of ZFC and false, since it is equivalent to a Π_1 sentence (Proposition 6.5). For Σ_1 -sound theories non-refutability of RH is itself equivalent to RH, so this does not by itself establish RH.

7.2 Positive forms and receivers are supported on the critical zeros

The programme builds many positive forms, Hilbert completions and receivers on the zeta quotient, and the exact results form one picture. A positive construction compatible with the scaling transfer is supported on the critical zeros; this is a theorem, by the classifications of Section 3.5. Each of the constructions listed below that registers the off-line zeros is zero, or injective, or positive, exactly when RH holds. And the residue duality B_{ζ} , which is nondegenerate unconditionally, is not positive.

N-Pos1 (positive transfer forms) [6, 26]. Every continuous positive form compatible with the scaling transfer vanishes on the off-line part $R = Q/N_O$ (CPS4.2); the classification of such forms (register S12, S22, S35, S48; Lemma 3.13, which in version 2 classifies all bounded Hermitian forms compatible with the transfer for $n = 2, 3$ and shows that positivity is exactly what removes the off-line blocks) makes this exact. CFP, CPS, TWC10 and SPF9 are related results, not independent ones; for tensor degree $k \geq 2$ reflected pairs survive, with modulus exactly n . So an argument through such receivers needs an additional input that controls R .

N-Pos2 (the L^2 norm, harmonic sweeping, Hilbert quotients) [24, 25, 30, 33, 35]. The summation image is dense in $L^2(du)$ by Wiener's L^2 Tauberian theorem, so the naive L^2 quotient seminorm is zero (ASD10, OPD3). Poisson-sweeping the zeros onto the line gives a positive distribution that descends to Q if and only if there are no off-line zeros (HSW6). The Hilbert receivers are automatically pure (prime spectra on $|z| = \sqrt{p}$, unconditionally), and the kernel of the map to their inverse limit is exactly the off-line jet classes, zero if and only if RH holds (WHR8–WHR12). My assessment: the spectral-radius argument there proves the purity of an image that retains only the critical-line jets, so to reach RH it would need an input showing that the kernel, the space of off-line jet classes, is zero. A seminorm invariant under T_p and T_p^{-1} vanishes on every primary block with $0 < \text{Re } \rho < 1$, and with the half-density action it vanishes on every off-line block (ATG6.5; 14_§2). The off-line functional $\Theta_{\text{off}} = \sum_{\text{Re } \rho \neq 1/2} m_{\rho} F(\rho)$, continuous on Q and zero exactly when there are no off-line zeros, is never a nonzero multiple of $b \mapsto b(1)$: an explicit b_* in the summation image has $b_*(1) = 2$ and $\Theta_{\text{off}}(b_*) = 0$ (VWR10.5–10.9).

N-Pos3 (RH-equivalent objects) [47, 48, 53, 54, 56]. The following are equivalent to RH, or to RH with simplicity, by their definitions, each being equivalent to the absence of off-line zeros (or of multiple zeros): AST's $\mathcal{R}$, GDC's N_O , PTQ's H_{off} and the descent of the Weil form, the traces and detectors of PTQ8.3 and ECR5.5 (eight constructions, three conditions); $W \geq 0$, W positive definite (with simplicity), $U_a = D_a^*$ for one $a \neq 1$, density of the trace image (simplicity) (RTT, RZ); the ordinary trace of GDE's energy, normality of $B_{r,t}$, the vanishing of the degree-adjusted trace, of ABH's $\mathcal{L}_{r,t}$, of HSR's $D_{r,t}$ and Ξ , of HBW's singularity-class map, unitarity of $n^{-1/2}A_n$, weight- $\frac{1}{2}$ counting moduli, monodromy phases -1 , and $D_{r,t} \subset H^2$. The two numerical detectors are not holomorphic in ρ , so the explicit formula does not compute them directly. Every attempted cancellation (cyclicity, the degree relation, the cover relation) leaves the off-line sector intact, and the unconditional results among them (NHI5.4, NHI8.4, HSR4.1, HSR5, HSR6, HSR10.6, HBW3, HBW8, HBW9) hold whether or not RH does. On the Hardy boundary receiver the adjoint-domain condition holds only for $x = 0$: $\mathcal{L}_{r,t}x \in \mathcal{D}_{\delta_1}$ if and only if $x = 0$ (Noor's unconditional theorem), so it cannot serve as the proposed test. (25_§5, 27_§2, 30_§2, 32_§2) The zero-set condition $D_{r,t_2} = D_{r,t_1}$ (finitely many off-line zeros) is weaker than RH, and the two conditions read off the zero-time Gaussian energy (compactness, and the trace-class condition $\sum m_{\rho}(\beta - \frac{1}{2})^2 < \infty$) are implied by RH (register S58); their status is not claimed.

N-Pos4 (the divisor potential) [29]. Under $s \mapsto 1 - \bar{s}$ the odd part of $\log |\zeta|$ is exactly $\frac{1}{2} \log |\chi|$,

and its contribution to the off-line detector $\mathcal{E}_r(t)$ is accounted for by the pole and the trivial zeros; the detector is carried entirely by the even part, which the functional equation does not constrain further (OZD5, OZD7). My assessment: for fixed (r, t) the Gaussian weight $e^{-2t\gamma^2}$ makes each such identity numerically insensitive to zeros above modest heights.

7.3 Constructions whose conclusions hold for a class of divisors

N-Unif1 (completeness, integrality, clocks) [14, 15, 16]. The inference that an off-critical zero would go undetected is false: the test function $\varphi(v)e^{-(\rho-1/2)v}$ detects any zero. An off-critical quartet would appear in the absorption measure as $-2m(e^{\beta t} + e^{(1-\beta)t}) \cos \gamma t$, and completeness supplies no sign or growth bound that would exclude it (IH2–IH4). Clock changes preserving order and composition are linear and move the critical line with the zeros (PL2–PL8). No finite-rank integral model of the dilations exists (IH5.3).

N-Unif2 (lifting and the restriction row) [18, 19, 20, 22]. No eigenvector maps onto the source character at any zero; the least lift of the k -th jet has order $m + k + 1$, which depends on the multiplicity, not on the position (MCL6). In the ζ restriction row, kernel, middle term and target carry the same character $1 - \rho$, so any weight truncation keeps or kills the whole extension, and the fixed-order class is nonzero at every zero, on or off the line (ORE5.8, ORE6.5); the residue pushout removes the obstruction at every zero (RPC6.3). MCL3–MCL7, ORE6.2, ORE6.5, IH6.4, the first two laws of IH6.5 and PL2.1 hold word for word for the even lattice sum at $a = \frac{1}{5}$, whose factorization is not unique and which has zeros in $\operatorname{Re} s > \frac{1}{2}$ (08_ §2).

N-Unif3 (covers and sections) [36, 55]. For the transposed cover action on the dual of $\mathcal{B} + \mathcal{C}h_{p,t}$, the largest cover-invariant part of the zero-coordinate slice is exactly the divisor that the chosen residue section inserts: the zeta divisor for ζ , any finite prescribed set of nonzero points for a polynomial section, the empty divisor for the constant section (RSS3.2–RSS3.3, RSS6). Two residue sections of the same non-split extension have cocycle spans I_ζ and $\mathcal{B}$ (NJS10). So the cover-invariant part recovers the divisor that the section inserts, whatever it is, and locating the zeros this way requires an input about ζ beyond the choice of section.

N-Unif4 (the two-line splitting and the second copy) [38, 44, 46]. The splitting $\mathcal{B}/(I \cap I_-) \cong \mathcal{B}/I \oplus \mathcal{B}/I_-$ and the derived separation of Q and Q_+ hold for every primitive Dirichlet L -function (Proposition 4.12) and, by the same argument with standard inputs not re-read here, for every Dedekind zeta function (the paragraph after that proposition); their proofs never use where the zeros lie inside the open strip, and the Euler product enters only through zero-freeness of $\operatorname{Re} s \geq 1$. The derived separation is the general vanishing of Ext for coprime annihilators. $\mathcal{B}$ itself does not split along the two lines. The statements about the copies that involve RH are equivalent forms of it through $\mathcal{R} = Q/N_O$, which is determined by the values at off-line zeros, and the programme states that neither a vanishing nor a new pure weight follows from them (ADM9).

N-Unif5 (returns, windings and monodromy) [58, 59]. BML's winding difference is invertible for every zero set avoiding the integers with $e^{-2\pi i \rho}$ at positive distance from 1 (Lemma 6.3), and BML11's H^1 is produced by the truncation of the logarithmic tower. The iterated-return and reflected-return theorems of BRC, GJN and GJNR hold for every divisor in the open strip with their other hypotheses (Lemma 6.4); the norm-shift relation $NP_b = bP_bN$ is the chain rule and has solutions for every nilpotent.

N-Unif6 (the doubled source, the adelic complex, the jet map) [69, 70, 71, 72]. The programme's doubled source has the cohomology of Connes–Consani's base: $R\Gamma(X^{\text{dbl}}, f^{-1}\Omega) \cong R\Gamma(Y, \Omega)$ for every sheaf Ω , with supports, and in the trivial finite-unit sector for $\mathbb{Q}$ this is $H^*(\mathbb{P}_{\mathbb{F}_1}^1, \Omega)$, whose H^1 is Connes–Consani's spectral realisation of the zeros (Section 6). The source-space gluing and the doubled pullback (CGS4–CGS10, DCP4–DCP12) are uniform in the zero set, which enters only as the spectrum of the dilations on H^1 and through the jets: in finite models with zeros on and off the line all dimensions, ranks, signs and sections agree. The lift in DCP is Connes–Consani's step,

which vanishes by Poisson summation, not by a separation of weights as in Weil II §3.6 (DCP12 says the comparison has not been assumed). The Ext-vanishing between the zeta quotient and the endpoint modules sees only $\zeta(0) = \zeta(1) = \frac{1}{2} \neq 0$ (GEX9 states the limitation). OMS5–OMS7A use only $0 < \operatorname{Re} \rho < 1$, the multiplicities and the unboundedness of the heights (OMS8 says so).

N-Prim (the primary decomposition) [73]. Convergence of the primary decomposition of the zeta quotient for every class is equivalent to polynomial bounds on the principal parts of $1/\zeta$ at the zeros, and for simple zeros to $|\zeta'(\rho)| \geq c(1 + |\gamma|)^{-K}$, which would imply that all zeros are simple (Theorem 3.9); this is the input the construction needs. Q is not the product of its primary blocks (Corollary 3.7); the constant tuple 1, the obstruction behind the programme's nonzero class c_{ζ} (PGD4), is not a jet tuple.

7.4 Weights, tensor powers and the square

N-W1 (the weight separations proved so far) [17, 21, 65]. The programme's receivers separate weights using only the strip bounds $0 < \operatorname{Re} \rho < 1$, or Tate shifts built into the receiver, and TL8 records that a geometric source for a dividing line at weight 1 has not yet been constructed. Positivity of the timed-prime measure reaches exactly $\operatorname{Re} s = 1$: it gives $\zeta(1 + it) \neq 0$ and, with standard growth bounds, the classical zero-free region, and no zero-free strip (Beurling systems with positive prime measure can have zeros approaching $\operatorname{Re} s = 1$). The method of Weil II §2.1, specialised to ζ (Deligne's example 2.1.9), proves only these classical statements: the compact form, the cohomological proof of condition (C), the strict bound and Weil II §3 have no ζ analogue along this route (37_ 37.N2–37.N3), and they are the steps an extension of the route would need.

N-W2 (tensor powers and the square, as constructed) [23, 27, 31, 32, 34, 52, 66]. The exponent data return $\max \operatorname{Re} \rho$ in every tensor degree, positive or not, and what positivity of even tensor powers adds is a pole at the real point (Proposition 5.1, which version 2 extends to every degree). The missing input is a pole bound at $k + 1 + C$ uniform in k , which on the programme's finite data is equivalent to $\max \operatorname{Re} \rho \leq \frac{1}{2}$. The characters of source and target match in every tensor power; the transfer-compatible positive forms kill every tuple off the centred set; the tensor Weil form is indefinite on the reflected sector; the square's obstruction vanishes exactly when the original one does (TWC6, TWC9–TWC11, FST3, FST4.2, FST4.4, FST7). In degree 1 the square reduces to $Q \otimes Q$ (GMC4–GMC6), and reflected off-line pairs are centred ($\rho + \rho^{\#} = 1 + 2i\gamma$). On the Hilbert-closure defect, source and target of the connecting map have the same character (HCS7, HCS10–HCS11). The faithful norm on $Q^{\otimes k}$ is bounded with exponent k (sharpness is proved only for the source seminorms, ATG6.1), and the critical Hilbert components have exponent $k/2$ but common kernel exactly the tensors with an off-line factor; for finite reflection-stable S the exact finite-trace exponent is $k \max \operatorname{Re} \rho$, so the error against $k/2$ is linear in k (ATG). The tensor Euler products of an off-line quartet have their genuine pole at $1 + 2k \max(\operatorname{Re} \rho, 1 - \operatorname{Re} \rho)$ for every complex ρ , zero of ζ or not (checked at $\rho = 0.7 + 10i$), so the position of this pole is determined by ρ alone, whether or not ρ is a zero of ζ , and detecting zeros through these products would require an input that singles out the zeros of ζ (37_ 37.N5).

N-W3 (the average weight and determinant data) [60]. The average weight of an off-line quartet is exactly 1 (Proposition 5.2); only the constituent weights register an off-line zero, and Weil II 1.5.1 takes a pure determinant weight as its input and does not produce one. Determinant data at a point control only the sum of the weights (36_ 36.N1).

N-Tr (the product trace and the diagonal return) [39]. The product-trace machinery does not force weight 1. The reflected-pair classes have character p (weight 2) whether or not either zero is on the critical line, and the full global complex kills them with an explicit boundary (FEM7–FEM8). The diagonal return keeps the product trace exactly but pairs the weights $2 \operatorname{Re} \rho$ and $2 - 2 \operatorname{Re} \rho$ (DER10); in the programme's words it does not prove either individual weight to be one. Also exact: no degree-zero same-site lift exists (FTD9), no continuous single-valued lift of q fixes the

arithmetic diagonal (FSC2), and the comparison $R\Delta^1 K \rightarrow \Delta^{-1} K$ is zero (DER6) (19_).

N-W4 (Deligne’s local theory against the programme’s operators) [61, 62, 63, 51]. The boundary bounds on V and $V^\vee$ alone do not force the conclusion of Weil II 1.8.4; the tensor square is needed (Proposition 5.3). An operator commuting with N cannot carry Deligne’s local weights unless $N = 0$, which applies to the programme’s counting operators W_p and C_b on jet blocks of length ≥ 2 ; the ramification operator P_b has no eigenvector on its space, so the weight filtration that Deligne builds from a Frobenius is not defined for it, and only the relation $NP_b = bP_bN$ transfers. Tensor Euler products are the same for a non-split Jordan block and for its split sum, and the programme’s residue extension is such a block, at the pole $s = 0$ (ETR10, NPE4).

N-W5 (the recovered arithmetic base) [67, 68]. The programme’s recovered ring $R = \text{End}(G/G_{\text{tors}}) \cong \mathbb{Z}$ carries only the free rank of the winding group; every automorphism acts trivially on it, so $\text{Spec } R$ is $\text{Spec } \mathbb{Z}$ with its standard arithmetic, and the torsion, the chart twist, the involution and the support τ do not reach it. Every Frobenius eigenvalue of the objects built there from programme data (the constant sheaf, its Tate twists, $i_{p*}E$, the costalks at closed points of $\text{Spec } \mathbb{Z}[1/\ell]$) is an integral power of the residue characteristic, so all these weights are even; the purity circle $|a|^2 = p$ needs weight 1, which Deligne’s theory supplies on this base through, for example, H^1 of an elliptic curve over $\mathbb{F}_p$. A weight-one object built from programme data is the input this route would need, and none of the objects built there so far is one (38_).

N-W6 (finer points of Weil II §2) [64]. Positivity alone cannot exclude Deligne’s quadratic exception (Lemma 5.5); in Weil II the double cover does, and for ζ the absence of a nontrivial quadratic character of $\mathbb{R}$.

N-F1 (items of the $\mathbb{F}_1$ analysis) [7, 8, 9, 10, 11]. Doubling has no sign-preserving mirror-compatible lift on Connes–Consani’s signed data (a lift killing the sign does exist). The compact-completion modulus theorem holds for every compact group (P12 says so). The $\sqrt{n}$ dilation radius cancels in the product formula. The twistor involution does not fix the radius (equal radii only on $|z| = 1$). The weight-one sign sector is $L(E, s)$, of conductor 32, rather than $\zeta(s)$.

7.5 The two-line system

N-Two1 (parity and the four-direction anomaly) [40]. The $\mathbb{Z}_2$ anomaly of one line counts only on-line zeros; in the two-line system the $\mathbb{Z}_2$ anomaly vanishes identically and the $\mathbb{Z}_4$ anomaly equals the one-line parity; the integer count sees off-line pairs only together with $N_0(T)$, and the holonomy index does not separate off-line pairs from multiple zeros on the line (Propositions 4.8 and 4.9).

N-Two2 (the structure without the Euler product) [41, 45]. The Davenport–Heilbronn function has the completed functional equation, the reality, the nonnegativity of its two-line product on $\text{Re } s = 0$ (strict positivity is not established for it), and a size-four orbit for its off-line zero; the superposition of $L(s, \chi)$ and $L(s, \bar{\chi})$ cancels the root-number phase by addition and loses the Euler product, while the product $L(s, \chi)L(s, \bar{\chi})$ cancels it by multiplication and keeps it (22_ 22.10–22.11). The separator proof does not reach the Davenport–Heilbronn function (Section 4); whether the splitting fails for it is open.

N-Two3 (the second line on its own) [74]. For products Λ of completed L -functions of primitive Dirichlet characters in which χ and $\bar{\chi}$ occur equally often, the positivity $G_\Lambda(it) = |\Lambda(1 + it)|^2 > 0$ follows from Λ ’s functional equation, reality and nonvanishing on $\text{Re } s = 1$, and G_Λ and its explicit formula are functions of Λ , so a bound on the count of off-line zeros obtained through G_Λ is one obtained from Λ (whether one-line inputs can bound that count better than Turing’s method is open). The chiral cross form is hyperbolic whether or not RH holds, and its diagonal part, which is Weil’s form, is the part that registers off-line zeros (Proposition 4.6). The two-line prime race differs from the one-line race by a convergent series, so its limiting distribution (under GRH) and its logarithmic density (under GRH and linear independence) are unchanged (Proposition 4.10).

7.6 Cross-programme transfers

N-X (transfers between the workbenches) [42, 49, 57]. Joint averages do not survive restriction of the Navier–Stokes torus cover to unchanged finite torsion (13 against 4 at $p = 13$, $a = 4$); the complete preimage repairs this. The Navier–Stokes profile gives Yang–Mills a zero-quotient sequence only as the coupling tends to 0, and only as one admissible weight. The Jacobian $\rightarrow$ Yang–Mills chain gives statements along couplings $g_j < 1/j$ tending to 0, not at a fixed coupling or in the continuum: the Yang–Mills workbench’s Theorem 14.2 is correct conditional on standard fixed-box weak-coupling limits that the file only sketches, and its energy quotients are, up to $1 \pm 1/(10j)$, the lowest free two-gluon colour-singlet energy of an open box of side $j/50$ at couplings $g_j < 1/j$, so $Q_j \cdot (\text{box side}) \rightarrow 2\sqrt{2}\pi$, the scaling of a massless free field, whether or not the continuum theory has a gap; the Jacobian tensor enters the norm and the energy, not the limit of the quotient, and a statement about the continuum mass gap through this chain would need a fixed-coupling or continuum input (34_). The $S^6 \rightarrow$ Yang–Mills edge uses only $D > 0$ on its patch; the Navier–Stokes workbench certifies none of its early heat or S^6 transfers; prime-only clock seeds are not transfer-stable (21_, 28_).

8 Bridges to other areas, and the directions they open

Much of the programme consists of bridges between the zeta function and other parts of mathematics, and in the author’s view they are among its most valuable outcomes. This section treats each bridge in the same way. It first gives the typed map: what is mapped to what, and what crosses. It then says what is established, what the bridge suggests on each side, and what my assessment is. The assessment is a view held at this stage and is labelled as such. Each item says whether it is a theorem, an identity of objects, an analogy or a proposal.

8.1 Meyer’s quotient and interpolation theory

The jet map $J : Q \rightarrow \prod_{\rho} A_{\rho}$ carries Meyer’s quotient to the jets at the zeros, and what crosses is the interpolation problem at the zeros. It is established that J is injective with dense image in the space Λ_m of rapidly decreasing jet tuples (Corollary 3.7), and that it is onto exactly when the principal parts of $1/\zeta$ at the zeros are polynomially bounded (Theorem 3.9). The convergence of the primary decomposition is thereby the same kind of question as Berenstein and Taylor’s interpolating varieties [6] and Bondarenko, Radchenko and Seip’s Fourier interpolation at the zeta zeros [9]. The Gonek–Hejhal conjectures concern the averages $\sum |\zeta'(\rho)|^{-2k}$, while Theorem 3.9 asks the pointwise question (an analogy of questions; the theorem itself is proved).

For interpolation theory the bridge offers the zeta zeros as a test case for interpolating varieties; for ζ it offers a structural reason to care about pointwise lower bounds for $|\zeta'(\rho)|$. My assessment: this is the most precise translation in the programme, because it turns a structural property of its central object into a classical analytic condition. The case of multiple zeros, where in the weighted spaces A_{ρ} the leading coefficients alone suffice [6, Theorem 4], is the natural next question (Question 1).

8.2 The two-line system, Weil’s explicit formula and Turing’s method

Three maps make up this bridge. The decomposition $f \mapsto (u, v)$ into parts even and odd under the translation carries the chiral form to $Q_A(f) = 2\Omega(u) - 2\Omega(v)$. The loop $T^{-1}A$ is Weil’s reflection $\#$ through $\frac{1}{2}$. And the holonomy count goes to Turing’s index $(N(T) - V(T))/2$. All three are theorems (Section 4). The mirror through the pole at 0 followed by the passage between the lines is Weil’s reflection. On the diagonal of the even data the chiral form is twice Weil’s form (Proposition 4.6). The count of nontrivial holonomies plus $\sum \lfloor m_{\rho}/2 \rfloor$ over the on-line zeros is Turing’s index (Proposition 4.9). On Weil’s side the bridge suggests a test family $\{f_k\}$ that is even under the translation by construction and has the explicit prime side $-G'/G$ (Corollary 4.7). On

Turing's side it suggests that an index for off-line zeros alone needs a map that separates off-line pairs from multiple zeros on the line.

My assessment: the bridge recasts Weil's positivity as a signature statement about a form that is hyperbolic for every configuration of zeros. Since the whole form has the same signature with or without RH, I expect progress along this bridge to come from the diagonal family, where the positivity question is concentrated (Question 2).

8.3 The programme's weights and Deligne's Weil II

This bridge carries four maps:

- a zero ρ goes to the eigenvalue p^ρ , of weight $2 \operatorname{Re} \rho$;
- the even tensor powers go to the positivity step of Weil II 1.5;
- the timed-prime measure goes to Deligne's positive measure;
- the ramification operator P_b goes to a relation $FN F^{-1} = Q^{-1}N$ of Deligne's form.

Established are the results of Section 5, and further that Weil II §2.1 specialised to ζ is the classical proof of $\zeta(1 + it) \neq 0$, with the timed-prime measure mapping onto Deligne's positive measure (DB9, DR0–DR4; theorems). In the programme it is ramification, not counting, that satisfies a relation of Deligne's form ($P_b N P_b^{-1} = b^{-1}N$ on the image of P_b). Every solution of the relation on a jet block forces the weight shift $-2j$ of the programme's regrading, and such relations have solutions for every nilpotent, so what is established is a comparison of relations (35_ §3). Under that regrading the jet block of a zero of multiplicity m satisfies the conclusion of Weil II 1.8.4 with centre $2 \operatorname{Re} \rho - (m - 1)$. For $m \geq 2$ the block is mixed in the sense of Weil II (1.7.4), and it is pure in the sense later used for Weil–Deligne representations [44]. The regrading is chosen in the programme rather than derived from an arithmetic source, so this is a comparison of structures (36_ Remark 36.R1). Deligne's step $\alpha \mapsto \alpha^2$ ($H^1 \otimes H^1 \hookrightarrow H^2$) has as additive shadow Kurokawa's tensor product of zeta functions, with spectral parameters $\rho + \rho'$ (Manin [31], §1.4). Weil's proof uses $\bar{C} \times \bar{C}$, and Connes proposes the square of the arithmetic site as its analogue [17]; these are analogies, not identities (08_ §§3–4).

For Weil II the bridge produced a purity criterion, corrected lemmas and a list of misprints. For ζ it produced a precise form of the weight question (Question 8), which for the square is Manin's question of absolute Descartes powers of $\operatorname{Spec} \mathbb{Z}$ [31]. My assessment: the programme's even-power computations supply the positivity half of Deligne's mechanism in Weil II §1.5, and the other half is the uniform pole bound. Since that bound is equivalent to $\max \operatorname{Re} \rho \leq \frac{1}{2}$ on finite exponent data, I think the decisive work on this bridge is an independent source for it. The starting point is the programme's own statement (TWC9, TWC11) that its computations leave room for one.

8.4 Nyman–Beurling in three topologies

The same kind of family, the cover discrepancies or dilation tests, is studied in three settings: in $\mathcal{B}$, in $L^2(0, \infty)$ with two-sided tests, and in $L^2(0, 1)$ and Noor's Hardy space with one-sided tests. In $\mathcal{B}$ the family generates the zero-jet ideal unconditionally, already for the degrees $2^a 3^b$ when $t > (\log 2)(\log 3)/(4\pi) \approx 0.0606$ (Proposition 3.11). In $L^2(0, \infty)$ two-sided tests are dense unconditionally (Wiener's L^2 Tauberian theorem [51]). One-sided density in $L^2(0, 1)$ is equivalent to RH (Beurling [7], Báez-Duarte [3]; Burnol's co-Poisson theory [13]), and Noor's Hardy-space form is also equivalent to RH [34]. The three settings differ in topology, support and test class, and RH is the statement about the third (theorems; 09_ §4(c), 16_). The bridge suggests a comparison map between the settings, with its kernel identified, which would show where the content of RH enters. My assessment: the contrast is instructive, since the same family is dense unconditionally in one topology and exactly under RH in another. The sparse families of Question 5 probe which

features of the topology matter.

8.5 Connes, Meyer and Connes–Consani

This bridge carries several maps: from Meyer’s quotient Q [32] to Connes’ weighted Hilbert quotients [16]; from the programme’s doubled source to Connes–Consani’s three-point base Y [18, §5]; the programme’s adelic periodization complex; from Connes–Consani’s absolute twistor line [20] to the elliptic curve $y^2 = x^3 - x$; and from the reversed prime knots of [19] to the conjugate character. These are established:

- The kernel of the first map is exactly the classes vanishing on all critical-line jets, so it is injective if and only if RH holds; the loss happens in the Hausdorff quotient (Section 3.5; register S26, S28).
- For sheaves pulled back from the base, the second map induces an isomorphism on cohomology, with supports (Section 6; DCP7.2, 39_). In the trivial finite-unit sector it gives $H^*(\mathbb{P}_{\mathbb{F}_1}^1, \Omega)$, whose H^1 is Connes–Consani’s spectral realisation of the zeros.
- The adelic periodization complex is formal: after degree-zero coinvariants and in the algebraic representation category, $C \simeq \mathcal{B}_{\text{pr}}[1] \oplus Q[0]$ equivariantly (OMS7A.1; 40_ §4).
- The twistor line is linked to the CM curve through its $\mathbb{F}_{1^2}$ -points (Section 6).
- The prime knots carry the conjugate character when reversed, and for primitive χ the functional equation puts the anti-character on the other side at the phase $W(\chi)$, with $W(\chi)W(\bar{\chi}) = 1$ (22_ Proposition 22.8). These are theorems, and their reading as anti-numbers is a proposal.

Four further identifications also hold: Connes’ harmonic distribution of §VIII, done with the full Gram matrix, is the programme’s FGR; spectral synthesis on Q is a local-description theorem in Krasichkov-Ternovskii’s sense [29]; the positive transfer forms before the quotient are the Bochner–Schwartz theorem in Mellin coordinates [48]; and the divisor potential of Section 3.5 belongs to the Littlewood-lemma family [5, 43] (identities of objects and theorems, as stated in the programme and audited).

For Connes’ approach the bridge suggests an exact account of what the Hausdorff completion loses, and a jointly faithful family of quotients over all lines $\text{Re } s = \sigma$ (VWR). For the programme it suggests a dictionary with the established spectral realisations. My assessment: the comparison of Connes’ Hilbert quotients with Meyer’s Fréchet quotient is the cleanest structural statement here. The comparison of the classification of positive forms with Meyer’s work and with the literature on Weil positivity has still to be made, and I would make it first.

8.6 Clocks, Bost–Connes, Beurling and monoids

The programme’s normalised clock operators $S_n = \sqrt{n} V_n$ satisfy $S_m S_n = S_{mn}$ and $S_m^* S_n = S_{n/d} S_{m/d}^*$ ($d = \gcd(m, n)$), the relations of the semigroup crossed product $C(\widehat{\mathbb{Z}}) \rtimes \mathbb{N}^\times$ (Laca–Raeburn [30]). So the author’s global quotient of all prime clocks is the space on which the Bost–Connes system acts (P14; 01_ §4; theorem). The partition function of that system is $\zeta(\beta)$, with its phase transition at the pole $\beta = 1$ [10]. Its link with the ledger of the pole at 1 in Proposition 6.1 is an interpretation, for which a map remains to be constructed. The timed-prime reconstruction is Beurling’s generalised-number construction $D = \exp_*(R)$ specialised to the actual primes (TP9–TP13; identity). At $a = 1/q$ the copy’s Theorem E is the factoriality criterion for $\{n \equiv 1 \pmod q\}$, a Krull monoid with class group $(\mathbb{Z}/q)^\times$. The even-sheet test is the same criterion after dividing the class group by ± 1 , and it coincides with half-factoriality of the one-sided monoid, the Krull-monoid analogue of Carlitz’s class-number-two theorem ([4, 14]; Section 2; theorem).

For the theory of monoids the bridge yields a freeness test for every monoid of integers (Theorem 2.3). For generalised primes it shows that regularity of the counting function alone does

not force the analogue of RH (item N-Beurling). My assessment: the monoid results are of independent interest as elementary number theory, Theorem 2.3 and Corollary 2.4 most clearly. The novelty of the theorem was checked by one search, and that of the corollary was not searched.

8.7 Two-prime density and mean-periodic functions

MCL3.4, IH6 and PL3 rest on one fact: $(\log p)\mathbb{Z} + (\log q)\mathbb{Z}$ is dense in $\mathbb{R}$. This is the multiplicative form of the two-period phenomenon of mean-periodic functions [42]. The density holds because $\log p / \log q$ is irrational, and the link with mean-periodic functions is a comparison of mechanisms. The bridge explains why two primes determine every zero while one prime leaves the aliases $\rho + 2\pi ik / \log p$ (Section 6). My assessment: for mean-periodic functions spectral synthesis is a theorem: every closed translation-invariant subspace of $C(\mathbb{R})$ is the closed span of the exponential monomials $x^k e^{\lambda x}$ it contains [42]. Under $u = e^x$ these monomials become the functions $u^\lambda (\log u)^k$ that pair with the jets at the zeros, so I think the transfer of Schwartz's theorem to the programme's jet calculus is a direct next step, and one whose outcome can be checked.

8.8 The anomaly dictionary (a proposal)

The author's physical picture is matched term by term with exact objects (22_§1). The objects and the facts proved about them hold without assuming RH; the matching itself is a proposal.

Author's term	Exact object
anti-number	the clock run backwards with conjugation, $g \mapsto \overline{g(1/u)}/u$ (Mellin shadow $s \mapsto 1 - \bar{s}$)
chirality	the mirror $A(s) = -\bar{s}$ through the pole at 0
the doublet	the orbit $\{\rho, \rho^\#, \rho^\# - 1, \rho - 1\}$ of an off-line zero in the two-line system
$\mathbb{Z}_2$ anomaly	the two lines as the two sheets of one double cover, deck transformation the central -1 of $SU(2)$
$\mathbb{Z}_1$ anomaly	the root-number phase $W(\chi)$, cancelled by the anti-sector: $W(\chi)W(\bar{\chi}) = 1$
inflow	the bulk between the lines, which holds exactly the off-line zeros with $\operatorname{Re} \rho < \frac{1}{2}$ and their mirrors
the support τ	the pivot of the swing between 0 and 1: on the zeta side the reflection $s \mapsto 1 - \bar{s}$ about $\frac{1}{2}$

In these terms RH has three exact forms. Every zero is its own anti-number. The bulk is empty. And the Weil functional is nonnegative on every annihilation $g * g^\#$ (Weil's criterion); the value of the annihilation at the unit, $\int |g|^2 du$, is positive with or without RH. My assessment: the dictionary is useful because every entry is an exact object. Its test is whether a statement first seen in the physical language yields a new statement about these objects.

8.9 The $\mathbb{F}_1$ context

The author describes τ as carrying $\mathbb{Z}_1$ and no $\mathbb{Z}_2$. This is realised at Connes–Consani's generic point η , which every homeomorphism fixes and where no exchanged label can sit, although the invariant local data there still contain the sign ε . The absence of $\mathbb{Z}_2$ thus has two exact readings: no exchanged label, which holds at η ; or no sign at all, $\varepsilon = 1$, which holds among field-valued points exactly in characteristic 2, through the retraction $\mathbb{F}_{12} \rightarrow \mathbb{F}_1$. The labels $\mathbb{Z}_n$ follow the principle that an $\mathbb{F}_{1^n}$ -structure is a free $\mathbb{Z}/n$ -action (Kapranov–Smirnov, via Connes–Consani–Marcolli [21], §3; a reading). The points of $\mathbb{P}^1$ over $\mathbb{F}_{12}$ are $\{0, \infty, \pm 1\}$, and the double cover branched at them is $y^2 = x^3 - x$. The Eulerian sheets $a = 1/q$ of the even lattice family are exactly those with $(\mathbb{Z}/q)^\times = \{\pm 1\}$, the units of $\mathbb{F}_{12}$, and exactly those on which the one-sided monoid is half-factorial (Proposition 2.7); reading this as $\mathbb{F}_1 \rightarrow \mathbb{F}_{12}$ is an interpretation. The global quotient of all prime clocks is $\widehat{\mathbb{Z}}$ (Proposition 6.2), the Bost–Connes space, which Connes and Consani treat as an $\mathbb{F}_1$ object (a reading). The split-zero semiring $G(\mathbb{Z})$ adds a Boolean point below the generic point of $\operatorname{Spec} \mathbb{Z}$, where the sign is forgotten, while Connes and Consani add a closed point ∞ glued along the generic point (a reading). The question at which the programme's analysis of weights arrived

can be stated without its vocabulary. Does the reconstructed arithmetic carry either (a) a square with a Künneth map $H^1 \otimes H^1 \rightarrow H^2$ and an auxiliary object with discrete weights, or (b) an independent cohomological source for a pole bound uniform in the tensor power (Section 5)? For the square this is Manin’s question of absolute Descartes powers of $\text{Spec } \mathbb{Z}$ [31]. My assessment: the reading with mathematical content of its own is the coincidence of the Eulerian even sheets with the units of $\mathbb{F}_{12}$, which deserves a statement independent of the reading.

8.10 The author’s other workbenches

The author maintains workbenches on the Erdős–Straus conjecture (ES), on Yang–Mills (YM, which also holds the Navier–Stokes (NS), S^6 and Jacobian-map material), on Erdős problem 817, and on the Collatz problem. Companion records in the same folder treat these workbenches and the other sides of the edges below. The table lists a selection of the edges located from the zeta side, audited in the notes 21_, 28_ and 34_ and, where stated, in the companion records. The Collatz side of the last edge was located in an external record, the Clankers’ Zenodo record 19900461; the author’s Collatz workbench itself was not read for this record, and the companion record treats it. The status column says what the audits established: *verified* means read and checked, with the note or record cited; *located* means found and described, not checked.

Edge	Shared object or lemma	Status
NS $\rightarrow$ ES	the torus endomorphism $J = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}$ of the NS manuscript, used as an operator in 21 ES statements; exact aliasing on finite character groups;	verified (register S40); joint averages need the complete preimage (N-X)
NS $\rightarrow$ YM	sharp constant $1/\sqrt{4 + 2\sqrt{2}}$ for the $\sqrt{2}$ directions NS velocity as a reducible $SU(2)$ connection; $-2 \sum \text{tr } \mathcal{F}_{ij}^2 = \lambda^2 \text{curl } u ^2$	verified, conditional on imported NS bounds and on YM sections not read; zero-quotient sequence only as the coupling tends to 0
zeta $\rightarrow$ YM	the Gram sandwich from a relative column error	verified (register S41)
Jacobian map $\rightarrow$ NS	$U = DF^{-1}(V \circ F)$, the escaping curve, the material	verified: Lemma 6.6; Theorem 14.2 correct conditional
kinematics $\rightarrow$ YM	tensor; YM Theorem 14.2	on fixed-box weak-coupling limits (N-X)
$S^6 \rightarrow$ YM	the imaginary period block B of the S^6 period matrix enters only through $D = \det B$	verified (register S53)
$S^6 \rightarrow$ ES	the integral monodromies of the $(3, 4, \infty)$ period system give decorated affine-torsion covers of the ES divisor atlas	located in version 2; the orbits of the covers are now verified at every level in the Erdős–Straus record
ES $\leftrightarrow$ zeta	the split-zero semiring $G(R) = R \sqcup \{\tau\}$ with its unique Boolean character	verified (the zeta proof checked by hand)
zeta $\rightarrow$ ES	ES family-C results use zeta NG20–NG21 as theorem dependencies	located (a dependency, not re-proved in ES)
ES $\leftrightarrow$ zeta	a four-dimensional Keller map P with $\det DP = -2$, transported into the split-zero weighted conductor; the shared file is byte-identical in both repositories	$\det DP = -2$ verified; the rest located
NS $\rightarrow$ zeta	the released NS field as arithmetic input (Mellin transports, residue at $s = -1$)	located; depends on the imported NS existence theorem, whose independent verification the zeta side records as unfinished

Edge	Shared object or lemma	Status
817 $\rightarrow$ zeta	finite word sources of problem 817 mapped into the split-zero supported quotient (typed maps written; no zeta statement is entered)	located in version 2; one of the ten interface notes is now verified in the companion record
Collatz $\leftrightarrow$ ES $\leftrightarrow$ zeta	the support-idempotent homogenisation $(x, h) \mapsto (ax + bh, h)$	located in version 2 (same formula on all three sides); verified in part in the companion record

One Collatz item is elementary and can be checked here. With $T(n) = 4n + 1$, $\theta(n) = 3n + 1$ and the odd step $U(n) = (3n + 1)/2^{v_2(3n+1)}$, one has $\theta(T(n)) = 12n + 4 = 4\theta(n)$, hence $U(T^j n) = U(n)$ for all j (ES results bench, R05). The Erdős–Straus workbench states that it does not claim a proof or counterexample of the conjecture, and the Collatz workbench that it is not a claim that the Collatz conjecture has been solved; the Yang–Mills and Navier–Stokes material states its own scope limits, recorded in N-X.

My assessment: the shared linear algebra (Gram sandwiches, Schur complements) is the most reusable crossing, since it is exact and needs no identification of objects. The edges that carry a statement about ζ into another workbench, or back, still need typed maps with verified content.

9 Questions, and what was not verified here

9.1 Questions left open by the proved results

1. *The primary decomposition.* Do the principal parts of $1/\zeta$ at the zeros satisfy polynomial bounds (Theorem 3.9(c))? For multiple zeros, does the leading coefficient alone suffice?
2. *The diagonal family.* Is the T -even family $\{f_k\}$ of Corollary 4.7 easier to control on the prime side (the explicit formula for G'/G) than Weil’s own family?
3. *An index for off-line zeros alone.* The holonomy index $I(T)$ counts off-line pairs together with $\sum \lfloor m_\rho/2 \rfloor$ (Proposition 4.9); an index counting off-line zeros alone, obtained from a map on the strip, has not been constructed. Whether one-line inputs can bound the bulk count better than Turing’s method is open.
4. *The Davenport–Heilbronn function.* Does the splitting $\mathcal{B}/(I \cap I_-) \cong \mathcal{B}/I \oplus \mathcal{B}/I_-$ fail for it? The separator proof does not reach it (Section 4).
5. *Sparse cover degrees.* Are the cover discrepancies with n in $\{2^a 3^b\}$ dense in $\mathcal{B}$? Version 1 left this open. By Proposition 3.11 they are dense for every $t > (\log 2)(\log 3)/(4\pi) = 0.060598\dots$: Carleman’s formula in the half-plane gives $t > 0.121196\dots$, and in the angle $|\arg w| < \pi/4$ it gives $t > 0.060598\dots$ (the Jensen argument needs $t > (\log 2)(\log 3)/2 \approx 0.381$). It remains open for $0 < t \leq (\log 2)(\log 3)/(4\pi)$: Carleman’s formula in the sectors $|\arg w| < \pi/(2k)$, applied to $L(z^{1/k})$, gives the thresholds $(\log 2)(\log 3)/(2\pi k)$ for $0 < k \leq 2$ (from the identity $(4 - k^2) \int_{-\alpha}^{\alpha} \sin^2 \varphi \cos k\varphi d\varphi + 2k \sin^2 \alpha = 4/k$, $\alpha = \pi/(2k)$; check Z7), and for $k > 2$ its zero sum converges, so the argument of step 3 gives no contradiction. No finite combination of point evaluations and their derivatives annihilates the family (31_ Lemma 31.1, scope). More generally: which sets S of quadratic counting give density for every $t > 0$?
6. *Velocity sums.* The one-sided statements for the sharp velocity sums, and the lemma the copy calls U at rational non-Eulerian a (03_ §6); whether the sharp spectrum differs from the Dirichlet spectrum off the Eulerian sheets (numerically suggested at $a = 2$, not proved).

7. *Beurling systems*. Does $N(x) = cx + O(x^\theta)$ with $\theta < \frac{1}{2}$ force $\zeta_P \neq 0$ on $\sigma > \frac{1}{2}$ (02_§6)?
8. *The weight question*. Can the programme's reconstructed arithmetic supply either a square with a Künneth map and discrete weights, or an independent source for a pole bound uniform in the tensor degree (Section 5)? TWC9 and TWC11 state that the programme's computations do not rule such a construction out, and none has yet been assigned the fixed-curve cohomological denominator.
9. *The local model at τ* . A pole at τ , in the author's sense, would be computed from a local model of the test functions at τ (Proposition 6.1(b)). Which model does the programme intend?
10. *A ramified map*. The final reports of the programme's working sessions of 24–25 September describe a ramified map with the exact factor n^{-1} in the nilpotent relation, which the ordinary counting map lacks. How does it bear on the question of a uniform pole bound (Section 5)?

9.2 What was not read or not re-derived

- *Products of the programme's sessions*. The final reports of the programme's working sessions of 24–25 September name products that no audit note reads (41_§3): the global prime-clock comparison (GPC0–GPC12); a canonical weight transfer (CW1–CW8, a different file from the one read in 40_); a supported-cycle trace review; a three-point localisation derivation (TPL0–TPL13); an integral adjoint prism lift; a Deligne compact-bound subderivation and its review; the sections BQC0–BQC6, FT10 and PCJ10; a localised Cartesian proof check; ASQ0–ASQ12, CQF0–CQF10, CCR0–CCR10 and OZR10; and thirteen further named files (WR, DCA, VSD, GR, GTA, GTC, ESI, OEC, FGC, HSC, NHDA, GAPA and a review of the global Mellin synthesis). Two ancillary files named there are unread, and the TR0–TR6 wording correction could not be traced to a file (41_§1). The programme's TD, GHE beyond GHE1, TP and the global Mellin synthesis S1–S7 were read only in the content maps 01_–04_, not audited.
- *Programme blocks not read, or read in part*. NPE5–NPE10 and NER were not read; the appendix files GAP, GER, GEX, CTF, ASD, JTB, PGD, PGC, SCL and DPL were not read in full (only GEX1–GEX3, GEX7, GEX9, SCL5–SCL7 and PGD4 were checked); RSR and SSI10 were read in part. Each audit note lists, in addition, the inputs it did not check.
- *Inputs taken on trust*. Meyer's paper itself [32] (the comparison with Proposition 3.5 rests on the programme's description of its two theorems); the measure-theoretic steps of the programme's appendix CFPA.1–CFPA.7 (read in 04_, not re-derived; the division estimate itself was audited in 26_); GSP, CTS (read for correctness, not re-derived); the lattice spectral calculations of the Yang–Mills tensor transfer; the fixed-box weak-coupling limits used in Yang–Mills Theorem 14.2.
- *Independent checks still to do*. A line-by-line re-derivation of the copy's Theorem A; independent checks of Theorem A_a at $a = \frac{1}{2}$ and $a = 2$ and of the copy's Propositions G and H; an independent check of the reading of TWC and FST in 08_§3 (read only by claude-ab).
- *Novelty*. Novelty searches were made for some results only: Theorem 3.9, Theorem 2.3, the copy's Theorems A–F (by the copy), Corollary 2.2 (two quick searches) and the Weil II errata (a quick search); the catalogue records each result's status in the audit's words (*not searched*, *elementary*, *classical* and so on). The register lists novelty searches for S5–S9, S17–S18 and S23 as still to do, and the literature comparison of the classification of positive forms with Meyer and with Weil-positivity work as open.
- *The author's July 2026 notes*. Six were outlined only, and a seventh was outlined except for its Hopf-fibre section, which was read for Section 4; the note on the Giuga blind plane was read in full.

A Catalogue of the standalone results

This is the standalone part of the audit register (`00_RESULTS_REGISTER.md`, Goal 3): results that can be stated without the programme's vocabulary. *Programme* means proved in the programme's files and audited; *copy* and *claude-ab* are the two Claude instances (see the Introduction); a reference in the second column points to where the result is stated or proved in this reader. Novelty is reported only as the audit recorded it.

No.	Result	Status and source	Novelty
S1	Exact explicit formula for smoothed sums of zero velocities on every sheet $a > 0$ (Thm. 2.1)	proved; checked at $a = 1$ to $1.8 \cdot 10^{-18}$; copy, Theorems A, A_a	none found in the copy's search; standard technique as S1
S2	Smoothed velocity laws, horizontal and vertical, with explicit error at $a = 1$	proved; checked; copy, Corollaries B, B_a	
S3	Sharp velocity sums deviate at order T under (H); the smoothed error is almost periodic with explicit Bohr coefficients (unconditional)	parts 1–2 conditional on (H), part 3 unconditional; copy, Theorem C, proof completed by claude-ab (20_)	as S1
S4	Exact smoothed velocity spectrum on every sheet	proved; checked at $a = 2$ and $\frac{1}{2}$; copy, Theorem F	as S1
S5	The velocity spectrum factors over a $\mathbb{Q}$ -independent basis iff $a \in \{1, \frac{1}{2}\}$	proved; full referee pass (06_); copy, Theorem E	at $a = 1/q$ the criterion of [4]; other a open
S6	The velocity angle of the n -th zero is its Gram offset	proved assuming the zeros up to γ_n simple and on the line (which holds for $n \leq 1.2 \cdot 10^{13}$); copy, Theorem D	believed new; elementary
S7	Only two Hurwitz sheets are Eulerian (Cor. 2.2)	proved; claude-ab; also a consequence of S5	possibly known
S8	Sign-preserving mirror-commuting endomorphisms of $\mathbb{Z} \times \mathbb{Z}/4$ (§6)	proved; exhaustive for $ n \leq 9$; programme W5	not in Connes–Consani's paper as S8
S9	Parity is derived from the mirror; no equivariant section	proved; programme W2	
S10	The weight-one sign sector over the $\mathbb{F}_{12}$ -points is $y^2 = x^3 - x$ with Hecke character $\chi_{-4}(n)n$	proved (classical ingredients); copy R5	identification new to the programme
S11	Typed morphisms to the programme's twistor data; $L(\text{Sym}^2 E, s) = L(\psi^2, s)L(s - 1, \chi_{-4})$	proved (classical); checked for $n \leq 3000$; copy, Proposition H	classical
S12	Classification of positive forms compatible with the scaling transfer on $\mathcal{B}/I_\zeta$ (§3.5)	proved in the programme (CFP); core algebra checked	related to Weil's criterion; not searched
S13	Spectral synthesis for Connes' map: the closure of the image is the zero-jet ideal	proved in the programme (S5)	related to Meyer; not compared
S14	At most two primes have p^ρ algebraic, for nonreal ρ	proved (six exponentials theorem); programme IH5.3	elementary consequence of a known theorem
S15	Two primes suffice to see every zero	proved; programme IH6	not searched
S16	Deligne's specialisation-surjectivity argument (Weil II §3.6) with weight bookkeeping	reconstruction; programme TL1	classical
S17	Two primes suffice for generalised eigenvectors	proved; programme MCL3.4	case of [42]
S18	Least-order lifting at a zero of multiplicity m is $m + k + 1$; the fixed-order obstruction class	proved in the programme; verified for $m = 1, 2$ to 10^{-39}	not searched

No.	Result	Status and source	Novelty
S19	Even lattice sums at $a = 1/q$: free $\iff \varphi(q) \leq 2 \iff \text{formal log} \geq 0 \iff$ no zeros in $\text{Re } s > 1$ (Prop. 2.7)	proved (one step cites [40]); claude-ab; refereed	criterion classical; the sheet packaging not in [4]
S20	Source-pairing identity (§3.5)	proved; programme OPD4.2	elementary
S21	The harmonic (Poisson-swept) defect	proved in the programme (HSW5–HSW6B)	rational analogue of Connes §VIII
S22	Positive transfer forms before the quotient; descent iff the measure lives on critical ordinates	proved in the programme (SPF)	Bochner–Schwartz in Mellin coordinates
S23	The formal logarithm detects freeness; first negative coefficient (Thm. 2.3)	proved; programme (CEI, ETR1) and claude-ab independently	not located (one search)
S24	M_H half-factorial iff $[(\mathbb{Z}/q)^\times : H] \leq 2$ (Prop. 2.6)	proved; claude-ab	the Krull-monoid criterion (class group G/H); for $H = \{1\}$ [4, 14]
S25	Divisor-potential form of RH	proved in the programme (OZD); constants checked	Littlewood-lemma family; exact form not searched
S26	The swept translation trace	proved in the programme (FGR); checked	completes Connes' §VIII sketch
S27	Exact weighted dilation norm; prime spectra on $ z = \sqrt{p}$	proved; programme WHR8	elementary
S28	Connes' Hilbert quotients against Meyer's quotient: kernel the off-line jets	proved in the programme (WHR)	not searched
S29	Spectral synthesis on the zeta quotient	proved in the programme (GSP, CTS); not re-derived	Krasichkov-Ternovskii type
S30	The two poles differ in kind (Prop. 6.1)	derived; checked to 10^{-31} ; claude-ab	classical in content
S31	Connes' construction on every vertical line	proved in the programme (VWR)	other lines not checked
S32	The clock stack is ψ ; the completed prime clock is $\widehat{\mathbb{Z}}$ (Prop. 6.2)	proved; programme PMS	classical in content
S33	Nyman–Beurling in the Fréchet topology (Prop. 3.11; in version 2 with a Carleman criterion, see S75)	proved in the programme (NCl, RSS)	not searched
S34	The mirror line is the Hurwitz one-line shift (Thms. 4.1, 4.2; Props. 4.3, 4.4)	proved; claude-ab	not searched
S35	Positive measure receivers, all at once	proved in the programme (SMC)	not searched
S36	Residue pairing and diagonal return on the three-point base	proved in the programme (FTD–NHJ)	not searched
S37	The two-line product is real and positive on $\text{Re } s = 0$ (Prop. 4.5)	proved; claude-ab	not searched
S38	Two normalisations of running the clock backwards	proved; claude-ab	elementary
S39	The two lines as one double cover; the chiral pair	proved; claude-ab	not searched
S40	Exact aliasing for the NS torus endomorphism on finite character groups	proved in the ES workbench; checked	not searched

No.	Result	Status and source	Novelty
S41	Gram sandwich from a relative column error	proved; checked	elementary
S42	Every failure of RH is detectable (Prop. 6.5)	proved; claude-ab from the author's argument	standard
S43	Separator lemma with matched rates (Lemma 4.11)	proved (programme GSL, general constants by claude-ab)	standard $\bar{\delta}$ technique
S44	Two-line splitting for every primitive Dirichlet L -function (Prop. 4.12); for Dedekind zeta functions in outline (§4)	proved; claude-ab; the Dedekind form in version 2	not searched
S45	Exact range of the defect ratio (Lemma 4.13)	proved; claude-ab	elementary
S46	No equivariant maps across disjoint divisors (Lemma 3.3)	proved; programme GSL6, claude-ab	elementary
S47	The exact summation image (Prop. 3.5)	proved in the programme (SSI, OMS)	Meyer's range theorem made explicit, on the programme's account
S48	Positive transfer-adjoint forms are diagonal on the line zeros (Lemma 3.13; generalised in S74)	proved; programme PTQ4, shortened	elementary
S49	Global residue duality with explicit constants	proved in the programme (GZR)	classical residue pairing
S50	Spectral theory of multiplication by s on $\mathcal{B}/I_D$ (Prop. 3.2)	proved; programme RZ, claude-ab	elementary
S51	$\mathcal{B}$ is a Fréchet–Schwartz space (Lemma 3.1)	proved; claude-ab	standard
S52	Keller maps give incompressible polynomial flows; the escaping curve (Lemma 6.6)	proved; checked exactly; claude-ab	not searched
S53	The S^6 period block enters the YM background only through $D = Lq + 6m^2$	proved; claude-ab	elementary
S54	Signed clocks and the integral Bost–Connes ring	proved in the programme (CCT); finite-level index by claude-ab	consistent with [21]
S55	The first negative Euler-log coefficient above $-\frac{1}{2}$ (Cor. 2.4)	proved; checked on 286 generator sets; claude-ab	not searched
S56	Shifted zeta functions are multiplicatively independent (Lemma 2.9)	proved in the programme (ETR8); re-derived	not searched
S57	One tensor degree recovers the exponent measure	proved in the programme (ETR4–ETR9); re-derived	Deligne's §1.5 mechanism, credited
S58	Two conditions implied by RH, read off the zero-time Gaussian energy	proved; claude-ab	status of the conditions not claimed
S59	The corrected Noor tests are dense for every correction term	proved; claude-ab	elementary
S60	An infinite set of cover degrees need not suffice; in version 2 with infinite codimension (Prop. 3.12)	proved; claude-ab; codimension in version 2	elementary
S61	The residue extension survives evaluation only at the pole $s = 0$	proved; programme NPE4, claude-ab	elementary
S62	Tail annihilators of the corrected tests	proved; claude-ab	not searched
S63	Uniqueness of exponential sums on a divisor (Lemma 3.4)	proved; claude-ab (= programme HBW8.9–8.12)	elementary
S64	The winding at the zeros is hyperbolic (Lemma 6.3)	proved; claude-ab	elementary

No.	Result	Status and source	Novelty
S65	Escape under ramification (Lemma 6.4)	proved; claude-ab	elementary
S66	An algebraic purity criterion, the core of Weil II 1.8.4 (Prop. 5.3)	proved; checked on 142 random models; claude-ab	equivalence not claimed new
S67	Misprints in the printed Weil II (§5.4)	checked against the printed pages	a record; no novelty claimed
S68	The concentration lemma of Weil II 2.1.7 with its outside term (Lemma 5.4)	proved; checked; claude-ab	the repair is the programme reader's
S69	Nonnegative power sums; the simple pole of a curve zeta function (Lemma 5.6)	proved; claude-ab	no novelty claimed
S70	Pulling back to the doubled source adds no cohomology	proved in the programme (DCP5); every Ω and supports by claude-ab	elementary
S71	When the primary decomposition converges (Thm. 3.9)	proved; claude-ab; refereed	not found in the programme or the literature searched
S72	The chiral cross form is Weil's form on the diagonal; the holonomy index (Props. 4.6, 4.9)	proved; claude-ab	elementary; not searched
<i>Added in version 2 (48_ ; register S73–S79)</i>			
S73	The first Hurwitz jet on every sheet: its zero divisor is $\text{div } \zeta(\cdot, a) - 1$ (Thm. 4.1)	proved; the review's F1, re-derived by claude-ab	not searched
S74	Bounded Hermitian forms compatible with the transfer for $n = 2, 3$ are $\sum b_\rho m_\rho x_\rho \overline{y_\rho}$; positive exactly when b vanishes off the line and is ≥ 0 on it (Lemma 3.13)	proved; the review's F2, re-derived by claude-ab	elementary
S75	Carleman criteria for Nyman–Beurling density; $\{2^a 3^b\}$ is dense for $t > (\log 2)(\log 3)/(4\pi)$ (Prop. 3.11)	proved; the review's F3 (half-plane) and the version-2 referee's angle step, both re-derived by claude-ab	not searched
S76	Negative formal-log coefficients $1 - j + \varepsilon$ at every element whose proper M -divisors factor uniquely (Thm. 2.3, Cor. 2.4)	proved; the review's F4, re-derived by claude-ab	elementary
S77	The abscissa $1 + k\beta_{\max}$ in every tensor degree, for every finite S (Prop. 5.1)	proved; the review's F5, re-derived by claude-ab	elementary
S78	Zeros of $Z_{1/q}$ on both sides of the line and in $\text{Res} < 0$, for every q with $\varphi(q) > 2$ (Prop. 2.8)	proved using [40] as stated in its abstract; the review's F7 (prime q , quantifiers corrected by the verifier) and the version-2 referee's extension, re-derived by claude-ab	not searched
S79	The chiral form: unconditional orthogonal splitting and signature; $L(s, \chi)$ forms of Thm. 4.2 and Prop. 4.6	proved; the review's F17, corrected by the verifier	elementary

B Corrections recorded

Corrections are part of the record of any audit. This appendix lists them neutrally: first those to the audit's own notes and to the two Claude instances' earlier statements, then those made in the programme's files themselves, then the corrections of printed sources, then the corrections and generalisations of version 2 of this reader, and finally the changes of version 3. None of the corrections in the first three subsections changes a theorem stated in Sections 2–6; where a correction changed the scope of a statement, the statement in this reader is the corrected one.

B.1 Corrections to the audit's own statements

- *An earlier claim by claude-ab was false.* A note of 23 September claimed that zeros of $\zeta(s, 1+t)$ lie on vertical lines only at the Eulerian times. Certified crossings computed in the programme refuted it: zero 4 of ζ crosses $\operatorname{Re} s = \frac{1}{2}$ at $t = 0.95095300425$ and zero 5 at $t = 0.858704259442$ (located numerically at 25 digits). The erratum, `ERRATUM_FLIP_FABLE_ADDENDUM_4.md`, is among the withdrawn notes in the branch history cited in the introduction, and its check is `checks/zeta/hurwitz_crossing_check.py`.
- *A table of claude-ab labelled as the first 2000 zeros at $T = 2515$ contains 1999 zeros; the entries themselves are right for $T = 2515$.*
- *A zero count in 08_ was 13 where it should have been 24; found by a referee pass and by the programme's own reading, and corrected together with three citation fixes.*
- *Referee passes.* Nineteen passes over the audit notes and a two-part final pass over the summaries were made by Claude instances that had not written the notes. They found, among many wording and attribution points, the following statements false or overstated as written, all corrected before this reader was written: a Hurwitz flow described as translating the zero set (only its first jet vanishes on $Z - 1$); a pair of objects described as the two copies of the zero set when one of them holds only off-line data; a lemma false without the hypothesis $\delta \leq 1$ (the Gram sandwich); a heuristic about the Yang–Mills mass gap stated as a fact; the term *determinant weight* used for the average weight of a whole quartet, whereas Weil II's determinant weights belong to the constituents; the $\mathbb{Z}_4$ anomaly said to vanish identically (it equals the one-line parity); Turing's method said to give the bulk count exactly (Turing-type data give $P(T) + \sum \lfloor m/2 \rfloor$); the first Hurwitz jet's value at $s = 0$ (it is -1); and a numerical observation at $a = 2$ stated as a theorem. Each note lists the findings applied to it, and `10_RESULTS_DIGEST_DRAFT.md` §6 lists them pass by pass.
- *Priority.* The general formal-logarithm criterion and the sharp bound $-1/6$ were derived first in the programme (CEI Theorem 3.1 and ETR1); claude-ab's derivation is later and independent (29_ §1). The priority of the programme's NPE4.2 over part of 31_ Lemma 31.2 is likewise recorded.

B.2 Corrections in the programme's files

- *The programme's own late corrections* (final reports of its working sessions of 24–25 September) are carried by the versions audited here: the real boundary pole of the tensor Euler functions gives the position and dimension data only for conjugation-stable exponent data in even degrees (ETR9); zero multiplicity gives vanishing of the derivatives of order $< m_\rho$ at the endpoint and a nonzero m_ρ -th derivative, rather than vanishing precisely for $j < m_\rho$ (GJN0.8, GJNR8); a proof step in CS2 is done on differences before passing to classes. A wording correction to TR0–TR6 (scalar trace values test zero location; the full trace operators recover local jets) could not be traced to a file with certainty. One claim was withdrawn in the programme on 24 September, together with the matrix file on which it rested (41_ §§1–2).
- *Small points found by the audit* in the programme's files, each harmless to the result concerned: typographical slips in the heat-metric blocks HM10 and HM19; the expression $2j!$ in NH10, HSR0.2, HBW0.1 and HBWR0.1 must be read as $2 \cdot (j!)$; a stated file hash that differs by one character from the file's; and a corollary in one of the author's notes of July 2026 stating that the critical line, with the pairing $(s, 1 - \bar{s})$, maps onto the whole Hopf fibre, whereas its image is the open half-circle (the line at $-\frac{1}{2}$ gives the other half; 17_ Proposition 17.3(c)).
- *The programme's Deligne reader* flagged a number of displays of Weil II and repaired them. Its reading log records a comparison with the printed pages for §1, placing for example (1.3.10)(iv) and the twist in (1.6.14.3) in the original; its sections on §§2–3 make no claim about the printed

original and treat the discrepancies there as defects of its transcription witnesses. The printed pages show that these, too, are in the original (Section 5.4). The reader's mathematics stands.

B.3 Corrections of printed sources

The printed displays of Weil II listed in Section 5.4: six in §1 (two of them in (1.7.5)) and seven in §§2–3, among them the choice $\varepsilon_1 + \varepsilon_2 < \varepsilon$ in the proof of 2.1.7, which does not give Deligne's inequality, while $\varepsilon_1 + 2\varepsilon_2 < \varepsilon$ does (Lemma 5.4). None affects a conclusion of Weil II.

B.4 Version 2 of this reader

Version 1 of this reader (Zenodo 10.5281/zenodo.22970703) was refereed for errors and overstatement before publication. At the author's request it was then reviewed for understatement and missed generality as well as for overstatement, by a Claude instance that had written neither the reader nor the notes. The author of this reader verified its four corrections and five major findings against the sources, re-deriving each proof, with independent checks (Z1–Z6); a second independent instance verified the seventeen minor findings with checks of its own (48_; checks/zeta/zeta_reader_review/verify_minors/). Version 2 applies them; the status line of every changed statement says what changed. Version 2 then had its own referee pass (Appendix C), whose two new mathematical items, the angle condition of Proposition 3.11 and composite moduli in Proposition 2.8, were re-derived by the author (checks Z7–Z8) and applied. 48_ and the review use the numbering of version 1; version 1's Lemmas 2.8–2.9 and Corollary 2.10 are Lemmas 2.9–2.10 and Corollary 2.11 here.

Corrections.

- O1. Proposition 5.2 stated the rank $4m$ and the determinant p^{2m} for every zero. At an on-line zero the four points $\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}$ are two, the rank is $2m$ and the determinant p^m . The conclusion, average weight 1, is unchanged.
- O2. Corollary 2.2 said that $\zeta(s, 1+t)$ has an Euler product of Euler type only for $t \in \{0, -\frac{1}{2}\}$. At $t = -\frac{1}{2}$, $\zeta(s, \frac{1}{2}) = 2^s \prod_{p>2} (1 - p^{-s})^{-1}$, so the Euler product is that of the normalised sheet $2^{-s} \zeta(s, \frac{1}{2})$; the corollary now speaks of $a^s \zeta(s, a)$.
- O3. The source note 02_ §4.2 concludes that both error terms of Corollary B tend to 0 once $\Delta \geq (1 + \delta) \sqrt{2 \log T_2} / \log 2$ for a fixed $\delta > 0$; this also needs $T_1 / \Delta \rightarrow \infty$ (for example $T_1 \geq 4\Delta \sqrt{\log \Delta}$) and O -constants independent of Δ . The bullet on Corollaries B, B_a in Section 2 now says so.
- O4. The outline of the proof of Proposition 3.11 ended with a step that gives $\Lambda = 0$ only on $g_t \mathcal{B}$. The missing step, the density of $g_t \mathcal{B}$ in $\mathcal{B}$, is now proved, and the proposition has a complete proof.

Stronger statements. Theorem 4.1 on every sheet $a > 0$ with its whole-plane divisors; Lemma 3.13 for all bounded Hermitian forms compatible with the transfer for $n = 2, 3$; Proposition 3.11 with Carleman criteria in the half-plane and in the angle $|\arg w| < \pi/4$, which answer open question 5 for $t > (\log 2)(\log 3)/(4\pi)$ (the angle step is the version-2 referee's); Theorem 2.3 and Corollary 2.4 at every element whose proper M -divisors factor uniquely; and Proposition 5.1 in every tensor degree, without positivity. These are register items S73–S77.

Smaller additions and omissions made good. Proposition 2.6 without Dirichlet's theorem and Example 2.5 on the whole half-plane of absolute convergence; Proposition 2.8 (zeros of $Z_{1/q}$ on both sides of the line for every q with $\varphi(q) > 2$, using Saias–Weingartner as stated in its abstract; S78); Lemmas 2.9 and 2.10 for complex data; Theorem D's hypothesis in the verified range; Proposition 3.2 and Lemma 3.3 for $D \leq \text{div } \Phi$; Lemma 3.4 with polynomial factors; Lemma 3.6 with a sharper factor and Corollary 3.7 with density in Λ_m ; Theorem 3.9(b') and its form for $L(s, \chi)$; Proposition 3.12 with infinite codimension; the residue-duality constant 8.79; Proposition 4.3 with

one dilation; Proposition 4.6(c) unconditionally, with the $L(s, \chi)$ forms of Theorem 4.2 and Proposition 4.6 (S79); Proposition 4.10 for every modulus; the separator for Dedekind zeta functions; the direction of monotonicity in Lemma 4.13; Lemmas 5.5 and 5.6 in real-valued and exponential forms; Proposition 6.2 for every integer $a \geq 2$; and Lemmas 6.4 and 6.6 for real $a > 1$ and for C^1 fields. The independent verification corrected the review's wording in six of these (Proposition 2.8, Theorem D, Theorem 3.9, Proposition 4.6, the Dedekind separator and Lemma 6.4), and rejected one proposed item, for Proposition 6.5, as not an understatement.

None of these is a new statement about the location of the zeros.

B.5 Version 3 of this reader

Version 2 was withdrawn on 27 September 2026, together with the audit notes, because of its framing. Version 3 keeps every theorem, proof, figure and citation of version 2 and corrects the framing; the only changes of content are the two decimal renderings and the three edge statuses listed at the end of this subsection. The changes of framing are of four kinds.

Scoped negative results instead of verdicts. Where version 2 turned a scoped negative result into a general statement that a route does not decide the location of the zeros, version 3 states the scoped result and, where it is known, the input the construction would need. This concerns Section 7, now titled *Proved negative results, with their exact scope*, and in particular:

- the items N-Mirror, N-Logic, N-Pos1–N-Pos4, N-Unif1 and N-Unif3–N-Unif6, N-Prim, N-W1–N-W5, N-F1, N-Two3 and N-X;
- the closing paragraph of Section 3.5;
- the discussion after Proposition 5.1 and of Weil II 1.5.1;
- the status of Lemma 6.4;
- the discussion after Proposition 6.5.

The item N-Unif6 now states that the cohomology of the doubled source is that of Connes–Consani's base, and what this gives for the spectral realisation of the zeros, instead of a summary that the doubled source adds nothing.

Bridges. Section 8 is restructured. Each bridge now gives its typed map, what is established, what it suggests on each side and my assessment, and the introduction presents the bridges among the main outcomes. The mathematical content of each item of version 2's list is kept.

Front matter. The abstract and introduction are rewritten: they state the main results as Theorems I–V, explain what they mean, describe the bridges, and set out the four statuses. They also say how the research was carried out and where the withdrawn notes and the republished checks are.

Wording. Quotation marks around the author's terms and around standard terms are removed. The word *pure* in Section 8 is used in its standard sense for Weil–Deligne representations. Session reports are cited as the programme's. Question 9 and the other items of Section 9 are stated as questions.

Changes of content.

- Two decimal renderings were wrong. One has $(\log 2)(\log 3)/(4\pi) = 0.0605982\dots$ and $(\log 2)(\log 3)/(2\pi) = 0.1211965\dots$, which version 2 printed as 0.06060... and 0.12120... (in Section 3, after Proposition 3.11, and in Question 5); version 3 prints 0.060598... and 0.121196.... The thresholds themselves, stated as the closed forms, are unchanged.
- Three statuses in the table of Section 8.10 are updated from the companion records: the edge from S^6 to the Erdős–Straus workbench (the orbits of the covers are verified at every level in the Erdős–Straus record), the edge from problem 817 to zeta (one of the ten interface notes is verified in the companion record), and the Collatz edge (verified in part in the companion

record). Each keeps the status it had in version 2, marked as such.

Apart from the two decimal renderings and the three updated edge statuses, no statement changed its mathematical content, and none was removed.

C How the audit was verified

C.1 What was read

The audit read the programme's files at the commits recorded in the notes, and in particular: the second attempt's fixed-point, clock, positive-quotient, integrality, weight-separation and character-lifting blocks (content maps 01_, 04_, 05_, 07_); the continuations of 24–25 September (OZD, SPF, GSP, CTS, FGR, WHR, HCS, GMC, VWR, ATG, PMS, NCI, RSS, NHJ, SMC, GAP, ADM, FTD, PRS, DER, FSC, FEM); the analytic core of the three-strand edition (GSL, GMS, AST, GDC, PTQ, SSI, ECR, ECI, GZR, OMS, CGS, DCP, RTT, RZ, SDT, FOD, NEA); the bulletin of 25 September (ETR, CEI, GDE, ABH, NJS, SSR, NPE0–NPE4, NHI, HSR, HBW, BML, BRC, GJN, BNS); the programme's reconstruction of Weil II (DW) and its full Deligne reader (DP, DMG, DLM, DB, DBC, DW0–DW4, MDB); the copy's two rounds of results; the cross-programme material of the Erdős–Straus, Yang–Mills, Navier–Stokes and S^6 workbenches used in Sections 6 and 8, including the proof of the Yang–Mills workbench's Theorem 14.2; and, in outline, the final reports of the programme's 94 working sessions of 24–25 September. What was not read is listed in Section 9. Printed sources were checked where the notes say so: Weil II on the numdam scan (pp. 150–203), Titchmarsh's Theorems 9.2, 9.6(A) and 9.7 (by a referee, on page images), and the published pages of the cited theorems of Berenstein–Taylor and Abanin–Le Hai Khoi–Nalbandyan.

C.2 Referee passes

The audit notes 08_–42_ were each refereed by a Claude instance that had not written them; the referee re-ran the note's check script, wrote checks of its own, and reported findings as major (a statement false or overstated as written) or minor. The content maps 01_, 04_, 05_, 07_ and the copy's reports 02_, 03_ had no separate pass (their summaries in the register were checked in final pass A, and the copy's Theorem E had a full referee pass, 06_). All findings were checked by the author of the note against the sources before they were applied.

Pass	Notes	Findings
1–2	08_ ; 09_ and digest draft 1	a zero count, three citations, overstatements
3	11_–13_	25 (wording, labels, hypotheses; no mathematical error)
4	14_–16_ , digest draft 2	2 errors of statement, 21 further
5	17_	16
6	18_–20_	1 major, several overstatements, 3 gaps in 20_
7	21_	14
8	24_–26_	1 major, 8 minor
9	27_–28_	1 major, minor
10	29_–31_	5 major (attribution and record-keeping), 18 minor
11	32_–33_	2 major, 16 minor
12	34_–35_	3 major, 26 minor
13	36_	2 major, 17 minor
14	37_	2 major, 18 minor
15	38_	2 major, 13 minor
16	39_	3 major, 9 minor
17	40_	2 major (wording), 14 minor
18	22_–23_	4 major, 18 minor
19	41_–42_	3 major, 11 minor
final A	register S1–S72, digest §§1–2, cross-references	6 major (all in summaries), 54 minor
final B	negative results, bridges, $\mathbb{F}_1$ items, digest §§3–6	7 major (all in summaries), 36 minor

The copy's Theorem E had a separate full referee pass (06_), and the proof of its Theorem C was completed and refereed (20_). This reader was compiled from the refereed notes and then had a referee pass of its own, by a Claude instance that had not written it: it found no gap in the proofs, 12 major findings (scope, credit, verification claims, and hypotheses dropped in summarising) and 28 minor ones, and all were applied before publication.

Version 2. After publication the author asked that reviews also look for understatement and missed generality. A further Claude instance, which had written neither the reader nor the notes, reviewed version 1 in both directions with ten check scripts of its own; it reported four overclaims or gaps (O1–O4), five major understatements (F1–F5) and seventeen minor ones (F6–F22). The author of this reader verified O1–O4 and F1–F5 against the sources, re-deriving each proof, with independent checks (Z1–Z6); a second independent instance verified F6–F22 with eleven scripts of its own (eleven confirmed, six confirmed with a corrected wording, none rejected; one sub-item, for Proposition 6.5, was found not to be an understatement). The outcome is 48_, and Appendix B lists what changed. Version 2 then had a referee pass of its own, by a further Claude instance that had written none of it, in both directions and with sixteen scripts of its own. It re-derived every new or changed proof, found no false theorem and no missing hypothesis in a theorem, and found nothing true in version 1 lost. It reported 3 major findings: the density threshold for $\{2^a 3^b\}$ can be halved with Carleman's formula in the angle $|\arg w| < \pi/4$ (an understatement; now condition (C2) of Proposition 3.11); the credit for the version-2 results, which the review found and proved; and Appendix B's account of who verified what. It also reported 13 minor ones, among them the extension of Proposition 2.8 to every q with $\varphi(q) > 2$, a numerical slip in the proof of Lemma 3.6, an overstated clause after Theorem 4.1, and missing caveats where a source was not re-read. The author re-derived the two new mathematical items, with checks Z7–Z8, and applied all the findings. Version 2 adds to the checks `zeta_reader_addendum_checks.py` (Z1–Z8) and the review's, the verification's and this referee's scripts in `zeta_reader_review/`, both in the same folder.

C.3 Checks

Every numerical or symbolic claim in the notes has a script with recorded output. The scripts are republished in the folder `contrib/claude-ab/rewrite/checks/zeta/` of the branch `claude/claude-ab-grind-20260925`: for version 1, 48 scripts at its top level (47 of them with a recorded `_OUTPUT.txt`), four further scripts of the first referee pass in `referee08/` with their argument-principle outputs, five figure scripts in `figures/`, and the copy's code in `copy_round2_code/` (which reads its zero data from the directory named by the environment variable `ZERO_DATA_DIR`). They use `mpmath`, `sympy` and, for the argument-principle counts, `Arb` ball arithmetic; runtimes are seconds to minutes. Checks marked *expected fail* in a referee's script document a finding (for example the counterexample of Lemma 5.4) and are not failures of the note. Every check is a check of a stated identity or bound; no statement of this reader rests on a numerical check alone unless it is labelled as numerical.

C.4 The audit notes

The notes were withdrawn from the branch `claude/claude-ab-grind-20260925` on 27 September 2026 because of their framing. Their mathematics is unchanged, and they remain in the branch's history at commit `a90d9e5b`, in the folder `contrib/claude-ab/grind_20260925/`.

Note	Subject
00_	results register (all results, negative results, bridges, $\mathbb{F}_1$ items, open items)
01_, 04_, 05_, 07_	content maps of the second attempt: fixed point and clocks; positive quotient; integrality, loops and weight separation; character lifting and the Deligne cross
02_, 03_, 06_, 20_	the copy's results (rounds 1 and 2), the referee report on Theorem E, Theorem C re-proved
08_, 35_–38_	Deligne's mechanism; the programme's Weil II reconstruction and Deligne reader
09_, 12_, 14_–16_, 18_, 19_	continuations: harmonic sweep and source pairing; divisor potential, synthesis and Hilbert returns; weight receivers and tensor growth; clock stack; Nyman–Beurling in the Fréchet topology; the two lines in the programme; product trace and diagonal return
10_	results digest
11_, 13_	free monoids, Eulerian sheets and $\mathbb{F}_{12}$ units; the ledger of the zeros and ones
17_, 22_, 23_, 42_	the mirror line and the Hurwitz jet; the anomaly picture; the detectability argument; the two-line directions
21_, 28_, 34_	cross-programme bridges (parts 1 and 2); the Yang–Mills Theorem 14.2 audit
24_–27_, 29_–33_, 39_, 40_	audits of the programme's blocks (separator, summation image, residue duality, spectral theory; Euler coefficients and tensor products; Gaussian energy and Hardy receivers; residue sections; holomorphic receivers; windings and ramification; the doubled source; the jet map and the primary decomposition)
41_	the logs of the programme's working sessions: corrections carried, one withdrawal, what remains unaudited
43_–47_	audits of the Erdős–Straus, Collatz, Erdős Problem 817 and Yang–Mills workbenches and of their intersections (the basis of the companion records)
48_	the addendum to version 1 of this reader: corrections O1–O4 and the generalisations applied in version 2

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